

Matchmaking Strategies for Maximizing Player Engagement in Video Games

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Managing player engagement is vital to the online gaming industry, given that many games generate revenue through subscription models and microtransactions. We scrutinize engagement management in the prevalent category of competitive video games, where players are frequently matched against one another, and matchmaking systems substantially impact engagement. We propose a dynamic model to analyze player dynamics and optimize matchmaking policies for maximum engagement. Our model takes into account two essential factors in competitive games: heterogeneous skill levels and players' aversion to losing. Additionally, the model enables us to consider pay-to-win strategies and AI-powered bots, which are contentious methods of influencing player engagement and endogenously affect the optimal matchmaking policy.

To provide sharp insights, we analyze a specific case where there are two skill levels, and players churn only after experiencing a losing streak. The optimal matchmaking policy considers both short-term rewards by matching players myopically and long-term rewards by adjusting skill distribution. The pay-to-win system can positively impact player engagement when the majority of players are low-skilled, as adopting pay-to-win also affects skill distribution. This result challenges the conventional wisdom that typically regards pay-to-win as trading player experience for revenue. When incorporating AI-powered bots, we demonstrate that optimizing the matchmaking policy can significantly reduce the number of required bots. We then extend our model and conduct a case study with real data from an online chess platform. The optimal policy can improve engagement by 4-6% or reduce the percentage of bots by 3% in comparison to skill-based matchmaking.

Key words: matchmaking; video games; engagement; pay-to-win; customer retention

1. Introduction

The global revenue of the gaming industry is estimated to reach a record-high of \$187.7 billion in 2024, involving more than 3.4 billion players over the world (Newzoo 2024). One significant category of online games is competitive games, in which players engage in matches against one another, either individually or as part of teams. Popular competitive games often involve millions of players and generate substantial revenue. For instance, in 2022, the online competitive games

Honors of Kings and *PUBG: Mobile* amassed revenues of 2.2 and 1.7 billion dollars, respectively, making them the highest-grossing mobile games.¹ Additionally, the daily active users for such games can be as high as 100 million.² These games (and many others) adopt the games-as-a-service (GaaS) approach, providing game content through a “continuing revenue model”. These games are typically free-to-play, with revenue generated from in-game advertisements, microtransactions for virtual items such as loot boxes, and subscriptions for seasonal premium passes (sometimes called “battle passes,” which offer premium content for subscribers). Managing player engagement is vital for such free-to-play games, as the total revenue is directly linked to the number of active players. Daily active users (DAU) and concurrent users (CCU) serve as key performance indicators in the gaming industry, with high DAU/CCU values generally indicating strong product performance.

Various factors determined before the release can influence player engagement, such as game content, game mechanics, and user interfaces. In addition to these factors, the gameplay experience, as reflected by match outcomes, is crucial in affecting engagement in competitive video games. Studies by Chen et al. (2017b) and Huang et al. (2019) have documented the significant impact of match outcomes, including wins, losses, and scores, on player engagement. One natural way to intervene in players’ match outcomes is through the matchmaking system, which determines the opponents for each player in every match, directly affecting the current match’s outcome and indirectly influencing players’ future engagement behavior. The prevalent approach, known as *skill-based matchmaking* (SBMM), matches players by finding opponents with the closest skill level (Park 2021) and disregards the outcomes of previous matches. While SBMM is straightforward and ensures fair gameplay, it has received considerable criticism from media and players for making games less casual.³ To illustrate this issue, consider a casual player who participates in three matches every night. Under SBMM, a player loses about half of their matches. Consequently, there is a substantial probability (about one in eight) that three consecutive losses ruin their entire night. Such an outcome is highly undesirable for players and may lead to permanent disengagement (churn).

A recent update by popular game *Dota 2* acknowledged the challenge of matchmaking and confirmed players’ desires to occasionally play with weaker opponents (and win): “*Matchmaking can be a challenge, especially if you’re good at Dota, and like playing with people who are almost but hopefully not quite as good at Dota as you are.*”⁴ Industry practitioners have recently begun exploring how to leverage matchmaking systems to improve player engagement (thereby increasing

¹ <https://mobilegamer.biz/2022s-top-grossing-mobile-games-honor-of-kings-pubg-mobile-genshin-impact-and-more/>

² <https://techcrunch.com/2020/11/01/tencent-claims-record-100m-daily-users-on-mobile-game-honor-of-kings/>.

³ <https://www.dexerto.com/gaming/what-is-sbmm-in-gaming-and-why-is-it-so-controversial-1367022/>

⁴ <https://www.dota2.com/newfrontiers?l=english>

revenue). For example, Electronics Arts has filed a patent for an engagement-oriented matchmaking system, which largely follows Chen et al. (2017b).⁵ Although they recognize the dynamic nature of the problem, their patent is based on a static one-shot model. Modeling active players’ evolution is challenging, as they interact repeatedly over time. Furthermore, providing a theoretical analysis of how to design the optimal policy for such a complex system, as well as the value of such an effort compared to the status quo, is even more difficult. At first glance, the system appears to be a zero-sum game between players, as only half of the players can win a competitive game under any matchmaking policies. The potential benefits arise from the fact that players may exhibit a memory effect in relation to their recent gameplay. However, solving a dynamic problem with a memory effect is notoriously difficult. In this paper, we provide a dynamic model and analytical results that capture the impact of matchmaking policies on player behavior and engagement.

In addition to selecting matching partners, video game companies also influence engagement and the matchmaking system in other ways. One controversial practice is introducing a *pay-to-win system* (PTW). PTW allows players to improve their competence by paying real currency for better performance. For example, the popular competitive game *Dota 2* offers a subscription service called “Dota Plus”. At the start of each match, ten players pick their characters one-by-one, and then subscribed players receive analytical suggestions on which character to pick based on the existing picks so far and winrate data from recent weeks. Moreover, subscribed players receive data-driven suggestions during the match on items to buy and skills to learn. Such information may not be necessary for professional players, but it provides substantial help to most amateur players. While PTW provides a direct revenue stream, PTW also influences engagement of subscribed and unsubscribed players through match outcomes. In our model and results, we capture the interplay between PTW and the matchmaking system and show how it affects overall revenue and the engagement of players.

Another controversial way to influence engagement is to add AI-powered bots into the matchmaking pool. AI-powered bots imitate real human players in competitive games, and players are not notified when they are matched with a bot. Many competitive games are suspected and criticized of having a high bot ratio (Gelbart 2021), and some games such as *Fortnite* have publicly confirmed the existence of bots in human matches.⁶ Such practices are actually double-edged swords. On the one hand, the outcome of a match with bots can be easily controlled because bots can be made to perform very poorly or strongly as needed. Thus, AI-powered bots can influence player engagement positively by manipulating the match outcomes properly. On the other hand, bots can be clumsy or

⁵ <https://www.pcgamesn.com/ea-matchmaking-microtransactions-eomm-engagement-patent>

⁶ <https://www.epicgames.com/fortnite/en-US/news/matchmaking-bots-controls-and-the-combine-update>

predictable at times and are often recognized by human players after repetitive encounters. Players are not happy when they intend to compete with other human players but find out that they are frequently in a match with bots. Players who have made in-game purchases and want to use (and parade) them on other players instead of bots can be particularly upset about this. In practice, the value of either PTW or AI-powered bots is hard to evaluate through field experiments since the impact is confounded with the matchmaking policies used. Our model allows us to analyze these interventions in conjunction with an optimal matchmaking strategy to maximize engagement.

To analyze the intricate player dynamics, we propose a novel infinite-horizon dynamic program that aims to maximize the cumulative active players, which is our metric for engagement. We assume that players have heterogeneous skill levels, and their state depends on the win-loss outcomes of the most recent past matches. In each period, the game decides whom to match with each other, which directly influences the outcome of the current match and indirectly influences the long-term engagement. Under the same framework, we consider PTW by separating players into subscribed and unsubscribed players, and investigate AI-powered bots by matching players with bots. To our knowledge, we are the first to study engagement maximization by directly monitoring the interaction of players in a dynamic setting. We are also the first to investigate the economic value of PTW and AI-powered bots through the lens of player engagement. Below we provide a summary of our contributions and findings.

- (1) We analyze a stylized variant of our model with only two skill levels and where players leave if and only if they lose two games in a row. We fully characterize the optimal dynamic policy under the baseline model. The optimal policy matches players myopically for short-term rewards but also adjusts player distribution with tie-breaking for long-term rewards. We also show that the benefit of optimal policy does not snowball over time: compared to the benchmark SBMM, the maximum benefit in one period is 33.3% and increases to 50% over an infinite time horizon.
- (2) Using the stylized model, we then consider the interplay between PTW strategies and the matchmaking system. We find that contrary to the conventional wisdom that PTW is simply to increase revenue, it can also be an effective lever to change the distribution of player states and increase engagement. The potential positive externality of player engagement may transform the public perception of PTW strategies.
- (3) Next, we investigate the value of optimizing the matchmaking system in the presence of AI-powered bots. We assume that a bot match may lead to a satisfactory outcome when players do not realize they are playing with bots, and the more bots there are, the more likely players will detect the bots (and thus get frustrated). For a platform using SBMM with a relatively small bot ratio, it may achieve the same engagement level with no bots at all if it transitions to

an optimal matchmaking policy. Interestingly, even when the platform prioritizes maximizing engagement by synergizing bots with an optimal matchmaking policy, the optimal bot ratio under this optimal matchmaking remains lower than that under SBMM.

- (4) Finally, we generalize our stylized model to a more comprehensive model and conduct a case study with data from Lichess (a large online chess platform) to validate our findings. After fitting a player behavior model within our framework, we demonstrate that the optimal policy could augment engagement by 4-6% compared to SBMM. Heuristic policies, guided by insights from the primary model, also yield satisfactory results. The optimal bot ratio under the optimal policy is less by 3% than that under SBMM. While the optimal policy does introduce disparities in user experience across skill levels, our modeling framework enables us to modify the matching policy and attain an almost Pareto improvement over SBMM.

2. Literature Review

Our work contributes to the emerging literature on operations management in video games. Closest to our work, Chen et al. (2017b) and Huang et al. (2019) also investigate the problem of maximizing player engagement in video games through matchmaking. Chen et al. (2017b) proposes a model that estimates the churn risk of every pair of possible matches through logistic regression and myopically minimizes the churn risk in the next round. Their numerical study considers one-period problems and shows 0.7% improvement over SBMM in one period. On the other hand, Huang et al. (2019) estimates user engagement with a hidden Markov model and proposes a heuristic algorithm that assigns a selected player to one of the pre-specified candidate matches (assuming all the other players are fixed). In contrast, our paper investigates how to optimally solve the dynamic matchmaking problem through a fluid model, taking both the myopic reward as well as the long-term player engagement into account. We also explicitly solve for the optimal policy in special cases and provide insights on how the policy looks. Our new framework also enables us to analytically investigate the value of optimal dynamic matchmaking, the pay-to-win system, and AI-powered bots. Recent works also investigate player engagement empirically (Goetz and Lu 2022, Jiao et al. 2022, Avoyan et al. 2024), but provide little guidance to the matchmaking policy.

Aside from matchmaking, several papers in this field consider the monetization of video games (Turner et al. 2011). For example, Sheng et al. (2020) and Sheng et al. (2025) consider the monetization problems in freemium games through incentivized advertisements and bonus actions, respectively. Chen et al. (2020) considers how to optimally price and design ‘loot boxes’, which is a popular randomized selling scheme for virtual items in video games. Mai and Hu (2022) considers the introduction and pricing of premium content in freemium games. Jiao et al. (2021) considers whether the seller should disclose an opponent’s skill level when selling PTW items, and Deng

et al. (2024) considers the pricing and quality decisions for the PTW services. Our paper also investigates PTW, but focuses on the interplay between matchmaking and PTW. While previous papers usually assume random matching or skill-based matching for tractability, our model enables us to check the joint value of PTW with the optimal (endogenous) matchmaking policy. As a result, our insights on the role of PTW in shifting players' skills are new to the literature. Apart from monetization, there is increasing interest in the design of video games. Hanguir et al. (2024) recently considered a problem on how to design the game loadouts to maximize the strategies' diversity. Li et al. (2022) and Li et al. (2023) consider how to maximize player utility by optimizing the topological structure of gaming elements in a single-player game.

Our work also connects to the growing literature on dynamically managing user engagement and user lifetime value in a service system. In such systems, the revenue is usually proportional to the amount of cumulative active users, and users' subscription behavior is based on the service history up to date. Aflaki and Popescu (2014) considers how to maximize user engagement by dynamically adjusting the service quality. Kanoria et al. (2023) considers how a fund manager should switch between risk mode and safe mode to maximize customer lifetime value. Bernstein et al. (2022) and Caro and Martínez-de Albéniz (2020) consider how a content provider (e.g., video streaming services) could maximize the subscription revenue by dynamically changing their content. Our work contributes to this field by considering how to manage user retention with dynamic matchmaking in video games. While the decision variable in most of the above papers is a single variable (service quality or service mode), the decision in our paper is multi-dimensional (matchmaking flows), which brings fundamentally new technical challenges and insights.

More broadly, our work relates to dynamic decision-making when players have a memory of the decision history. Research in this field has considered dynamic pricing with reference effects (Popescu and Wu 2007, Hu et al. 2016, Chen et al. 2017a, Cohen-Hillel et al. 2023), dynamic capacity allocation with customer memory effects (Adelman and Mersereau 2013), network revenue management with repeated customer interactions (Calmon et al. 2020), improving matching rates in dating markets (Rios et al. 2022), and dynamic personalized pricing with service quality variability (DeCroix et al. 2021). When customer preferences are not fully specified, Bastani et al. (2022), and Cao et al. (2019) consider how to learn customer preferences on the fly in the context of product recommendation or promotion, with the risk that customers may leave the system permanently upon consecutive bad decisions. In our work, user churn decision depends on the outcomes of the most recent previous matches.

Finally, our work contributes to the broad literature on dynamic matching. Here we only review papers where agents may only stay in the system for certain periods before leaving. Ashlagi et al. (2023) provides an approximation algorithm for a setting where customers will stay a fixed number

of periods before leaving. Aouad and Saritaç (2022) propose approximate algorithms for dynamic matching over edge-weighted graphs, where the arrival and abandonment of agents are stochastic. Hu and Zhou (2022) considers dynamic matching over a bipartite graph with finite types of nodes on both sides. Unmatched supply and demand may incur waiting or holding costs and will be partially carried over to the next period. Our paper considers dynamic matching with finite types of players in a fluid model (see, e.g., Azevedo and Leshno 2016), and player churn risk evolves dynamically based on outcomes of past matches.

3. Baseline Model

In this section, we introduce a simple baseline model to illustrate our key assumptions and the problem formulation. We provide sharp characterizations of this model in Section 4.1, and derive managerial insights on pay-to-win and AI bots based on this model in Sections 4.2 and 4.3, respectively. We then generalize the player model in Section 5 before setting up the numerical studies.

Consider a matchmaking system for a 1-versus-1 competitive video game. We assume that each player has a skill level that describes their relative competence in the game. In the baseline model, there are 2 ordered skill levels, where level 2 is more skilled than level 1. For each match, exactly one of the two players will be the winner, and the outcome of each game is either a win or a loss (no draw). Players of the same skill level are equally likely to win. A level 2 player with a higher skill level than their opponent has a strictly larger than 0.5 probability of winning, and for simplicity, we assume that level 2 players always win when matched with level 1 players. We assume that a player’s skill level is fixed over their lifespan in the matching system. In practice, most players are casual players, and it is reasonable to assume that they cannot significantly improve their relative competence once they are familiar with the game. In our case study (Section 5), we also provide supporting evidence of this assumption.

Next, we model the players’ churn behavior. In practice, many factors may influence the players’ engagement. The outcome of past matches has been shown to have a significant impact on player engagement. For example, Chen et al. (2017b) documented that players’ churn risk varies significantly with the win-loss outcomes of the last three matches. We assume that a player’s *engagement* state is determined by the win-loss record, and players favor wins over losses. This assumption follows the cognitive evolution theory in psychology that the *intrinsic* motivation to an activity (by the underlying need for competence and self-determination) would increase (decrease) when perceiving oneself as competent (incompetent) (Deci and Ryan 1980). Winning and losing are natural signals of competence and incompetence, and the relative benefit of winning over losing on the intrinsic motivation (measured by subsequent time spent on the game) has been widely supported in the literature (Reeve et al. 1985, Vansteenkiste and Deci 2003). Despite the outcome itself, game

designers also offer various *extrinsic* incentives contingent on match outcomes, such as badges and in-game currencies, which further reinforce players' desires to win (Richter et al. 2015). In our baseline model, we use the simplest setting that captures the aversion of losses: players leave the game if and only if they have experienced two consecutive losses. We assume that the players may leave only upon the second loss for two reasons: first, people generally accept the fact that both wins and losses happen in a competitive game, and one loss should be acceptable. Second, if people leave upon their first loss, then the system would be memoryless, and all the policies are effectively the same (regardless of how we match, half of people lose the game and leave). Thus, the memory effect has to be modeled with at least two matches.

Under this stylized model, we can classify active players into 4 *demographics*: high (level 2) and low-skilled (level 1) players who won the past game, denoted by $2w$ and $1w$, respectively; high and low-skilled players who lost the past game, denoted by class 2ℓ and 1ℓ , respectively. For convenience, we define $\mathcal{P} := \{1w, 1\ell, 2w, 2\ell\}$.

Now we describe the problem from the perspective of the firm/matchmaker. In practice, players request matches randomly, and the matchmaker reviews the matchmaking pool periodically and formulates matches based on specific constraints. For simplicity, we assume that for each time period, all the active players request a match, and each player is assigned to an opponent by the matchmaker. The outcomes of all matches are then revealed simultaneously, and players update their engagement states upon completion. Then the next time period starts, and all players that have not churned request a match, and so on.

In practice, a popular competitive game usually has millions of concurrent online players. Motivated by this, we follow the literature of fluid matching (see, e.g., Azevedo and Leshno 2016) and assume that players are infinitely divisible. Let $\mathbf{s}^t = (s_{1w}^t, s_{1\ell}^t, s_{2w}^t, s_{2\ell}^t)$ denote the size of the four demographics in period $t = 0, 1, 2, \dots$. Furthermore, denote $f_{i,j}^t$ as the amount of players in demographic i that are assigned to match with players in demographic j in period t , where $i, j \in \mathcal{P}$. Note that it is possible to use flow variables that only consider one of $f_{i,j}^t$ and $f_{j,i}^t$. However, the current formulation allows an easier presentation of the evolution of demographics' sizes over time. A feasible match given $\mathbf{s}^t$ is a set of matching flows $f_{i,j}^t$ that satisfies:

$$\begin{aligned}
\sum_{j \in \mathcal{P}} f_{i,j}^t &= s_i^t, \forall i \in \mathcal{P}, \\
\sum_{i \in \mathcal{P}} f_{i,j}^t &= s_j^t, \forall j \in \mathcal{P}, \\
f_{i,j}^t &= f_{j,i}^t, \forall i \neq j, i, j \in \mathcal{P} \\
f_{i,j}^t &\geq 0, \forall i \neq j, i, j \in \mathcal{P}.
\end{aligned} \tag{FBs}$$

Namely, (FB_S) are *flow balance* constraints that make sure every active player is matched. The first equation ensures that every type i player is matched with some opponents, and the second equation ensures that the total amount of matches against type j players equals the number of such players. The third equation makes sure that for every pair of demographics, a match results in an equal effect on supply and demand.

Next, we depict the evolution of the system. At time t , type i players who are matched with type j players know their outcome immediately, and update their state at the start of $t+1$ accordingly. For example, for $2w$ players who are matched with $2l$ players, half of them win the match and stay in $2w$ state, while the other half lose the match and become $2l$ in the next period. On the other hand, for $2l$ players who are matched with $2w$ players, half of them win the match and become $2w$ in the next period, while the other half lose the match and leave the system permanently as they have experienced two losses in a row. Given the match assignment $f_{i,j}^t$ in period t , the *evolution of demographics* is given by

$$\begin{aligned} s_{2w}^{t+1} &= \frac{1}{2}(f_{2w,2w}^t + f_{2w,2l}^t + f_{2l,2w}^t + f_{2l,2l}^t) + f_{2w,1w}^t + f_{2w,1l}^t + f_{2l,1w}^t + f_{2l,1l}^t, \\ s_{2l}^{t+1} &= \frac{1}{2}(f_{2w,2w}^t + f_{2w,2l}^t), \\ s_{1w}^{t+1} &= \frac{1}{2}(f_{1w,1w}^t + f_{1w,1l}^t + f_{1l,1w}^t + f_{1l,1l}^t), \\ s_{1l}^{t+1} &= \frac{1}{2}(f_{1w,1w}^t + f_{1w,1l}^t) + f_{1w,2w}^t + f_{1w,2l}^t. \end{aligned} \tag{ED}_S$$

The evolution in (ED_S) reflects that high-skilled players always beat low-skilled players and players from the same skill level have equal chances to win a match. Furthermore, since churn is guaranteed after two losses, only players in demographic $2l$ and $1l$ may potentially leave the system in the next period. It is worth pointing out that when matching players within the same skill level, the matchmaker only needs to keep track of the total flow in this skill level, instead of each individual flow. For example, to induce one unit of high-skilled players to match with each other, a unit of $f_{2w,2l}^t$ is equivalent to half unit of $f_{2w,2w}^t$ plus half unit of $f_{2l,2l}^t$ according to (ED_S) since they induce the same evolution of players' demographics in the next period.

The engagement at period t is given by $s_{1w}^t + s_{1l}^t + s_{2w}^t + s_{2l}^t$, the total amount of active players. The matchmaker is interested in maximizing players' engagement by designing matching flows satisfying both (FB_S) and (ED_S) constraints over an infinite time horizon. The engagement maximization problem can be formulated as a Markov decision process, where the states are $\mathbf{s}^t$, the amount of active players in each demographic. Let V^π be the value function of a feasible policy π . The value-to-go function is given by

$$\begin{aligned} V^\pi(\mathbf{s}^t) &= (s_{1w}^{t+1} + s_{1l}^{t+1} + s_{2w}^{t+1} + s_{2l}^{t+1}) + V^\pi(\mathbf{s}^{t+1}) \\ &\text{subject to } (\text{FB}_S), (\text{ED}_S). \end{aligned}$$

Our goal is to find the optimal policy with a value function $V^*(\cdot)$, such that $V^*(\mathbf{s}^t) \geq V^\pi(\mathbf{s}^t)$ for any feasible policy π . Note that the system dynamics are all linear, so when the initial size of each demographic is given, maximizing the engagement is equivalent to the following infinite linear program:

$$\begin{aligned} V^*(\mathbf{s}^0) = & \max_{\{f_{i,j}^t\}_{t=0}^\infty, \{s_i^t\}_{t=1}^\infty} \sum_{t=1}^\infty \sum_{i \in \mathcal{P}} s_i^t \\ \text{s.t. (FB}_S\text{) and (ED}_S\text{)} & \forall t = 0, 1, 2, \dots, \text{ and } i, j \in \mathcal{P}, \end{aligned} \quad (\text{P})$$

Since there are no new players in our system, it is not hard to imagine that all the players eventually leave, and the total engagement is finite. We formalize the finiteness of engagement under a more general situation in Appendix A (Lemma EC.2).

While the model in this section is highly stylized, it does capture the essence and fundamental trade-off of the engagement maximization problem: heterogeneous skills and aversion to frequent losses. The assumptions of two skill levels and the 100% high-versus-low winrate are simplifications of heterogeneous skill levels. The dependency on the most recent two matches captures the players' bounded memory, which is common in the literature to deliver tractable results (e.g., Hu et al. 2016, Cohen-Hillel et al. 2019). In the context of pricing, empirical evidence also suggests that customers may only remember the current and most recent prices (see, e.g., Krishnamurthi et al. (1992) and the numerical part of Hu et al. (2016)). It is also consistent with industry practitioners' opinions that players have very limited memory capacity (Hiwiler 2015, Hodent 2020). In Section 5, we numerically show that looking at losing streaks of more than two matches does not help much in estimating churn risk, and our theoretical insights hold in the case study when memory is longer than two matches. The assumption of no new players is reasonable for franchises that publish new titles regularly or those games that cannot afford continuous user acquisition campaigns. For franchises that publish new titles annually (e.g., *NBA2k* and *Call of Duty*), most new players arrive in the first few days. For example, *NBA2k21* sold 8 million copies by the first month of the 2020-2021 season, but by the end of the season, the total sales were just 10 million.⁷ For companies that cannot afford continuous user acquisition campaigns, their visibility on distribution channels (e.g., *Steam*) drops significantly after the initial launch, especially if the game is not a blockbuster.⁸

⁷ The sales by Dec 2020 is given by <https://www.vg247.com/nba-2k21-8-million-ps5-xbox-series-x-s-price-hike>. The sales by May 2021 are given by <https://www.gamesindustry.biz/take-two-hits-usd3-37bn-revenue-in-record-year>.

⁸ <https://partner.steamgames.com/doc/marketing/visibility>

4. Analyzing Optimal Policies, Pay-to-Win, and AI Bots

4.1. Matchmaking Policies

Although we have already stripped the matching model to its simplest form, it still possesses intricate dynamics. In this section, we describe several properties of this system and finally characterize the optimal policy.

We start by setting a benchmark policy. As mentioned in Section 1, the status quo in the industry is SBMM, the skill-based matchmaking policy. Hence, it is important to characterize its performance, and we will use SBMM as a benchmark in our following analysis. In our setting, SBMM refers to any policy that lets players in the same skill level match with each other. Note that SBMM is always feasible by letting players in the same state match with each other, i.e., $f_{i,i}^t = s_i^t$. Let $V^{\text{SBMM}}(\cdot)$ be the value function of the SBMM, which we characterize in Lemma 1.

LEMMA 1 (SBMM Value). *Under SBMM, we have $V^{\text{SBMM}}(\mathbf{s}^t) = 5(s_{2w}^t + s_{1w}^t) + 3(s_{2\ell}^t + s_{1\ell}^t)$.*

The proof is based on the memoryless property of Markov chains. Lemma 1 shows that, under SBMM, the ‘shadow price’ of a player is 5 if they just won the last match, and 3 otherwise. That is, upon winning (losing), an average player would stay in the system for five (three) more rounds before permanently leaving.

Next, we consider how to solve the engagement maximization problem. While traditional literature on Markov decision processes usually considers steady-state solutions, our problem (P) cannot have a steady-state solution since the total engagement is finite. In fact, we show in the Appendix (Lemma EC.4) that our matching system does not emit a steady-state even under a relaxed definition. As we show next, SBMM is actually sub-optimal as long as there are positive amounts of players in both levels. On the other hand, searching and conducting analysis for steady states is doomed to fail. Hence, even under this simplest setting, finding the optimal matching policy is highly non-trivial and needs to take the dynamics of player demographics across time into consideration.

Before presenting the optimal matching policy that maximizes the matchmaker’s problem in (P), we first consider a one-period matching, which provides insights into the structure of the optimal matching flows. Suppose the matchmaker only needs to design the optimal matching flows for the initial period. That is, we can modify the matchmaker’s problem in (P) to

$$\max_{f_{i,j}^0} \sum_i s_i^1, \quad \text{s.t. (FB}_S) \text{ and (ED}_S) \text{ when } t = 0. \quad (\text{P}_1)$$

In other words, by designing the matching flows in the initial period $t = 0$, the matchmaker wants to retain as many players in the next period $t = 1$ as possible. Lemma 2 summarizes the optimal matching flows in the one-period matching problem (P₁).

LEMMA 2 (Myopic Policy). *Consider the one-period matching problem in (P_1) . The optimal matching policy maximizes flow between demographics 2ℓ and $1w$. The rest of the players can be matched arbitrarily as long as $2w$ and 1ℓ are not matched to each other.*

Lemma 2 states that the *myopic* decision that maximizes player engagement is to maximize the matching flow between demographics 2ℓ and $1w$. In other words, the matchmaker saves as many as possible high-skilled players who are about to leave the system using low-skilled players that are not at risk of churning. Other players can simply be matched with same-level opponents or any other ways as long as $2w$ and 1ℓ are not matched. The intuition behind the optimal matching policy in this one-period setting is fairly intuitive. It preserves as many high-skilled players as possible without sacrificing low-skilled players at all.

The above intuition may not hold true anymore if we consider the infinite-horizon matchmaking problem, because of the trade-off between short-term and long-term reward. Compared to SBMM, the myopic policy increases the number of players in demographic 1ℓ , leading to a higher churn rate in the subsequent period, and thus decreases the possible $2\ell - 1w$ matching in the future. It turns out, however, the optimal matchmaking policy in the infinite-horizon problem (P) resembles the one in Lemma 2. In the optimal matchmaking policy, the matchmaker still maximizes the flow between 2ℓ and $1w$ demographics in order to maximize the overall player engagement, despite the forward-looking nature of the problem. However, the matchmaker also needs to balance the populations in each demographic at the same time, which is not a concern in the one-period matching problem (P_1) . We present the formal result in Theorem 1.

THEOREM 1 (Optimal Matchmaking Policy). *The optimal policy always maximizes matching flows between demographics 2ℓ and $1w$. The rest of the players are matched via SBMM in most scenarios, except when there are fewer high-skilled players compared to low-skilled players and too many 1ℓ players. In this case, some players in demographic 2ℓ are matched with players in 1ℓ .*

We provide detailed descriptions of the optimal matching flows in Appendix B.1.1 Table EC.5. Our proof provides a complete characterization of the evolution of $\mathbf{s}^t$ under the optimal policy, and we provide a simplified version of this evolution in Figure 1. This figure separates the four-dimensional state space into four regions and $\mathbf{s}^t$ transits following the arrows. A complete version of the evolution, which contains seven regions, is presented in Appendix B.1. To prove optimality, we also characterize the shadow price of each demographic in each of the four regions of Figure 1. We then rely on strong duality and show that complementary slackness holds for all the possible state transitions (i.e., all the arrows in Figure 1).

Next, we illustrate the intuition behind the optimal policy. According to Theorem 1, the optimal matching policy always maximizes the flows between demographic 2ℓ and $1w$. Thus, matching

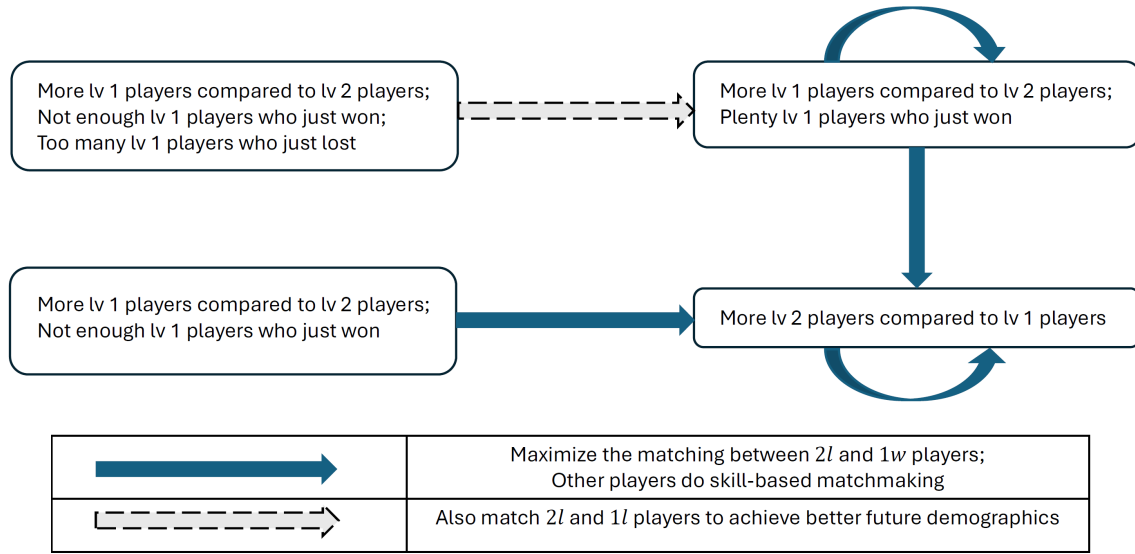


Figure 1 Simplified Evolution of Players' Demographics

players from different skill levels is beneficial, just like the one-period example. On the one hand, the platform should always “save” as many high-skilled players ($2l$, who are about to leave the platform) as possible by matching them with low-skilled players ($1w$, without losing streaks). When matched with low-skilled players, high-skilled players ($2l$) can enjoy a free win to break the losing streak and participate in future matchings. On the other hand, since low-skilled players can potentially prevent high-skilled players from leaving when matched, low-skilled players are very valuable to the matchmaker, so they should not be completely exhausted. Note that low-skilled players used to “save” high-skilled players are those from $1w$, so they may potentially recover from the losing streak in the next period. In most cases, the matchmaker does not match remaining players in $2l$ to players in $1l$ since players from the latter leave the matching process afterward, resulting in no immediate improvements and losing the potential for them to come back to demographic $1w$.

One exception is when there are not enough low-skilled players in $1w$ but too many players in $1l$ compared to the size of $2l$ (the top-left scenario in Figure 1). Then, after exhausting players in $1w$, the matchmaker also matches low-skilled players in $1l$ to high-skilled players in $2l$. Here, the matchmaker is essentially adjusting the overall player distribution among the demographics. Although they have the potential to increase other players' engagement, low-skilled players are more valuable if there are enough high-skilled players that need to be saved. The adjustment preserves high-skilled players in the process who may need to be saved by low-skilled players in the future. As we can see from Figure 1, under the optimal matching policy, in the immediate next period, we still have more low-skilled players due to their abundance. Although we sacrifice some $1l$ players to save $2l$ players, we still have enough $1w$ players for the next period.

On the contrary, we should not match 1ℓ players to 2ℓ in a similar scenario but without an abundance of 1ℓ players (bottom-left in Figure 1). By doing so, we exhaust too many low-skilled players. Instead, besides maximizing the flow between 2ℓ and $1w$, all other players are matched via SBMM, and this scenario readily transits naturally into scenarios where there are either enough $1w$ players to save 2ℓ players in the next period, or there are more high-skilled players. Once the players' demographics transit into the two scenarios on the right in Figure 1, it will stay there under the optimal matching policy. In other words, the matchmaking only need to maximizing flow between 2ℓ and $1w$ and let all other players do SBMM for all subsequent periods.

Insights from the Dual Problem Our proof is based on duality, and another way to gain insights is to look at the shadow prices of each demographic, which can be informally understood as the value of a specific demographic to the matchmaker. We reserve the detailed descriptions of shadow prices in Appendix B.1, and illustrate the main insights here. Generally, the shadow prices for players not on a losing streak are higher than those who have just lost a game. On the other hand, when we evaluate the shadow prices in relation to skills, we make several noteworthy observations. Firstly, when contrasting different levels in the same state (e.g., $2w$ versus $1w$), the shadow prices hinge on the relative population of high and low-skilled players. If there is a significant number of high-skilled players compared to low-skilled ones, the shadow prices of low-skilled players are higher but remain capped by a constant. However, as the ratio of high to low-skilled players approaches 0, the shadow prices for high-skilled players increase since these players can effectively survive ‘indefinitely’ if introduced into the system. This also explains why the matchmaker is willing to sacrifice players in 1ℓ to preserve high-skilled players. Secondly, when we examine the difference between winning and losing players, we find that the disparity between $1w$ and 1ℓ is always greater than that between $2w$ and 2ℓ . The reasoning is that transitioning a high-skilled player from a winning to a losing state is less detrimental to overall engagement, as they can potentially be rescued by low-skilled players. Understanding these structural properties of the shadow prices is important because, if the shadow price for each demographic can be accurately inferred, the matchmaker may recover the optimal matching flow in each period by solving a one-period matching problem as given below:

$$\max_{f_{i,j}^t} \sum_i (1 + \lambda_i^t) s_i^{t+1}, \quad \text{s.t. (FB}_S) \text{ and (ED}_S),$$

where λ_i^t is the optimal shadow price in hindsight. We demonstrate later in Section 5 that heuristic policies based on dual properties perform reasonably well in general scenarios. It should be noted that, a similar dual analysis is also used in Rios et al. (2022), where they consider a matching problem of dating platforms, and the matching probability in subsequent periods will be decreased

when more matches are made in the current period. They propose a heuristic by penalizing the reward of matching a user in the next period, and then matching the users myopically. Such a method is essentially equivalent to guessing the shadow price of each user in the next period. Our analysis provides a complete characterization of the dual dynamics for a matching problem with memory effects, although in a different context. In summary, our analysis from both primal and dual perspectives indicates that to maximize long-term engagement, the optimal policy must not only myopically collect short-term rewards but also maintain a focus on demographic distribution.

Connection with Literature Before moving on, we discuss the relevance of our results to existing literature. It is worth noting that our optimal policy resembles the inventory control policy with substitution and rationing in Shumsky and Zhang (2009) and Yu et al. (2015), as well as the optimal type matching policy in Hu and Zhou (2022). The main driver behind the optimal matching structures in these papers is that either the rationing or the match-down-to structure provides the decision maker the opportunity to wait for better matching pairs in the future. More specifically, because the reward of each matching pair is fixed in each period, it is possible to generalize the ideas of pairwise comparison and augmenting paths into intertemporal settings to obtain structural properties. However, in our paper, the matchmaker cannot strategically wait for better matching pairs since the reward structure is endogenously determined and constantly evolving. If we regard each player as a matching resource similar to a user or driver in Hu and Zhou (2022), then each player has two dimensions: skill and win/loss history. While their skills are exogenously given, just like in the papers mentioned above, players' matching history (and therefore their matching reward in the future) is endogenously decided by the matching policy and constantly evolves over periods. As a result, it is challenging to conduct a matching-flow-based analysis similar to the above papers. In addition, even though two matching pairs generate the same current period reward, they may have different long-term effects. As a result, to pin down the exact priority structure in our model, we have to leverage the duality approach similar to Rios et al. (2022), since the complex future dynamics of matching pairs are captured by their shadow prices.

Benefits of Optimal Matchmaking Besides the optimal policy, another interesting question is the benefit it provides compared with the SBMM benchmark. We analyze the performance of the optimal matchmaking policy in Theorem 2 below.

THEOREM 2 (Engagement Improvement). *(a) For one period, the myopic decision (from Lemma 2) that maximizes engagement garners at most $4/3$ engagement as SBMM. (b) In the infinite-horizon setting, we have that the optimal policy from Theorem 1 garners at most $3/2$ engagement as SBMM. In other words,*

$$\frac{V^*(\mathbf{s}^0)}{V^{SBMM}(\mathbf{s}^0)} \leq \frac{3}{2}, \quad (1)$$

for any $\mathbf{s}^0 \geq 0$, where V^* is the value function under the optimal policy. Furthermore, the upper bound is achieved for some $s_{2\ell}^0 = s_{1w}^0 > 0$ and $s_{2w}^0 = s_{1\ell}^0 = 0$.

Theorem 2 first states that the value of the myopic policy over SBMM is at most $4/3$ in a single period. While Chen et al. (2017b) conjectured that the power of myopic matchmaking grows exponentially as the time horizon increases, we resolve this in the negative: over the whole time horizon, the benefit of optimal policy over SBMM is at most 50%. As discussed, the optimal matching policy utilizes matches between 2ℓ and $1w$ players to improve engagement. Thus, as we can see from Theorem 2(b), the upper bound of the performance ratio is attained when the initial demographics only have an equal amount of 2ℓ and $1w$ players. In this scenario, the matchmaker improves the most compared to SBMM by using low-skilled players who are not in danger of churning to save high-skilled players who are about to churn. Although the benefit does not grow with time, because the benefit of optimal policy comes from cross-level matching, we may expect that the optimal policy becomes more beneficial when there are more than two skill levels.

COROLLARY 1. *When the winrate between high and low-skilled players decreases from 1 to 0.5, the ratio in the inequality (1) decreases monotonically to 1.*

Corollary 1 states that the power of optimal matching decreases when the winrate between high and low-skilled players decreases. Recall that the benefit of optimal matching over SBMM comes from cross-skill level matchmaking. As the winrate decreases, it becomes ineffective to use low-skilled players to “save” high-skilled players, decreasing the reward from cross-skill level matchmaking. We shall validate these ideas in the case study in Section 5.

We use the following two subsections to discuss two important interventions to increase engagement: pay-to-win strategies in Section 4.2 and AI-powered bots in Section 4.3.

4.2. Pay-to-win system

Building upon the baseline model, suppose the matchmaker can offer low-skilled players a chance to purchase items or information to gain an advantage in the next match. Specifically, we follow the practice in popular games such as *Dota 2* (introduced in Section 1) and consider a subscription service that provides additional features to players. Such features are largely useless for high-skilled players but may help low-skilled players substantially. Therefore, in the model of consideration, we only consider a pay-to-win (PTW) option for low-skilled players in every period.⁹ We assume that the subscription fee is r per period. Low-skilled players who just experience a loss would subscribe with a fixed probability $\beta \in (0, 1]$, and keep their subscription for all matches until they churn.

⁹ In practice, such membership is bundled with other perks, e.g., decorative staffs, so high-skilled players may also pay for it. However, our qualitative insights are consistent as long as high-skilled players do not improve their skill level further.

This assumption is a reasonable reflection of reality since players may be more willing to pay upon losing a match. We leave r and β as exogenous and focus on the interplay between PTW and the matchmaker. For simplicity, we normalize a unit of player engagement translates to one unit of revenue generated for the seller.

Adopting the notation from Section 3, suppose the matchmaker faces an initial state of demographics $\mathbf{s}^0 = \{s_{1w}^0, s_{1\ell}^0, s_{2w}^0, s_{2\ell}^0\}$. By having the subscription, some low-skilled (level 1) players elevate their gaming skills to the high level (level 2), denoted by $\bar{2}$, so that they behave exactly like high-skilled players in Section 3. However, purchasing the subscription feature does not (and should not) reset a player's losing streak. Thus, instead of having 4 demographics, with the addition of the subscription feature, there are 6 demographics in each period, denoted by $s_{2w}^t, s_{2\ell}^t, s_{\bar{2}w}^t, s_{\bar{2}\ell}^t, s_{1w}^t, s_{1\ell}^t$. Note that $s_{2w}^t, s_{2\ell}^t$ represent players with subscriptions who just won and lost a game, respectively. To differentiate from the original demographics, we use $s_{\bar{2}w}^t, s_{\bar{2}\ell}^t$ to denote unsubscribed high-skilled players, and $s_{1w}^t, s_{1\ell}^t$ to denote unsubscribed low-skilled players. In the initial period $t = 0$, we have $s_{2w}^0 = 0, s_{2\ell}^0 = 0, s_{\bar{2}w}^0 = s_{2w}^0, s_{\bar{2}\ell}^0 = s_{2\ell}^0, s_{1w}^0 = s_{1w}^0, s_{1\ell}^0 = s_{1\ell}^0$, and the subscription starts from $t = 1$. In Appendix Section B.1.3, we introduce new flows between demographics (Eq. (FB_{ptw}), Eq. (ED_{ptw})) and the updated matchmaking problem (Eq. (P_{ptw})).

Our focus is on the interplay between PTW and matchmaking, as well as their aggregate value. We provide our main findings in Theorem 3 below.

THEOREM 3. *Consider $\beta \in [0, 1)$.*

- (a) *Subscribed players have a higher priority compared to unsubscribed high-skilled players. That is, unsubscribed high-skilled players in $\bar{2}w$ ($\bar{2}\ell$) would only be matched with any unsubscribed low-skilled players after all the subscribed players in $\bar{2}w$ ($\bar{2}\ell$) have matched with unsubscribed low-skilled players.*
- (b) *When there are more high-skilled players than low-skilled players, implementing PTW decreases engagement. However, in terms of the overall revenue (engagement plus subscription revenue) generated, implementing PTW is beneficial if and only if r is greater than a specific threshold.*
- (c) *When the ratio of high- over low-skilled players is sufficiently small, and r is sufficiently small, implementing PTW increases engagement and subscription revenue simultaneously.*

Theorem 3(a) depicts the impact of PTW on the matchmaking policy. We prove that the optimal policy gives subscribed players high priority and would save an unsubscribed high-skilled player from churning only when all the subscribed players within the same engagement state are saved. Theorem 3(b) and (c) describes the influence of PTW on user engagement and the total value function. When there are more high-skilled than low-skilled players, Theorem 3(b) claims that the

corresponding engagement inevitably falls, and it is only worth introducing PTW when the unit profit r is high enough. This is because, in this case, low-skilled players are already scarce resources, and a PTW system further increases their scarcity. On the other hand, Theorem 3(c) claims that PTW can increase engagement while providing direct revenue. This happens when there are too few high-skilled players. Intuitively, this is because now the high-skilled players are scarce resources, and a few more high-skilled players can facilitate cross-level matching significantly and thus improve engagement when the revenue from PTW is not too large. Together, Theorem 3(b) and (c) provide important managerial insights on the value of a PTW system when taking matchmaking and player engagement into account: besides its direct revenue, it also works as a lever to change the skill distribution and may influence the total engagement positively. This happens when the majority of players are low-skilled, which is reasonable for most games. On the other hand, if the majority of players are high-skilled, PTW will hurt engagement, and the seller should be careful to introduce it. Our finding echos the discussion in Section 4 on the importance of managing player distribution.

4.3. AI-powered bots

Game designers routinely develop AI-powered bots to mimic human player behavior closely. Ideally, a matchmaker can use bots whenever a human player is about to churn if they experience one more loss. The bot is designed so that it is competitive but still loses to the human player, resulting in the human breaking their losing streak and remaining in the system longer. On the other hand, due to the limitations of technology, AI-powered bots can be identified by experienced players. If a human player is frequently matched with bots, they may find out that their opponents are not human and perhaps be discouraged from playing the game. This section provides a framework to analyze the value of adding a percentage of AI-powered bots to the matching pool.

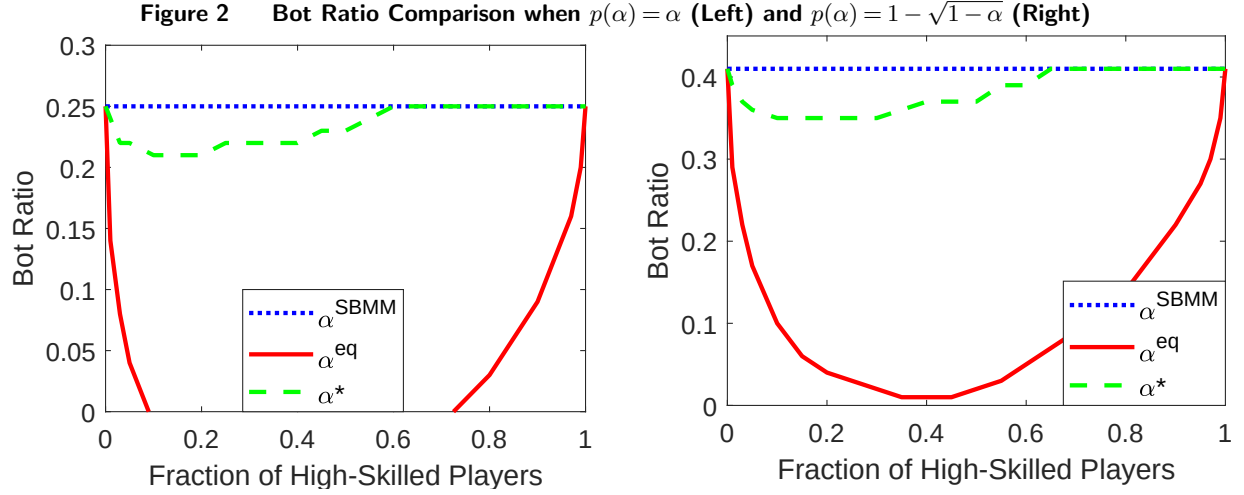
We focus on the impact on players' engagement when a certain percentage of the demographics are bots. We consider the same stylized model with the addition that each active player who just lost a game is matched to a bot with an independent and fixed probability, or *bot ratio*, $\alpha \in [0, 1]$ in each period. With probability $1 - \alpha$, the players who just lost are matched with another human player. This setup tailors toward players who are on losing streaks and focuses on using bots to improve player engagement. To model the negative impact of a high bot ratio, we assume that if players are matched with bots, they may recognize the bots with probability $p(\alpha)$. In this case, players would be frustrated and regard such matches as a losses. We assume that $p(\alpha)$ increases with α , i.e., the more bots in the system, the more likely they are able to detect the bots. Note that in the presence of AI-powered bots, the optimal matchmaking policy is also changed endogenously. To avoid repetition, we leave the formulation details in Appendix B.1.4 (Eq. (EC.44)). With slight abuse of notation, denote $V^{\text{SBMM}}(\mathbf{s}, \alpha)$ and $V^*(\mathbf{s}, \alpha)$ as the value functions of SBMM and the optimal matchmaking policy when the probability of matching with bots is α , respectively.

Under this setting, a natural trade-off emerges for deciding the optimal bot ratio: on one hand, if the players cannot detect the bots, they may enjoy a free victory, and it is beneficial to match more players with bots; on the other hand, players are more likely to detect bots under a high bot ratio, and bots cannot effectively save players in such cases. We first consider the optimal bot ratio under SBMM in Proposition 1.

PROPOSITION 1. *Under SBMM, the optimal bot ratio maximizes the winrate of losing players, i.e., $\alpha^{SBMM} \in \arg \max_{\alpha} \alpha(1 - p(\alpha)) + (1 - \alpha)/2$.*

Proposition 1 states that the optimal bot ratio under SBMM can be decided by solving a one-dimensional optimization problem. Adding bots may only influence the winrate of losing players, and the higher the winrate, the larger engagement they may have. Since players regard bot detection as equivalent to a loss, the expected winrate of a losing player in SBMM can be expressed as $\mathbb{P}(\text{bot})\mathbb{P}(\text{win}|\text{bot}) + \mathbb{P}(\text{not bot})\mathbb{P}(\text{win}|\text{not bot}) = \alpha(1 - p(\alpha)) + (1 - \alpha) \cdot \frac{1}{2}$, maximizing this winrate leads to the optimal bot ratio. For example, if $p(\alpha) = \alpha$, then $\alpha^{SBMM} = 25\%$; if $p(\alpha) = 1 - \sqrt{1 - \alpha}$, then $\alpha^{SBMM} = 41.1\%$.

Engagement under SBMM can be increased by adding bots, but it may also be improved by moving to the optimal matchmaking policy. As a result, bots and improved matchmaking policy can be viewed as substitute strategies. In other words, the bot ratio needed under the optimal policy can be much lower than SBMM in order to achieve the same level of engagement. To be specific, we find α^{eq} such that $V^*(\mathbf{s}, \alpha^{eq}) = V^{SBMM}(\mathbf{s}, \alpha^{SBMM})$, and compare it with α^{SBMM} from Proposition 1. In Figure 2, we consider the scenario where a firm adopts SBMM with optimal bot ratios. We assume $p(\alpha)$ equals α (left) and $1 - \sqrt{1 - \alpha}$ (right), so that α^{SBMM} is 25% and 41.1%, respectively. In each figure, we also vary the initial skill distribution: the x -axis represent the fraction of high-skilled players, and the corresponding initial state is $s_{2w} = s_{2\ell} = x, s_{1w} = s_{1\ell} = 1 - x$. The red solid line represents α^{eq} . As we can see, in both scenarios, the optimal policy needs less bots to achieve the same level of engagement. The bot reduction is most effective when the fraction of high-skilled players is medium, which is consistent with our results in Section 4.1 that shows the benefit of optimal matching policy comes from cross-level matching. In the best case, we can achieve the same engagement without any bots at all. When there is only one skill level (the fraction of high-skilled players is either 0 or 1), the optimal policy has to use the same bot ratio as SBMM, as policies are effectively the same in this setting. Thus, for companies that are criticized for using high bot ratios (Gelbart 2021) and that use SBMM, optimizing the matchmaking system may significantly alleviate such problems without loss of engagement (see Figure 2). In Section 5, our case study based on real data also shows that the optimal policy has the power to reduce the bot ratio significantly compared to SBMM.



Next, we consider the optimal bot ratio under the optimal matchmaking policy. That is, we would like to find $\alpha^* = \arg \max_{\alpha} V^*(\mathbf{s}, \alpha)$. Understanding the value of α^* is important: for firms whose goal is to maximize engagement solely, they will choose the best combination of α and the matchmaking policy. It is not clear whether the players may suffer from an even higher bot rate when firms can optimize the matchmaking policy. The next theorem answers this question in negative.

THEOREM 4 (Optimal Matchmaking Policy Needs Less Bots). *Suppose $\alpha(1 - p(\alpha)) + (1 - \alpha)/2$ is concave. Then under optimal matching, $\alpha^* \leq \alpha^{SBMM}$.*

Theorem 4 states that under the optimal matchmaking policy, the optimal bot ratio is always no larger than that under SBMM. This result is not immediately intuitive, and we provide a brief outline of the proof. For one unit of players who are matched with bots, their outcomes are equivalent to the combination of two scenarios: (a) a fraction of them are matched with undetectable bots that invariably result in wins; (b) the remainder are matched with bots that can be detected with a 50% probability, leading to outcomes akin to skill-based matchmaking. Under optimal matchmaking, the advantage of bots stems from scenario (a), whereas the existence of scenario (b) actually assigns a subset of players to engage in skill-based matching, thereby capping the maximum amount of cross-level matching. When we increment α from 0, we grapple with the trade-off between adding more perfect bots and restricting cross-level matching. The optimal α^* mediates these two influences. On the other hand, under SBMM, the system purely reaps the benefits of scenario (a), but scenario (b) mirrors skill-based matching. Consequently, the optimal α under SBMM does not grapple with this tradeoff and is larger than α^* . We formalize the above intuition in Appendix B.1.4. Theorem 4 underscores a crucial implication for the value of optimal matchmaking: even when we endogenously choose the bot ratio, optimal matchmaking can still

diminish the bot ratio. In Figure 2, we denote α^* with green dashed lines. When the proportion of high-skilled players is relatively low, α^* is distinctly less than α^{SBMM} . In such instances, the system derives more benefits from cross-level matching, thereby necessitating a lower bot ratio.

To summarize, optimizing matchmaking policy can be effective in reducing bots. When the goal is to reduce the bot ratio, we can lower down α significantly without loss of engagement. Even when the firm’s target is maximizing engagement, the endogenous bot ratio is still less under optimal matchmaking compared to SBMM, highlighting an unexpected externality of the optimal policy.

5. Numerical Insights

In this section, we conduct a numerical case study with the Lichess Open Database (Lichess 2021) to verify our insights in a realistic setting where real people play chess against one another. Before diving into the details of the case study, we briefly discuss how we can extend our model to accommodate more realistic and generalized features. To be more specific, we extend the baseline model in three directions: multiple skill levels, general winrate between players with different skill levels, and general churn behaviors. In particular, we provide a general formulation with K ordered skill levels, where match outcomes between players with different skill levels are modeled as Bernoulli random variables. Moreover, we consider that a player’s churn behavior is determined by the win-loss record of the last m matches. We provide detailed discussions on the general formulation in Appendix A. We also show that with all the added features, we can still formulate the problem into an infinite linear program. Structural results such as strong duality and complementary slackness continue to hold true, and the model serves as the foundation of the following numerical study.

5.1. Data Description

Lichess is a free and open-source Internet chess server, and all the match data since 2013 are available to the public. For each match, the data includes the game mode, starting time, players’ IDs, the match’s outcome, and the players’ ratings before and after the match. The platform adopts the Glicko-2 rating, a generalization of the Elo rating (Glickman 2012). We collected all the matches in the most popular mode “Rated Blitz Game” during 2013-2014, which includes 5.41 million matches and 135,073 unique players. To simplify our problem, we remove all the matches where the outcomes are draws (3.7% of total). We say a player is churned after a match if they stop playing for 14 days after the match. Hence, a player may churn more than once.

We focus on players who played for at least 5 matches for three reasons. First, players who churned in less than 5 matches may largely churn due to other factors (e.g., user interface, Internet connection). Second, new players’ ratings change wildly after the first few matches, making it difficult to estimate their skill levels. Finally, we would like to test models with a range of game memory m (up to 4), which means that the players need to play at least $m + 1$ matches. In our dataset, there

are $N = 60,334$ qualified players. For each player, we take at most 500 matches on their record (92% of the 60,334 players play less than 500 matches). This is done to avoid underestimating the churn probability since a few dedicated players never quit under any circumstance.

Table 1 Summary of Considered Players ($N = 60334$)

	Mean	Median	Min	Max
Number of Matches	171	21	5	18764
Estimated Ratings	1525	1518	764	2663

Because the ratings oscillate, we use the average of the ratings after the last 3 matches to estimate the rating, which is then used to determine their skill level. A summary of the players' engagement and ratings is in Table 1. Compared to the median rating (1518), the fluctuation is not large in general: for the focal players, the average of their lifetime standard deviations is 57, and in a calendar month the average standard deviation is 39. Thus, it is reasonable to assume that the skill levels are stationary. Based on the estimated ratings, we divide players into 13 skill levels: for the 96% of the considered players whose rating is between 1000 and 2100, we separate them into 11 levels (level 2 to 12) based on intervals of 100; for players with a rating less than 1000 and greater than 2100, we group them into two new levels, level 1 and level 13, respectively. To compute the winning probability between different skill levels, we use the realized winrate when the level difference is less than or equal to 8. When the level difference is greater than 8, we have too little data for accurate estimation and simply assume that the player with a higher skill level wins with probability 1. Table 2 describes the winrate we used in the case study.

Table 2 Winrate Estimation

Level Difference	0	1	2	3	4	5	6	7	8	9-12
Winrate	0.500	0.596	0.690	0.776	0.844	0.893	0.924	0.936	0.948	1.000

5.2. Players' Churn Behavior Estimation

To adapt our model with real data, we generalize the baseline model in Section 3 to a general losing streak model, where the players' churn probability depends on the length of the losing streak they experience.¹⁰ Let $g = 0, 1, 2, \dots, m$, record the length of the current losing streak. A player is fully characterized by their skill level and losing streak (state) g . Our goal is then to estimate ρ_g^{win} (ρ_g^{lose}), the churn probability of a player in state $g \in \mathcal{G}$ after winning (losing) the next match. Note that ρ_g^{win} is the same for all $g = 0, \dots, m$, because all such players have a 0 losing streak upon

¹⁰ A more generalized model, where players' churn probability depends on the outcomes of the last m matches they experience, is provided in Appendix A.

winning the next game. On the other hand, ρ_g^{lose} represents the churn probability upon losing the next game, given that the players have already lost g consecutive games (thus, they experience a $(g+1)$ -match losing streak). Our baseline model in Section 3 can be viewed as a special case of the losing streak model presented here, with $m = 1$, $\rho_g^{win} = 0$, $\rho_0^{lose} = 0$, and $\rho_1^{lose} = 1$. Using maximum likelihood estimation, these churn parameters are estimated by

$$\rho_g^{win} = \frac{\text{Number of churn decisions after a win}}{\text{Number of wins}}, \quad g = 0, \dots, m,$$

$$\rho_g^{lose} = \frac{\text{Number of churn decisions after } g+1 \text{ consecutive losses}}{\text{Number of times that players experience } g+1 \text{ consecutive losses}}, \quad g = 0, \dots, m.$$

We randomly sample 70% of the players to train the model and use the rest of them to validate the model. We test models ranging m from 0 to 4. When $m = 0$, the churn probability only depends on the outcome of the current game. In addition, we also estimate a null model as a benchmark, which assumes a uniform churn probability.

We summarize the out-of-sample negative log-likelihood (NLL) of the candidate models in Table 3. Notably, compared with the null model, even when $m = 0$, our model improves the fit by more than 4000 in NLL. The increase from $m = 0$ to $m = 1$ is another 381 in NLL, while increasing m from 1 to 4 has rapidly diminishing returns. Thus, assuming $m = 1$ does not result in much loss in the goodness-of-fit of the models. Furthermore, when $m = 1$, the parameter of the losing streak model is given by $\rho_g^{win} = 1.32\%$, $\rho_0^{lose} = 1.62\%$, and $\rho_1^{lose} = 2.54\%$. The surge in churn probability from one loss to two consecutive losses reflects the players' negative sentiment toward a losing streak. Our baseline stylized model in Section 3 reasonably captures the essence of this behavioral pattern. In Appendix A.5, we also present the estimation result of another model, where the churn probability is based on the winrate in the last m matches instead of the losing streak. The losing streak model is slightly better than the winrate model in terms of out-of-sample log-likelihood.

Table 3 Out-of-sample Negative Log-likelihood (NLL) for the Candidate Models

Models	Null	$m = 0$	$m = 1$	$m = 2$	$m = 3$	$m = 4$
NLL	144395.3	139949.6	139568.8	139415.8	139291.2	139210.8

5.3. Performance Comparison

In this section, we investigate the performance of optimal matchmaking, as well as several heuristics based on the optimal policy in Section 4. We consider a scenario where, in each period, all the active players join the matchmaking pool and the outcome is revealed immediately. We use the realized skill levels (13 total) and the realized engagement states after the fifth match as the initial input. In the following discussion, we denote $s_{k,g}^t$ as the population of level k players with losing streak g at time t , and $\lambda_{k,g}^t$ as the corresponding shadow price. We consider the following policies:

- (1) *SBMM*: Always match players from the same skill level together. The total engagement can be computed efficiently by Lemma EC.1 in Appendix A.
- (2) *Random Matching*: A player has a uniform chance to be matched with any other active player.
- (3) *Optimal Policy*: We solve the LP formulation corresponding to the comprehensive model (formally presented in Appendix A.3 Eq. (EC.2)) with a large enough T . In our case, we set $T = 1000$ and use Gurobi to solve the formulation.

In addition, we also consider heuristics based on the structure of optimal policy for the baseline model. As discussed in Section 4, the policy can be interpreted from two perspectives. From the primal perspective, the optimal policy is essentially a myopic policy with careful tiebreaking. From the dual perspective, we recover several interesting structural properties of the shadow prices. Therefore, we construct the following heuristic policies:

- (4) *Low-skill Favored Myopic Matching (LF Myopic)*: In this policy, we maximize the active players in the next period, but give priorities to low-skilled players when doing tiebreaking. To achieve this, in each period t , we solve two linear programs: in the first linear program, we solve the one-period myopic matching problem to find out V_t^M , the maximum total active players in the next period. Then in the second linear program, we solve a similar linear program, but make two modifications: first, we add a constraint that the total population in the next period is no smaller than V_t^M , which ensures that the resulting matching flow is still a myopic solution; second, we change the objective to be $\sum_{k=1}^{13} (13-k) \sum_{i=1}^m s_{k,g}^t$, so that the low-skilled players are more valued compared to high-skilled players.
- (5) *High-skill Favored Myopic Matching (HF Myopic)*: Compared to LF Myopic, we set the objective in the second linear program to be $\sum_{k=1}^{13} k \sum_{i=1}^m s_{k,g}^t$, so that the high-skilled players are more valued.
- (6) *Dual-based Policy*: Finally, we consider a dual-based policy, which first guesses the shadow prices of each player type, and then solves a one-period matching problem using the shadow prices as the reward. Such a policy has been considered in the context of dating platforms, where users also stay for extended periods and exhibit history effects (Rios et al. 2022, Rios and Torrico 2024). To be specific, let $\lambda_{k,g}^t$ be the optimal dual variables in the infinite linear program. If $\lambda_{k,g}^t$ is known a priori, then the optimal decision at t can be obtained by solving a one-period matching problem, i.e.,

$$\max_{f_{kg,k'g'}^t} \sum_k \sum_g (1 + \lambda_{k,g}^{t+1}) s_{k,g}^{t+1} \quad \text{s.t. flow balance and evolution of demographics,}$$

where $(1 + \lambda_{k,g}^{t+1})$ represents the reward in the current period plus all the future rewards for having one unit of type k, g players. Our dual-based policy estimates $\hat{\lambda}_{k,g}^{t+1}$ based on the current

population $s_{k,g}^t$, and then solve the above one-period matching problem by replacing $\lambda_{k,g}^{t+1}$ with $\hat{\lambda}_{k,g}^{t+1}$. To provide reasonable estimations, we rely on two structural properties of the shadow prices in our baseline model. First, a more scarce skill level usually has a higher shadow price, that is, if $s_{k,g}^t$ is much less than $s_{k',g}^t$, then $\lambda_{k,g}^t$ should be greater than $\lambda_{k',g}^t$. Second, for $g = 0, \dots, m$, $\lambda_{k,g}^t$ decreases with g , and the difference between $\lambda_{k,g}^t$ and $\lambda_{k,m}^t$ is decreasing with the skill level k . Intuitively, when a level 1 player experiences m consecutive losses, it leads to more losses to the total engagement compared to a level 13 player, because we cannot ‘save’ the level 1 player through matching. As a result, $\lambda_{k_1,g}^t - \lambda_{k_1,m}^t > \lambda_{k_2,g}^t - \lambda_{k_2,m}^t$ for $k_1 < k_2$. To reflect these two structural properties, we estimate $\hat{\lambda}_{k,g}^{t+1}$ as follows:

$$\hat{\lambda}_{k,m}^{t+1} = \lambda_m^{SBMM} + \log \left(\frac{\text{total population at } t}{\text{level } k \text{ population at } t} \right), \quad (2)$$

$$\hat{\lambda}_{k,g}^{t+1} = \hat{\lambda}_{k,m}^{t+1} + (0.1 + 0.15(13 - k))(\lambda_g^{SBMM} - \lambda_m^{SBMM}), \quad g = 0, \dots, m-1, \quad (3)$$

where λ_g^{SBMM} are the shadow prices under SBMM policy, computed by Lemma EC.1 in Appendix A. Because $\hat{\lambda}_{k,m}^{t+1}$ must be higher than the shadow price under SBMM, Eq. (2) construct $\hat{\lambda}_{k,m}^{t+1}$ using λ_m^{SBMM} plus an adjustment based on the current skill level distribution. The rarer the level k players are, the more valuable they are. For $g < m$, in Eq. (3) we construct $\hat{\lambda}_{k,g}^{t+1}$ using $\hat{\lambda}_{k,m}^{t+1}$ plus an adjustment term $(0.1 + 0.15(13 - k))(\lambda_g^{SBMM} - \lambda_m^{SBMM})$. The adjustment reflects the structural property that $\lambda_{k,g}^t - \lambda_{k,m}^t$ decreases with k . We also tried changing the coefficient (i.e., 0.1 and 0.15) slightly, and the performance is robust.

In Table 4, we show the power of various policies compared to SBMM. SBMM is better than the random policy, which shows that the status quo of SBMM is better than doing nothing. Notably, the optimal policy may improve the total engagement by 4.20-6.08%, depending on the choice of churn model and the memory parameter m . On the other hand, the heuristic policies based on primal and dual results in the baseline model also perform well. The LF myopic policy has a small gap compared to the optimal policy (up to 0.23%). When comparing LF myopic with HF myopic, we can see the value of tiebreaking: a tiebreaking that favors low-skilled players improves the performance by 0.20-0.66%. The dual-based policy achieves the best performance among all the heuristics: it has at most a 0.03% gap with the optimal policy. These results suggest that our insights from the baseline model can be carried over to more generalized situations, and we can construct effective heuristics when considering a user behavior model with prohibitive complexity. We also verify our findings on alternative initial states in Appendix A.6, and the performance is robust across all the scenarios.

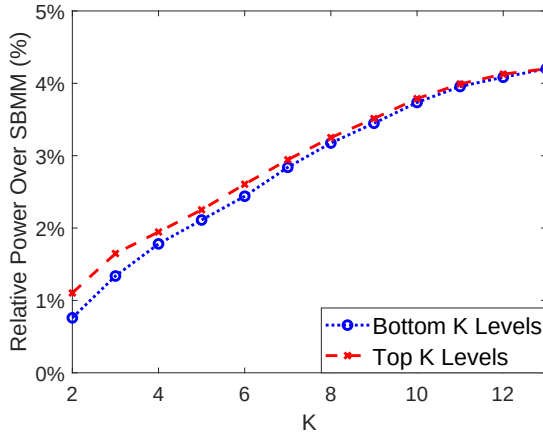
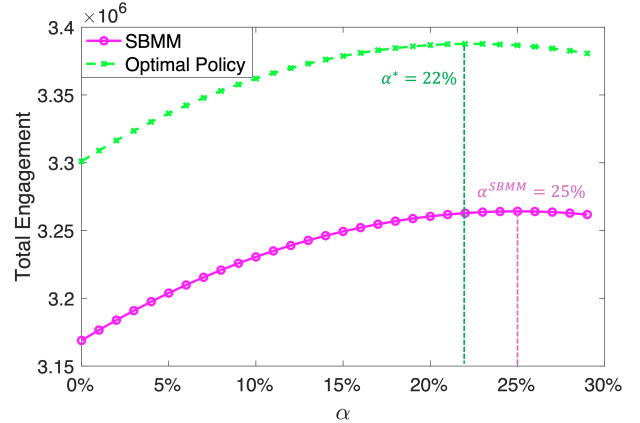
To further understand the policy structure, we scrutinize the matching flows in the first period when $m = 1$. Interestingly, we find that the optimal policy does not aim to maximize the number

Table 4 Relative Power of Candidate Policies Compared to SBMM

	Random	Optimal	LF Myopic	HF Myopic	Dual Policy
$m = 1$	-2.07%	4.20%	4.13%	3.47%	4.20%
$m = 2$	-3.78%	5.86%	5.66%	5.37%	5.85%
$m = 3$	-5.02%	6.04%	5.81%	5.58%	6.01%
$m = 4$	-5.79%	5.78%	5.56%	5.38%	5.77%

of active players in the subsequent period. In other words, it no longer adheres to a myopic policy under more complex circumstances. For instance, level 1 players in state 0 (who have just won their last match) are matched with level 6 players in state 1 under the optimal policy. However, both LF myopic and HF myopic pair them with level 7 players in state 1. This deviation in the optimal policy can be perceived as a protective measure for level 1 players, given that they will lose fewer matches when paired with players of a lower skill level, thereby preserving them for future cross-level matchings. This observation suggests that it may be advantageous to protect low-skilled players even at the cost of short-term rewards. Interestingly, our dual-based policy accurately matches level 1 players with level 6 players, indicating that the constructed shadow prices correctly encapsulate the trade-off inherent in the optimal policy.

Next, we check the power of optimal matchmaking with a varying number of skill levels. Recall that in Section 4.1, we conjecture that the benefit of optimal policy should increase as there are more skill levels. To verify this insight in a realistic setting, we now compute the relative power of optimal matchmaking when the top or bottom K skill levels are considered for $K = 2, \dots, 13$. In Figure 3, we show a representative example when $m = 1$. It turns out that the power of the optimal policy monotonically increases with K regardless of how we add the levels (bottom-up or top-down), which confirms our conjecture.

Figure 3 Power of Optimal Matchmaking with K Skill Levels ($m = 1$)**Figure 4** Total Engagement with Different Fractions of Bot Matches ($m = 1$)

Finally, we investigate scenarios with AI-powered bots. We assume that every player who just experienced a loss has an α chance to be matched with a bot, and the probability of detecting bots is $p(\alpha) = \alpha$. In Figure 4, we show the total engagement under SBMM and the optimal policy changes with α when $m = 1$. By Proposition 1, the optimal bot ratio under SBMM is 25%. Notably, the optimal policy with zero bots can achieve even higher engagement compared with SBMM plus 25% bots. Furthermore, the optimal bot ratio under the optimal policy is 22%, which is 3% less than 25% and consistent with Theorem 4. Thus, the optimal policy can not only achieve higher engagement, but also reduce the bot ratio endogenously, even the firm solely seeks to maximize the engagement.

5.4. Discussions on Other Metrics of Player Experiences

So far, our attention has been centered on total engagement, a metric integral to both a firm’s revenue and players’ experiences. However, it’s worth noting that player experiences can also be assessed by other metrics, and maximizing total engagement may generate heterogeneous effects across different skill levels. In this section, we discuss how various user experience metrics may fluctuate under the optimal policy.

We consider two metrics. The first metric is the lifetime winrate, with a high winrate denoting a more enjoyable experience. The second metric is a lifetime utility based on match history. For each period, we assume that players gain one unit of utility if they do not experience consecutive losses, and zero otherwise. Note that in our case study, players may remain in the game with a certain probability even after experiencing consecutive losses. We compute these two lifetime metrics when players churn from the system, grouping them by skill levels. Under SBMM, these two metrics are virtually identical across all skill levels, with a minimal difference attributable to the initial state distribution.

In Figure 5, we present the winrate by skill levels, where the size of the node represents the population of each group. Notably, the weighted average winrate of all players is 49.15%, which is virtually the same as that under SBMM (49.13%). In Figure 6, we depict the lifetime utility by skill levels. While the utility under SBMM is 44.04, the average utility under the optimal policy is 50.32, reflecting a 14.2% improvement. This enhancement exceeds our improvement in total engagement (4.20%). For both metrics, we observe an uneven distribution of user experiences: high-skilled players enjoy a higher winrate and utilities under the optimal policy, while low-skilled players fare worse compared to SBMM. Therefore, firms should be aware of the potential downside of maximizing total engagement and might consider balancing different metrics when designing a matchmaking system. One straightforward way to safeguard low-skilled players is to employ our dual-based policy, but with adjustments for low-skilled players: based on the constructed shadow

prices in Eq. (2)-Eq. (3), we add an additional term $\rho(12 - k)(2 - g)$, for all k and $g = 0, 1$, where $\rho > 0$ is a weight parameter. In other words, we presume that preserving low-skilled players is more beneficial, and maintaining players in state 0 is more rewarding than in state 1. When we set $\rho = 0.2$, the engagement improvement decreases from 4.20% to 2.25%, but the disparity between players is significantly reduced: as shown in Figure 7, all players have a winrate within the 45%-55% range, and the average winrate is slightly improved. Regarding player utility, Figure 8 illustrates that the variation among players is also significantly diminished. Notably, under the adjusted matching, all skill levels except for level 1 are better off compared to SBMM.

Figure 5 Winrates by Skill Levels

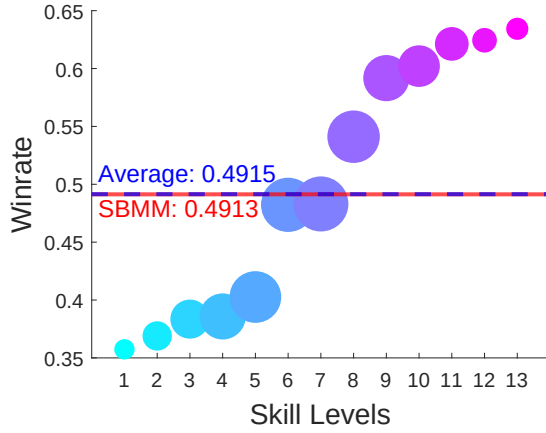


Figure 6 Utilities by Skill Levels

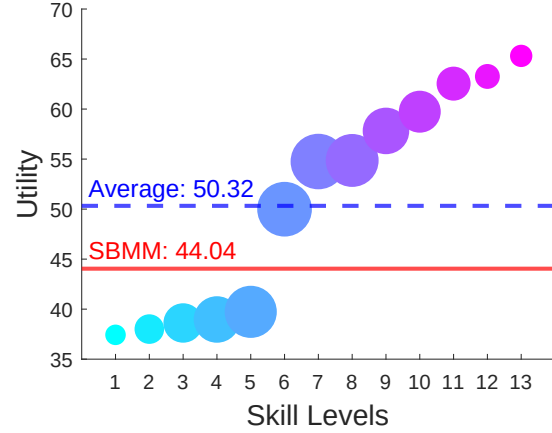


Figure 7 Winrates under Adjusted Dual Policy

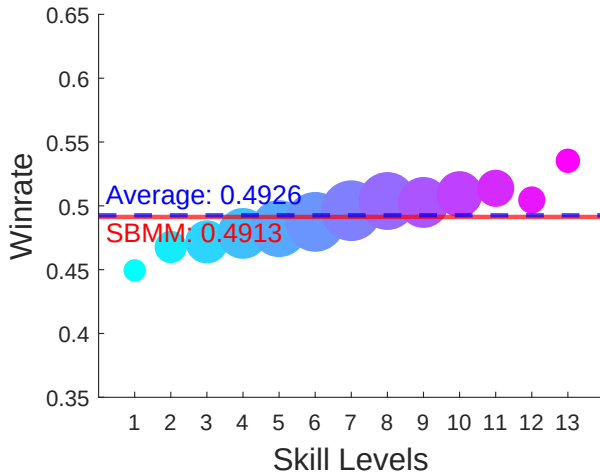
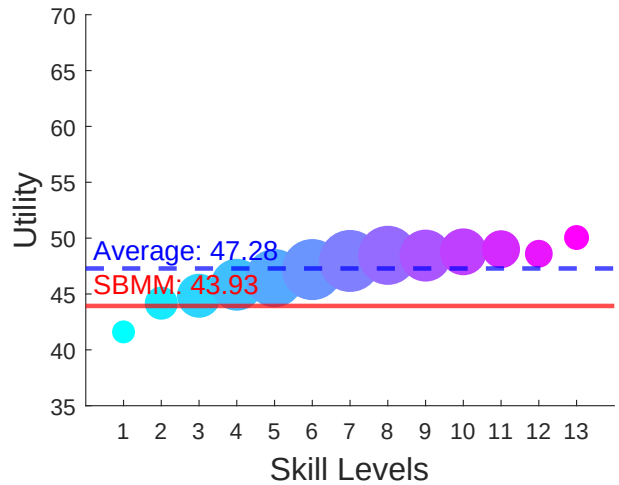


Figure 8 Utilities under Adjusted Dual Policy



6. Conclusions

This paper presents a modeling framework for managing player engagement through matchmaking in competitive games. The central challenge is to allocate limited satisfying experiences (wins) to players through various levers, such as matching, PTW, and AI-powered bots. We provide several novel insights to industry practitioners through sharp characterizations of a baseline model and a numerical case study based on real data. In terms of matchmaking, the current standard is SBMM, which matches players with similar skill levels. Although SBMM is intuitive and easy to implement, it does not maximize long-term engagement. For the engagement-optimal matchmaking rule, we show that in the special case, the optimal policy myopically maximizes the short-term reward in the next period but also adjusts player distribution for the long-term reward. For more general cases, we provide a comprehensive framework and propose several promising heuristic policies.

In addition, we provide new perspectives on controversial topics such as pay-to-win systems and AI-powered bots in competitive video games. By investigating the interplay between pay-to-win strategies and the optimal matchmaking policy, we provide a novel viewpoint of PTW as a lever to control the distribution over the demographics of players. Importantly, we show that PTW may even increase engagement. This potential positive externality on user engagement contrasts with the negative public image of PTW strategies. For the bot issues, our results show that using an optimal matchmaking policy instead of SBMM may reduce the required bot ratio significantly while maintaining the same level of engagement. Finally, using real data from an online chess platform, we show that players do indeed churn based on recent match outcomes, and the optimal policy can significantly improve over SBMM.

We acknowledge some limitations and suggest possible directions for future research. First, our model primarily focuses on players' skill levels during matchmaking, and our proofs are based on primal-dual analysis. Future work could consider other features, such as players' preferences for specific playing styles or strategies. Also, it is interesting to explore whether the dynamic programming in our problem exhibits any structural properties under general user behaviors, which may lead to simple policies for general situations. Second, while our work considers how to avoid consecutive bad experiences for players, we must also consider the social responsibility issues related to video games and other forms of digital addiction (Allcott et al. 2022, Chang et al. 2024). Third, we note that video games can have positive benefits, such as improving cognitive skills (Bader et al. 2022). Applying gaming elements to learning is a promising research direction. Lastly, our analysis could potentially inform engagement management strategies on other platforms. While our primary focus is on video game engagement, our modeling framework and insights could be readily applicable to other industries. For instance, educational platforms like *Prodigy*¹¹, and gig

¹¹ <https://www.prodigygame.com/main-en/prodigy-math/>

economy platforms such as *DiDi* (Chengi et al. 2021), utilize gamified 1-vs-1 competition to engage students and drivers. Our insights can be directly transferred to such contexts. More broadly, our modeling framework could be extended to other industries that aim to satisfy users with limited satisfying experiences. For instance, on online dating platforms, successful matches are rare. To boost user engagement, these platforms could optimize their matching policy, provide premium services (similar to PTW), or pair users with popular but inactive accounts (akin to bots).

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Online Appendix: Matchmaking Strategies for Maximizing Player Engagement in Video Games

Appendix A: Omitted Discussions in Section 5

In this appendix, we discuss how we can generalize our baseline model to fit more realistic and generalized settings, serving as a setup to the case study.

A.1. Player Behavior

We extend the baseline model in three directions: Having 1) multiple skill levels, 2) general winrate between players, and 3) general churn behaviors.

First, instead of having only two skill levels in the baseline model, here we consider there are K ordered skill levels, where level 1 is the lowest and level K is the highest.

Second, we extend the outcome of a match as a Bernoulli random variable depending on the skill levels of the two players. Let p_{kj} be the winrate of a level k player versus a level j player, implying that $p_{kj} = 1 - p_{jk}$. Players of the same skill level are equally likely to win, i.e., $p_{kk} = 0.5$. A player with a higher skill level than their opponent has a strictly larger than 0.5 probability of winning, i.e., $p_{kj} > 0.5$ if $k > j$. We still assume that a player's skill level is fixed over their lifespan in the matching system. In Huang et al. (2019), players' skill level is an evolving metric that monotonically increases as the players play more. In our setting, skill levels reflect *relative* competence and are more stable because others are also getting more familiar with the game.

Third, we generalize players' churning behaviors. In particular, we assume that a player's engagement state is determined by the win-loss record of the last m matches and transitions according to a Markov chain when the player plays a new match. We use q to denote the 'churn state', i.e., a player quitting the game permanently. Let $\mathcal{G}$ be the set of all possible states of a player, which has cardinality at most $2^m + 1$ (history of wins/losses and the churn state). A player is then fully characterized by their skill level k and engagement state $g \in \mathcal{G}$, which we refer to as the demographic of players. Let $P_{win}^k, P_{lose}^k \in [0, 1]^{|\mathcal{G}| \times |\mathcal{G}|}$ be the transition matrix of a level k player's engagement state, given that they win/lose the next match. Hence, if they are matched with a level j player, their aggregate transition matrix is given by $M_{kj} = p_{kj}P_{win}^k + (1 - p_{kj})P_{lose}^k$. For ease of notation, we also define $\bar{\mathcal{G}}$ to be the set of all the active states except the churn state q and $\bar{M}_{kj}$ be the reduced aggregate transition matrix without the churn state. We define an active player as one who has not churned and is thus in one of the states in $\bar{\mathcal{G}}$. We use a simple example to illustrate the notations introduced in this section.

EXAMPLE EC.1. Suppose that players have two skill levels, either high or low (denoted by level 2 and 1, respectively). They quit with a probability 0.2 if they experience two consecutive losses and with a probability 0.5 if they experience three consecutive losses. This implies that $m = 2$, as only two previous matches plus the current match outcome affect the transition state. We further assume that a high-skilled player wins against a low-skill player with probability $p_{21} = 0.8$. Hence, for each skill level, there are 4 engagement states in $\mathcal{G}$: the player may experience 0, 1 or 2 consecutive losses or reach state q . We use 20, 21, 22, $2q$ and 10, 11,

12, 1q to denote the states of high- and low-skilled players, respectively. For $k = 1, 2$, the transition matrix P_{win}^k and P_{lose}^k is given by

$$P_{win}^k = \begin{matrix} & k0 & k1 & k2 & kq \\ \begin{matrix} k0 \\ k1 \\ k2 \\ kq \end{matrix} & \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \end{matrix}, \quad P_{lose}^k = \begin{matrix} & k0 & k1 & k2 & kq \\ \begin{matrix} k0 \\ k1 \\ k2 \\ kq \end{matrix} & \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0.8 & 0.2 \\ 0 & 0 & 0.5 & 0.5 \\ 0 & 0 & 0 & 1 \end{pmatrix} \end{matrix}.$$

The aggregate transition matrix is given by

$$M_{kk} = \begin{matrix} & k0 & k1 & k2 & kq \\ \begin{matrix} k0 \\ k1 \\ k2 \\ kq \end{matrix} & \begin{pmatrix} 0.5 & 0.5 & 0 & 0 \\ 0.5 & 0 & 0.4 & 0.1 \\ 0.5 & 0 & 0.25 & 0.25 \\ 0 & 0 & 0 & 1 \end{pmatrix} \end{matrix}, \quad M_{21} = \begin{matrix} & k0 & k1 & k2 & kq \\ \begin{matrix} k0 \\ k1 \\ k2 \\ kq \end{matrix} & \begin{pmatrix} 0.8 & 0.2 & 0 & 0 \\ 0.8 & 0 & 0.16 & 0.04 \\ 0.8 & 0 & 0.1 & 0.1 \\ 0 & 0 & 0 & 1 \end{pmatrix} \end{matrix}, \quad M_{12} = \begin{matrix} & k0 & k1 & k2 & kq \\ \begin{matrix} k0 \\ k1 \\ k2 \\ kq \end{matrix} & \begin{pmatrix} 0.2 & 0.8 & 0 & 0 \\ 0.2 & 0 & 0.64 & 0.16 \\ 0.2 & 0 & 0.4 & 0.4 \\ 0 & 0 & 0 & 1 \end{pmatrix} \end{matrix}.$$

and the reduced transition matrix $\bar{M}_{kj}$ is defined as

$$\bar{M}_{kk} = \begin{matrix} & k0 & k1 & k2 \\ \begin{matrix} k0 \\ k1 \\ k2 \end{matrix} & \begin{pmatrix} 0.5 & 0.5 & 0 \\ 0.5 & 0 & 0.4 \\ 0.5 & 0 & 0.25 \end{pmatrix} \end{matrix}, \quad \bar{M}_{21} = \begin{matrix} & k0 & k1 & k2 \\ \begin{matrix} k0 \\ k1 \\ k2 \end{matrix} & \begin{pmatrix} 0.8 & 0.2 & 0 \\ 0.8 & 0 & 0.16 \\ 0.8 & 0 & 0.1 \end{pmatrix} \end{matrix}, \quad \bar{M}_{12} = \begin{matrix} & k0 & k1 & k2 \\ \begin{matrix} k0 \\ k1 \\ k2 \end{matrix} & \begin{pmatrix} 0.2 & 0.8 & 0 \\ 0.2 & 0 & 0.64 \\ 0.2 & 0 & 0.4 \end{pmatrix} \end{matrix}.$$

□

A.2. Firm's Dynamic Optimization Problem

Now we describe the firm/matchmaker's problem in the generalized comprehensive model.

Let s_{kg}^t be the number of players at time t in the demographic with skill level k and engagement state g . The population of level k active players in period t is given by the vector $\mathbf{s}_k^t \in \mathbb{R}^{|\bar{\mathcal{G}}|}$, and we use $\mathbf{s}^t = [\mathbf{s}_1^t, \dots, \mathbf{s}_K^t]$ to denote the system state. Let $f_{kg,jg'}^t \geq 0$ be the amount of level k players in state g that are matched to a level j opponent in state g' in time t . A feasible match given $\mathbf{s}^t$ is a set of matching flows $f_{kg,jg'}^t$ that satisfies:

$$\begin{aligned} \sum_{j=1}^K \sum_{g' \in \bar{\mathcal{G}}} f_{kg,jg'}^t &= s_{kg}^t, \quad k = 1, \dots, K, \forall g \in \bar{\mathcal{G}}, \\ \sum_{j=1}^K \sum_{g' \in \bar{\mathcal{G}}} f_{jg',kg}^t &= s_{kg}^t, \quad k = 1, \dots, K, \forall g \in \bar{\mathcal{G}}, \\ f_{kg,jg'}^t &= f_{jg',kg}^t, \quad j = 1, \dots, K, k = 1, \dots, K, \forall g \in \bar{\mathcal{G}}, g' \in \bar{\mathcal{G}} \\ f_{kg,jg'}^t &\geq 0, \quad j = 1, \dots, K, k = 1, \dots, K, \forall g \in \bar{\mathcal{G}}, g' \in \bar{\mathcal{G}} \end{aligned} \tag{FB}$$

Namely, (FB) are *flow balance* constraints that make sure every active player is matched. The first equation ensures that every level k player in state g is matched with some opponents, and the second equation ensures that the total amount of matches against level k players in state g equals the number of such players. The third equation makes sure that for every pair of demographics, a match results in an equal effect on supply and demand.

Next, we depict the evolution of the system. Using $\mathbf{f}_{kj}^t = \{f_{kg,jg'}^t\} \in \mathbb{R}_{\geq 0}^{\bar{\mathcal{G}} \times \bar{\mathcal{G}}}$ to denote the flow matrix between level k and j , the *evolution of demographics* is given by

$$\mathbf{s}_k^{t+1} = \sum_{j=1, \dots, K} (\mathbf{f}_{kj}^t \mathbf{1})^\top \bar{M}_{kj}, \quad k = 1, \dots, K, \tag{ED}$$

where $\mathbf{1}$ is a $|\bar{\mathcal{G}}| \times 1$ unit vector. Note that in (ED), $\mathbf{f}_{kj}^t \mathbf{1}$ is the vector describing how many level k players are matched to level j players for all states in $\bar{\mathcal{G}}$, and recall that $\bar{M}_{kj}$ is the state transition matrix for level k players matched to level j players. The engagement at period t is given by $\sum_{k=1}^K \sum_{g \in \bar{\mathcal{G}}} s_{kg}^t$, the total amount of active players.

The firm's objective is to maximize engagement, which we measure by the cumulative amount of active players across all periods, $\sum_{t=1}^{\infty} \gamma^{t-1} \sum_{k=1}^K \sum_{g \in \bar{\mathcal{G}}} s_{kg}^t$, where $\gamma \in (0, 1]$ is the discount factor. The engagement maximization problem can be formulated as a Markov decision process, where the states are $\mathbf{s}^t$, the amount of active players in each demographic. Let V^π be the value function of a feasible policy π . The value-to-go function is given by

$$V^\pi(\mathbf{s}^t) = \sum_{k=1}^K \sum_{g \in \bar{\mathcal{G}}} s_{kg}^{t+1} + \gamma V^\pi(\mathbf{s}^{t+1}) \quad (\text{EC.1})$$

subject to (FB), (ED).

Our goal is to find the optimal policy with a value function $V^*(\cdot)$, such that $V^*(\mathbf{s}^t) \geq V^\pi(\mathbf{s}^t)$ for any feasible policy π . Note that the system dynamics are again all linear, just like the baseline model, so when the initial size of each demographic is given, maximizing the engagement is equivalent to solving an infinite linear program, which we describe in details in Appendix A.3.

Also in Appendix A.3, we discuss another four more possible extensions of our model: 1) adding a draw outcome 2) only a fraction of active players join the matchmaking pool 3) the game may last multiple (and possibly random) periods; 4) new players arrive depending on the entire history of active players. All of the above extensions can be incorporated into the LP formulation.

We conclude this section by outlining a few technical results. First, with the comprehensive model, we can still derive the value function under SBMM where players are only matched to others in the same skill level. Second, we can guarantee that the total engagement under any policy is finite. Finally, we can show that strong duality and complementary slackness hold for the infinite linear program associating to (EC.1) (formally presented in Appendix A.4 (EC.2)). We leave the detailed technical discussions to Appendix A.4.

A.3. Linear Program Formulation of the Comprehensive model and Possible Extensions

First, we provide the complete linear program of the engagement maximization problem with the comprehensive model:

$$\begin{aligned} V^*(\mathbf{s}^0) = \max & \sum_{t=1}^{\infty} \gamma^{t-1} \sum_k \sum_{g \in \bar{\mathcal{G}}} s_{kg}^t \\ \text{s.t.} & \sum_{j=1}^K \sum_{g' \in \bar{\mathcal{G}}} f_{kg',jg'}^t = s_{kg}^t, \forall k, \forall g \in \bar{\mathcal{G}}, t = 0, 1, \dots \\ & \sum_{j=1}^K \sum_{g' \in \bar{\mathcal{G}}} f_{jg',kg}^t = s_{kg}^t, \forall k, \forall g \in \bar{\mathcal{G}}, t = 0, 1, \dots \\ & f_{kg,jg'}^t = f_{jg',kg}^t, j = 1, \dots, K, k = 1, \dots, K, \forall g \in \bar{\mathcal{G}}, g' \in \bar{\mathcal{G}}, t = 0, 1, \dots \\ & f_{kg,jg'}^t \geq 0, j = 1, \dots, K, k = 1, \dots, K, \forall g \in \bar{\mathcal{G}}, g' \in \bar{\mathcal{G}}, t = 0, 1, \dots \\ & \mathbf{s}_k^{t+1} = \sum_{j=1, \dots, K} (\mathbf{f}_{kj}^t \mathbf{1})^\top \bar{M}_{kj}, \forall k, t = 0, 1, \dots \end{aligned} \quad (\text{EC.2})$$

Next, we point out that our framework (EC.2) is flexible enough to allow for the following various practical extensions while still resulting in a nice LP formulation.

1. A draw/tie outcome can be easily added, since our model only depends on the aggregate transition matrix M_{kk} . Let $P_{tie}^k \in [0, 1]^{|\mathcal{G}| \times |\mathcal{G}|}$ be the transition matrix of a level k player's engagement state, given that they experienced a tie. Let $p_{kj}^{win}, p_{kj}^{lose}$ and p_{kj}^{tie} be the probability of win, lose and tie for a level k player who faces a level j player. Then the aggregate transition matrix is given by $M_{kj} = p_{kj}^{win} P_{win}^k + p_{kj}^{lose} P_{lose}^k + p_{kj}^{tie} P_{tie}^k$. Then the computation follows Eq. (EC.2).
2. If in each period, only *part* fraction of the idle players want to play, then we can simply multiply *part* on the right-hand-side of (FB) and add $(1 - part)s_k^t$ on the right-hand-side of (ED). For Eq. (FB), it now becomes

$$\begin{aligned}
\sum_{j=1}^K \sum_{g' \in \bar{\mathcal{G}}} f_{kg, jg'}^t &= part \cdot s_{kg}^t, \quad k = 1, \dots, K, \forall g \in \bar{\mathcal{G}}, \\
\sum_{j=1}^K \sum_{g' \in \bar{\mathcal{G}}} f_{jg', kg}^t &= part \cdot s_{kg}^t, \quad k = 1, \dots, K, \forall g \in \bar{\mathcal{G}}, \\
f_{kg, jg'}^t &= f_{jg', kg}^t, \quad j = 1, \dots, K, k = 1, \dots, K, \forall g \in \bar{\mathcal{G}}, g' \in \bar{\mathcal{G}} \\
f_{kg, jg'}^t &\geq 0, \quad j = 1, \dots, K, k = 1, \dots, K, \forall g \in \bar{\mathcal{G}}, g' \in \bar{\mathcal{G}}
\end{aligned} \tag{EC.3}$$

For Eq. (ED), it now becomes

$$s_k^{t+1} = (1 - part)s_k^t + \sum_{j=1, \dots, K} (\mathbf{f}_{kj}^t \mathbf{1})^\top \bar{M}_{kj} \quad k = 1, \dots, K. \tag{EC.4}$$

3. If the match duration is not one period, we can modify (ED) so that the match flow returns to the demographics after a positive and random delay. Suppose a match may last at most D periods. Let w_d be the probability that a match lasts d periods, and $\sum_{d=1}^D w_d = 1$. Then Eq. (ED) can be modified as

$$s_k^{t+1} = \sum_{d=1}^D w_d \sum_{j=1, \dots, K} (\mathbf{f}_{kj}^{t+1-d} \mathbf{1})^\top \bar{M}_{kj} \quad k = 1, \dots, K. \tag{EC.5}$$

4. New players whose amounts are linear functions of past history can be introduced easily by modifying (ED). Let $New_k^t(\{\mathbf{f}_{kj}^\tau\}_{\tau=1}^{t-1})$ be a linear function of $\mathbf{f}_{kj}^\tau$, $\tau = 1, \dots, t-1$. Then we can modify Eq. (ED) as

$$s_k^{t+1} = New_k^t(\{\mathbf{f}_{kj}^\tau\}_{\tau=1}^{t-1}) + \sum_{j=1, \dots, K} (\mathbf{f}_{kj}^t \mathbf{1})^\top \bar{M}_{kj} \quad k = 1, \dots, K. \tag{EC.6}$$

A.4. SBMM and Technical Results

To make our problem well-defined, we make the following mild assumption: there exists a positive integer m such that the probability of churning after $m+1$ consecutive losses is strictly positive. This assumption clearly holds in reality. Similar to our baseline model, in the generalized comprehensive model, we use SBMM to refer to any policy that lets players in the same skill level match with each other (they are all equivalent

with respect to engagement). Once again, SBMM is always feasible by letting players in the same state match each other, i.e., $f_{kg,kg}^t = s_{kg}^t$. Let $V^{SBMM}(\cdot)$ be the value function of SBMM. We first characterize the value of SBMM in Lemma EC.1, which is analogous to Lemma 1. Next, we show the finiteness of the value function under any policy.

LEMMA EC.1 (Value of Skill-based Policy). *For any $\gamma \in (0, 1]$ and any initial state of demographics $\mathbf{s}^0$, the value function of SBMM is*

$$V^{SBMM}(\mathbf{s}^0) = \sum_{k=1}^K \mathbf{v}_k^\top \mathbf{s}_k^0,$$

where $\mathbf{v}_k$ is given by

$$\mathbf{v}_k = \gamma^{-1} \left((I - \gamma \bar{M}_{kk})^{-1} - I \right) \mathbf{1}.$$

Under SBMM, a level k player starts from an active state g and transitions to the next engagement state according to the Markov chain M_{kk} . Note that each unit of players generates one unit of engagement in each period. Thus, the total engagement a unit of player can generate, which we refer to as their shadow price, is the average number of periods they stay active and is described by $\mathbf{v}_k$. The eigenvalue of $\bar{M}_{kk}$ is less than 1 and the total engagement is finite even when $\gamma = 1$.

LEMMA EC.2 (Finiteness of the Value Function). *For any policy π and initial state of demographics $\mathbf{s}^0$, $V^\pi(\mathbf{s}^0)$ is finite.*

Lemma EC.2 is somewhat counter-intuitive: because a myopic policy gains non-negative advantages in every period compared to SBMM, it is intuitive that the power of optimal policy should grow exponentially with time (also conjectured by Chen et al. (2017b)). However, Lemma EC.2 shows that the benefit of the optimal policy is finite.

Finally, because (EC.2) is an infinite LP, strong duality is not always guaranteed, although we prove it does hold in our setting.

LEMMA EC.3 (Strong Duality). *Strong duality and complementary slackness hold for (EC.2) for any initial demographics $\mathbf{s}^0$.*

Note that the stylized model presented in Section 3 is a special case of the comprehensive model in this section, so we also have strong duality and complementary slackness for (P). Proofs of all technical results in this Section will be provided in Appendix B.2.

A.5. General Model Estimation and Winrate model

We present a general estimation procedure that can accommodate more general player behaviors. To be more specific, we illustrate how to estimate P_{win}^k and P_{lose}^k through maximum log-likelihood estimation without assuming players' churn behavior is based on aversion to losing streaks. To simplify our problem and reduce the number of parameters, we assume that the players' churn behavior is independent of skill level and drop the superscript k . In the state transition matrix, the row that represents state g in P_{win} (P_{lose}) only has two positive entries: the probability of churning after winning (losing) the next match and the probability of moving to a certain non-churn state. Our goal is then to estimate ρ_g^{win} (ρ_g^{lose}), the churn probability of a

player in state $g \in \mathcal{G}$ after winning (losing) the next match (Note that $\rho_q^{win} = \rho_q^{lose} = 1$ since we assume that a churned player will stay churned.) Let g_t^i be the state player i reaches before playing t -th match, $\mathbb{W}_t^i$ be the indicator of whether player i wins the t -th match, and $\mathbb{I}_t^i$ be the indicator of whether the player i churns right after reaching g_t^i . For a player who played T^i total matches, we record his/her states and churn decisions starting from the 5th match because they only have a valid state right before the 5th match if $m = 4$. If the state sequence is given by $(g_5^i, g_6^i, \dots, g_{T^i}^i)$, the outcome sequence is $(\mathbb{W}_5^i, \mathbb{W}_6^i, \dots, \mathbb{W}_{T^i}^i)$, and churn decision sequence is $(\mathbb{I}_5^i, \mathbb{I}_6^i, \dots, \mathbb{I}_{T^i}^i)$, then the log-likelihood (LLH) function is given by

$$\sum_{i=1}^N \sum_{t=5}^{T^i} \log \left(\rho_{g_t^i}^{win} \mathbb{W}_t^i \mathbb{I}_t + (1 - \rho_{g_t^i}^{win}) \mathbb{W}_t^i (1 - \mathbb{I}_t) + \rho_{g_t^i}^{lose} (1 - \mathbb{W}_t^i) \mathbb{I}_t + (1 - \rho_{g_t^i}^{lose}) (1 - \mathbb{W}_t^i) (1 - \mathbb{I}_t) \right).$$

Because the state transition is exogenous, the estimation is computationally simple and usually has a closed form. We show the closed form of the parameters for a winrate model with $m + 2$ parameters below. *General Estimation Procedure and Winrate Model:* The winrate model assumes that the players' churn probability depends on the number of wins over the last m matches plus the next match (total $m+1$ matches). Hence, the model can be described by $m + 2$ parameters, the churn probabilities when the player has 0 to $m + 1$ wins. Note that although we only need to estimate $m + 2$ parameters, we still need $2^m + 1$ states. This is because, for the LP formulation, the customer transition matrix needs the full win-loss record. Hence, a state g is a length- m binary sequence denoting the win-loss record, and we use $|g|$ to denote the wins in g . The solution for the MLE is

$$\rho_g^{win} = \frac{\text{Number of churn decision when players won } |g| + 1 \text{ matches over last } m + 1 \text{ matches}}{\text{Number of times that players won } \tilde{g} \text{ matches over last } m + 1 \text{ matches}},$$

$$\rho_g^{lose} = \frac{\text{Number of churn decision when players won } |g| \text{ matches over the } m + 1 \text{ matches}}{\text{Number of times that players won } \tilde{g} \text{ matches over the } m + 1 \text{ matches}}.$$

The next table summarizes the out-of-sample negative LLH for some winrate models and their performance (relative power) compared to SBMM.

Table EC.1 Out-of-sample LLH and Relative Power Compared to SBMM for Winrate Models

	LLH	Optimal Policy	Random Policy
$m = 1$	139639.4	4.33%	-2.12%
$m = 2$	139496.3	5.92%	-4.18%
$m = 3$	139309.0	6.63%	-6.17%

A.6. Robustness Check of Alternative Initial Populations

In Section 5.3, we provide performances of different policies under an initial state calibrated from Lichess. To ensure that our insights are robust, we consider three alternative initial states and report the performance in this section. We consider the following representative states:

- A. Uniform: The initial state of all user types are the same, i.e., $s_{k,g}^0 = 1$ for all k, g .
- B. Mostly low-skilled: the low-skilled players are exponentially more than high-skilled players. Specifically, we set $s_{k,g}^0 = 2^{(13-k)/2}$ for all k, g .

C. Mostly high-skilled: the high-skilled players are exponentially more than low-skilled players. Specifically, we set $s_{k,g}^0 = 2^{(k-1)/2}$ for all k, g .

Table EC.2-EC.4 reports the results under these three initial states. The insights are exactly the same: The LF myopic performs better than HF myopic in all situations, highlighting the need for careful tiebreaking. The dual-based policy has almost zero gap with the optimal policy, suggesting that our constructed shadow prices capture the trade-off in the optimal policy.

Table EC.2 Performance Comparison under Uniform Initial State

	Random	Optimal	LF Myopic	HF Myopic	Dual Policy
$m = 1$	-2.70%	5.41%	5.28%	4.70%	5.41%
$m = 2$	-4.97%	6.91%	6.62%	6.39%	6.86%
$m = 3$	-6.67%	6.91%	6.62%	6.39%	6.86%
$m = 4$	-7.77%	6.62%	6.36%	6.14%	6.57%

Table EC.3 Performance Comparison under Mostly Low-skilled Initial State

	Random	Optimal	LF Myopic	HF Myopic	Dual Policy
$m = 1$	-1.78%	3.50%	3.46%	2.96%	3.49%
$m = 2$	-3.07%	4.33%	4.16%	3.82%	4.30%
$m = 3$	-3.89%	4.33%	4.14%	3.86%	4.29%
$m = 4$	-4.31%	4.19%	4.02%	3.75%	4.16%

Table EC.4 Performance Comparison under Mostly High-skilled Initial State

	Random	Optimal	LF Myopic	HF Myopic	Dual Policy
$m=1$	-1.45%	3.24%	3.13%	2.54%	3.24%
$m=2$	-2.80%	5.64%	5.34%	4.94%	5.64%
$m=3$	-3.88%	6.17%	5.89%	5.73%	6.15%
$m=4$	-4.62%	5.89%	5.67%	5.56%	5.89%

Appendix B: Omitted Proofs

B.1. Omitted Proofs from Section 4

Before proving results in Section 4, we first present a simplified formulation to the matchmaker's problem in (P). Recall that $\mathcal{P} := \{2w, 2\ell, 1w, 1\ell\}$. To begin, in any period $t = 0, 1, 2, \dots$, matching flows $f_{2w,2\ell}^t$, $f_{1w,1\ell}^t$, $f_{2\ell,2w}^t$, and $f_{1\ell,1w}^t$ can be set to zero without loss of generality. Taking flows $f_{2w,2\ell}^t = f_{2\ell,2w}^t = a \in (0, \min\{s_{2w}^t, s_{2\ell}^t\}]$ as an example, it can be represented by $f_{2w,2w}^t = a$ and $f_{2\ell,2\ell}^t = a$, since they induce the same evolution to players' demographics $\{s_{2w}^{t+1}, s_{2\ell}^{t+1}\}$ in the next period. We also use the fact that $f_{i,j}^t =$

$f_{j,i}^t, \forall i, j \in \mathcal{P}$ to reduce the problem to 8 flow variables for each period, which is half of the original described in (P). Thus, we can rewrite the matchmaker's problem in (P) as

$$\begin{aligned}
& \max_{\{\mathbf{f}^t\}_{t=0}^{\infty}} \sum_{t=1}^{\infty} \left(s_{2w}^t + s_{2\ell}^t + s_{1w}^t + s_{1\ell}^t \right) \\
& \text{s.t. } s_{2w}^0 = f_{2w,2w}^0 + f_{2w,1w}^0 + f_{2w,1\ell}^0, \\
& \quad s_{2\ell}^0 = f_{2\ell,2\ell}^0 + f_{2\ell,1w}^0 + f_{2\ell,1\ell}^0, \\
& \quad s_{1w}^0 = f_{1w,1w}^0 + f_{2\ell,1w}^0 + f_{2w,1w}^0, \\
& \quad s_{1\ell}^0 = f_{1\ell,1\ell}^0 + f_{2w,1\ell}^0 + f_{2\ell,1\ell}^0, \\
& \quad s_{2w}^t = f_{2w,2w}^t + f_{2w,1w}^t + f_{2w,1\ell}^t, \quad t = 1, 2, \dots, \\
& \quad s_{2\ell}^t = f_{2\ell,2\ell}^t + f_{2\ell,1w}^t + f_{2\ell,1\ell}^t, \quad t = 1, 2, \dots, \\
& \quad s_{1w}^t = f_{1w,1w}^t + f_{2\ell,1w}^t + f_{2w,1w}^t, \quad t = 1, 2, \dots, \\
& \quad s_{1\ell}^t = f_{1\ell,1\ell}^t + f_{2w,1\ell}^t + f_{2\ell,1\ell}^t, \quad t = 1, 2, \dots, \\
& \quad s_{2w}^t = \frac{1}{2} (f_{2w,2w}^{t-1} + f_{2\ell,2\ell}^{t-1}) + f_{2w,1w}^{t-1} + f_{2w,1\ell}^{t-1} + f_{2\ell,1w}^{t-1} + f_{2\ell,1\ell}^{t-1}, \quad t = 1, 2, \dots, \\
& \quad s_{2\ell}^t = \frac{1}{2} f_{2w,2w}^{t-1}, \quad t = 1, 2, \dots, \\
& \quad s_{1w}^t = \frac{1}{2} (f_{1w,1w}^{t-1} + f_{1\ell,1\ell}^{t-1}), \quad t = 1, 2, \dots, \\
& \quad s_{1\ell}^t = \frac{1}{2} f_{1w,1w}^{t-1} + f_{2w,1w}^{t-1} + f_{2\ell,1w}^{t-1}, \quad t = 1, 2, \dots, \\
& \quad f_{i,j}^t \geq 0, \quad \forall i, j, t \in \mathcal{P}.
\end{aligned} \tag{FB}$$

By merging the flow balance and evolution of demographic constraints, we can further remove s_i^t for $t > 1$, with only 8 decision variables per period:

$$\begin{aligned}
& \max_{\{\mathbf{f}^t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \left(f_{2w,2w}^t + \frac{1}{2} f_{2\ell,2\ell}^t + f_{1w,1w}^t + \frac{1}{2} f_{1\ell,1\ell}^t + 2f_{2w,1w}^t + f_{2\ell,1\ell}^t + f_{2w,1\ell}^t + 2f_{2\ell,1w}^t \right) \\
& \text{s.t.} \\
& \quad s_{2w}^0 = f_{2w,2w}^0 + f_{2w,1w}^0 + f_{2w,1\ell}^0, \\
& \quad s_{2\ell}^0 = f_{2\ell,2\ell}^0 + f_{2\ell,1w}^0 + f_{2\ell,1\ell}^0, \\
& \quad s_{1w}^0 = f_{1w,1w}^0 + f_{2\ell,1w}^0 + f_{2w,1w}^0, \\
& \quad s_{1\ell}^0 = f_{1\ell,1\ell}^0 + f_{2w,1\ell}^0 + f_{2\ell,1\ell}^0, \\
& \quad f_{2w,2w}^t + f_{2w,1w}^t + f_{2w,1\ell}^t = \frac{1}{2} (f_{2w,2w}^{t-1} + f_{2\ell,2\ell}^{t-1}) + f_{2w,1w}^{t-1} + f_{2w,1\ell}^{t-1} + f_{2\ell,1w}^{t-1} + f_{2\ell,1\ell}^{t-1}, \quad t = 1, 2, \dots, \\
& \quad f_{2\ell,2\ell}^t + f_{2\ell,1w}^t + f_{2\ell,1\ell}^t = \frac{1}{2} f_{2w,2w}^{t-1}, \quad t = 1, 2, \dots, \\
& \quad f_{1w,1w}^t + f_{2\ell,1w}^t + f_{2w,1w}^t = \frac{1}{2} (f_{1w,1w}^{t-1} + f_{1\ell,1\ell}^{t-1}), \quad t = 1, 2, \dots, \\
& \quad f_{1\ell,1\ell}^t + f_{2w,1\ell}^t + f_{2\ell,1\ell}^t = \frac{1}{2} f_{1w,1w}^{t-1} + f_{2w,1w}^{t-1} + f_{2\ell,1w}^{t-1}, \quad t = 1, 2, \dots, \\
& \quad f_{i,j}^t \geq 0, \quad \forall i, j, t \in \mathcal{P}.
\end{aligned} \tag{P'}$$

Proof of Lemma 1. Under SBMM, for each player, their state transition follows a Markov chain. After one round of match, a winning player garners one unit of engagement, and becomes either a winning or

losing player with the probability $1/2$. A losing player may become a winning player with probability $1/2$, or transfer to an absorbing state *quit* otherwise. The average engagement is the time before they transit to state *quit*. Let $v_{2w}, v_{2\ell}, v_{1w}, v_{1\ell}$ be the expected staying time of each type. For winning players, we have

$$v_{iw} = 1 + \frac{1}{2}v_{iw} + \frac{1}{2}v_{il}, \quad i = 1, 2.$$

For losing players, we have

$$v_{il} = \frac{1}{2} + \frac{1}{2}v_{iw}, \quad i = 1, 2.$$

Solving the above two equations, we have $v_{iw} = 5$ and $v_{il} = 3$, for $i = 1, 2$. The total engagement give state $\mathbf{s}^t$ is then $5(s_{2w}^t + s_{1w}^t) + 3(s_{2\ell}^t + s_{1\ell}^t)$. $\square$

Since our problem does not induce steady-states due to finiteness, we thus consider a generalization of steady-state referred to as *decaying steady-state*. A policy admits a decaying steady state $\mathbf{s}^t$, if there exists some $c \in (0, 1]$ such that under the given policy, we have $\mathbf{s}^{t+1} = c\mathbf{s}^t$. Unfortunately, Lemma EC.4 shows that no matching policy, besides SBMM, results in its demographics reaching a decaying steady state.

LEMMA EC.4 (No Steady State Exists). *Consider a fixed time period t .*

(i) *SBMM can induce a non-zero decaying steady-state, but only for $c = (1 + \sqrt{5})/4$ and $\mathbf{s}^t$ a positive multiple of the vector $((1 + \sqrt{5})/2, 1, (1 + \sqrt{5})/2, 1)$.*

(ii) *For any matching policy that involves matching flows between high-skilled and low-skilled players, there is no non-zero decaying steady-state for any $c \in (0, 1]$.*

Proof of Lemma EC.4. We work with the alternative formulation in (P') which has 8 flow variables each period.

(i) Let $\mathbf{f}^t = (f_{2w,2w}^t, f_{2\ell,2\ell}^t, f_{1w,1w}^t, f_{1\ell,1\ell}^t, f_{2w,1w}^t, f_{2w,1\ell}^t, f_{2\ell,1w}^t, f_{2\ell,1\ell}^t)$ be the flow vector at time t . With (FB_S) conditions, the state $\mathbf{s}^t$ can then be expressed as $B\mathbf{f}^t$, where

$$B = \begin{matrix} & f_{2w,2w}^t & f_{2\ell,2\ell}^t & f_{1w,1w}^t & f_{1\ell,1\ell}^t & f_{2w,1w}^t & f_{2w,1\ell}^t & f_{2\ell,1w}^t & f_{2\ell,1\ell}^t \\ \begin{matrix} s_{2w}^t \\ s_{2\ell}^t \\ s_{1w}^t \\ s_{1\ell}^t \end{matrix} & \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \end{pmatrix} \end{matrix}.$$

Similarly, using the (ED_S) conditions, the state $\mathbf{s}^{t+1}$ can be expressed as $A\mathbf{f}^t$, where

$$A = \begin{matrix} & f_{2w,2w}^t & f_{2\ell,2\ell}^t & f_{1w,1w}^t & f_{1\ell,1\ell}^t & f_{2w,1w}^t & f_{2w,1\ell}^t & f_{2\ell,1w}^t & f_{2\ell,1\ell}^t \\ \begin{matrix} s_{2w}^t \\ s_{2\ell}^t \\ s_{1w}^t \\ s_{1\ell}^t \end{matrix} & \begin{pmatrix} 0.5 & 0.5 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0.5 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.5 & 0.5 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.5 & 0 & 1 & 0 & 1 & 0 \end{pmatrix} \end{matrix}.$$

If there exists a decaying steady state for some $c \in [0, 1]$, then there exists vector $\mathbf{f}^t \geq 0$ such that

$$A\mathbf{f}^t = cB\mathbf{f}^t \iff (A - cB)\mathbf{f}^t = 0.$$

Now we provide the null space of $A - cB$. When $c \neq (1 + \sqrt{5})/4$, the null space is given by

$$\begin{pmatrix} 0.5c/g(c) & (-c^2 + 0.5c + 0.5)/g(c) & -0.5c/g(c) & (-c^2 + 0.5c)/g(c) & 0 & 0 & 0 & 1 \\ 0.5c/g(c) & (-c^2 + 0.5c + 0.5)/g(c) & (-c^2 + 0.5)/g(c) & (0.5c - 0.5)/g(c) & 0 & 0 & 1 & 0 \\ (-c^2 + c)/g(c) & (-0.5c + 0.5)/g(c) & -0.5c/g(c) & (-c^2 + 0.5c)/g(c) & 0 & 1 & 0 & 0 \\ (-c^2 + c)/g(c) & (-0.5c + 0.5)/g(c) & (-c^2 + 0.5)/g(c) & (0.5c - 0.5)/g(c) & 1 & 0 & 0 & 0 \end{pmatrix}, \quad (\text{EC.7})$$

where $g(c) = c^2 - 0.5c - 0.25$. Note that $g(c) = 0$ if $c = (1 + \sqrt{5})/4$. In that case, the null space of $A - cB$ is given by

$$\begin{pmatrix} \frac{1+\sqrt{5}}{2} & 0 & \frac{3+\sqrt{5}}{3-\sqrt{5}} & 0 & \frac{2}{-3+\sqrt{5}} & 0 & 0 & 1 \\ \frac{1+\sqrt{5}}{2} & 0 & \frac{3+\sqrt{5}}{3-\sqrt{5}} & 0 & \frac{-5+\sqrt{5}}{3-\sqrt{5}} & 1 & 1 & 0 \\ 0 & 0 & \frac{1+\sqrt{5}}{2} & 1 & 0 & 0 & 0 & 0 \\ \frac{1+\sqrt{5}}{2} & 1 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

If there exists a linear combination of rows in the null space resulting in all non-negative elements and at least one non-zero element, then we have found a valid flow vector $\mathbf{f}^t$ and thus a valid demographic $\mathbf{s}^t$ that decays steadily at a rate of c .

First, consider $c \in (\frac{1+\sqrt{5}}{4}, 1]$, which implies $g(c) > 0$. Then observe that elements in the third column of Eq. (EC.7), representing flow $f_{1w,1w}^t$ are all negative. Hence, for any linear combination of rows in Eq. (EC.7), as long as the flow $f_{1w,1w}$ is positive, at least one of the elements representing flows $f_{2w,1w}^t, f_{2w,1\ell}^t, f_{2\ell,1w}^t, f_{2\ell,1\ell}^t$ is negative. Thus, no steady state exists when $c \in (\frac{1+\sqrt{5}}{4}, 1]$.

Next, consider $c \in (0, \frac{1+\sqrt{5}}{4})$, which implies $g(c) < 0$. Note that elements in the second column of Eq. (EC.7), representing the flow $f_{2\ell,2\ell}^t$ are all positive. Hence, for any linear combination of rows in Eq. (EC.7), as long as the flow $f_{2\ell,2\ell}$ is positive, then at least one of the elements representing flows $f_{2w,1w}^t, f_{2w,1\ell}^t, f_{2\ell,1w}^t, f_{2\ell,1\ell}^t$ is negative. Thus, no steady state exists when $c \in (0, \frac{1+\sqrt{5}}{4})$.

Finally, consider $c = (1 + \sqrt{5})/4$. In this case, we can easily find a non-negative flow vector $\mathbf{f}^t$ by summing up the third and fourth row, which gives $((1 + \sqrt{5})/2, 1, (1 + \sqrt{5})/2, 1, 0, 0, 0, 0)$, representing SBMM, and any positive multiple of $\mathbf{s} = ((1 + \sqrt{5})/2, 1, (1 + \sqrt{5})/2, 1)$ can induce such flows. To see that no other policy can reach a steady state when $c = (1 + \sqrt{5})/4$, note that we have to find a non-zero non-negative flow vector $\mathbf{f}^t$ with the use of the first two rows. For the first row, the sign of the fifth entry is the opposite of the eighth entry; for the second row, the sign of the fifth entry is opposite of the sixth and seventh entries. Hence, there is no way to construct a feasible flow vector $\mathbf{f}^t$ with non-negative elements. Thus, no other policy besides SBMM can induce a decaying steady state when $c = (1 + \sqrt{5})/4$. $\square$

Proof of Lemma 2. For each flow that involves both high-skilled and low-skilled players, we can compare the outcome of one unit of such flow with SBMM flow that uses the same amount of players. A unit of $f_{2\ell,1w}$ uses the same amount of players as one unit of $f_{2\ell,2\ell}$ and $f_{1w,1w}$, but it leads to zero loss in the next period while SBMM loses a half unit of 2ℓ players. One unit of $f_{2\ell,1\ell}$ or $f_{2w,1w}$ leads to the same losses in the next period compared to the SBMM flow of one unit of $f_{2\ell,2\ell}, f_{1\ell,1\ell}, f_{2w,2w}$, and $f_{1w,1w}$. Finally, one unit of $f_{2w,1\ell}$ leads to a one unit loss of 1ℓ players in the next period, while the SBMM flow of one unit of $f_{1\ell,1\ell}$ and $f_{2w,2w}$ only leads to a half unit loss of 1ℓ players. Hence, to maximize the population in the next period, we should always maximize $f_{2\ell,1w}$ and set $f_{2w,1\ell} = 0$. For the rest of the players, they can be matched arbitrarily as the outcome in the next period is the same as SBMM. $\square$

B.1.1. Optimal Matching Policy The optimal matching flows in Theorem 1 are summarized in detail in the next table:

The optimal policy always maximizes matching flows between demographics 2ℓ and $1w$. The rest of the players are matched via SBMM in most scenarios (third row in Table EC.5), except when there are fewer

Table EC.5 Optimal Matchmaking Policy, $K_1 := \frac{18}{5}s_{2w}^t + \frac{9}{5}s_{2\ell}^t + \frac{3}{5}s_{1w}^t$, $K_2 := \frac{18}{5}s_{2w}^t + \frac{23}{5}s_{2\ell}^t - \frac{11}{5}s_{1w}^t$

States of demographics	Optimal matching policy
$s_{2w}^t + s_{2\ell}^t < s_{1w}^t + s_{1\ell}^t$, $s_{2\ell}^t \geq s_{1w}^t$, $s_{2w}^t < s_{1\ell}^t$, and $s_{1\ell}^t > K_2$	$f_{2w,2w}^t = s_{2w}^t$, $f_{2\ell,1w}^t = f_{1w,2\ell}^t = s_{1w}^t$, $f_{2\ell,1\ell}^t = f_{1\ell,2\ell}^t = s_{2\ell}^t - s_{1w}^t$, and $f_{1\ell,1\ell}^t = s_{1\ell}^t - f_{2\ell,1\ell}^t$
$s_{2w}^t + s_{2\ell}^t < s_{1w}^t + s_{1\ell}^t$, $s_{2\ell}^t \geq s_{1w}^t$, $s_{2w}^t < s_{1\ell}^t$, and $K_1 < s_{1\ell}^t \leq K_2$	$f_{2w,2w}^t = s_{2w}^t$, $f_{2\ell,1w}^t = f_{1w,2\ell}^t = s_{1w}^t$, $f_{2\ell,2\ell}^t = \frac{9}{7}s_{2w}^t + \frac{23}{14}s_{2\ell}^t - \frac{11}{14}s_{1w}^t - \frac{5}{14}s_{1\ell}^t$, $f_{2\ell,1\ell}^t = f_{1\ell,2\ell}^t = \frac{5}{14}s_{1\ell}^t - \frac{9}{7}s_{2w}^t - \frac{9}{14}s_{2\ell}^t - \frac{3}{14}s_{1w}^t$, and $f_{1\ell,1\ell}^t = s_{1\ell}^t - f_{2\ell,1\ell}^t$
Otherwise	$f_{2w,2w}^t = s_{2w}^t$, $f_{2\ell,2\ell}^t = s_{2\ell}^t - \min\{s_{2\ell}^t, s_{1w}^t\}$, $f_{1w,1w}^t = s_{1w}^t - \min\{s_{2\ell}^t, s_{1w}^t\}$, $f_{1\ell,1\ell}^t = s_{1\ell}^t$, and $f_{2\ell,1w}^t = f_{1w,2\ell}^t = \min\{s_{2\ell}^t, s_{1w}^t\}$

high-skilled players compared to low-skilled players and too many 1ℓ players (first and second rows in Table EC.5). In this case, the matchmaker also matches players in demographic 2ℓ with players in 1ℓ .

B.1.2. Proof of Theorem 1 and Theorem 2 Before proving Theorem 1, we write out the dual problem of (P') explicitly. Denote λ_i^t for $i \in \mathcal{P}$ as the dual variables (shadow price) for each demographic in period $t = 0, 1, 2, \dots$. Then we can write the dual problem of (P') as

$$\begin{aligned}
& \min_{\{\lambda^t\}} \sum_{i \in \mathcal{P}} s_i^0 \lambda_i^0 & (D') \\
& \text{s.t.} \\
& 1 \leq \lambda_{2w}^t - \frac{1}{2}\lambda_{2w}^{t+1} - \frac{1}{2}\lambda_{2\ell}^{t+1}, \quad t = 0, 1, 2, \dots, \\
& \frac{1}{2} \leq \lambda_{2\ell}^t - \frac{1}{2}\lambda_{2w}^{t+1}, \quad t = 0, 1, 2, \dots, \\
& 2 \leq \lambda_{2\ell}^t + \lambda_{1w}^t - \lambda_{2w}^{t+1} - \lambda_{1\ell}^{t+1}, \quad t = 0, 1, 2, \dots, \\
& \frac{1}{2} \leq \lambda_{1\ell}^t - \frac{1}{2}\lambda_{1w}^{t+1}, \quad t = 0, 1, 2, \dots, \\
& 2 \leq \lambda_{2w}^t + \lambda_{1w}^t - \lambda_{2w}^{t+1} - \lambda_{1\ell}^{t+1}, \quad t = 0, 1, 2, \dots, \\
& 1 \leq \lambda_{2w}^t + \lambda_{1\ell}^t - \lambda_{2w}^{t+1}, \quad t = 0, 1, 2, \dots, \\
& 1 \leq \lambda_{2\ell}^t + \lambda_{1\ell}^t - \lambda_{2w}^{t+1}, \quad t = 0, 1, 2, \dots, \\
& 1 \leq \lambda_{1w}^t - \frac{1}{2}\lambda_{1w}^{t+1} - \frac{1}{2}\lambda_{1\ell}^{t+1}, \quad t = 0, 1, 2, \dots
\end{aligned}$$

Proof of Theorem 1. We solve the problem in (P') by considering its dual problem (D'). In each period, there are four constraints in the primal problem (P'). Thus, we assign dual variable λ_i^t , where $i \in \{1w, 1\ell, 2w, 2\ell\}$, to each constraint representing the evolution of a players' demographics group in the primal problem. We will fully characterize the transition of primal and dual sequences, and show optimality by checking primal/dual feasibility and complementary slackness.

We break down the rest of this proof into 7 steps, corresponding to 7 scenarios of players' demographics. In each step, we analyze a scenario corresponding to a state of players' demographics. The states described by the 7 scenarios are mutually exclusive and collectively exhaustive. That is, in any period t , we have

- Scenario 1: $s_{2w}^t + s_{2\ell}^t \geq s_{1w}^t + s_{1\ell}^t$, $s_{2\ell}^t \geq s_{1w}^t$, and $s_{2w}^t \geq s_{1\ell}^t$; The optimal matching flows are: $f_{2w,2w}^t = s_{2w}^t$, $f_{2\ell,2\ell}^t = s_{2\ell}^t - s_{1w}^t$, $f_{1w,1w}^t = 0$, $f_{1\ell,1\ell}^t = s_{1\ell}^t$, and $f_{2\ell,1w}^t = f_{1w,2\ell}^t = s_{1w}^t$;

- Scenario 2: $s_{2w}^t + s_{2\ell}^t \geq s_{1w}^t + s_{1\ell}^t$, $s_{2\ell}^t < s_{1w}^t$, and $s_{2w}^t \geq s_{1\ell}^t$; The optimal matching flows are: $f_{2w,2\ell}^t = s_{2w}^t$, $f_{2\ell,2\ell}^t = 0$, $f_{1w,1w}^t = s_{1w}^t - s_{2\ell}^t$, $f_{1\ell,1\ell}^t = s_{1\ell}^t$, and $f_{2\ell,1w}^t = f_{1w,2\ell}^t = s_{2\ell}^t$;
- Scenario 3: $s_{2w}^t + s_{2\ell}^t \geq s_{1w}^t + s_{1\ell}^t$, $s_{2\ell}^t \geq s_{1w}^t$, and $s_{2w}^t < s_{1\ell}^t$; The optimal matching flows are the same as those in Scenario 1;
- Scenario 4: $s_{2w}^t + s_{2\ell}^t < s_{1w}^t + s_{1\ell}^t$ and $s_{2\ell}^t < s_{1w}^t$; The optimal matching flows are the same as those in Scenario 2;
- Scenario 5: $s_{2w}^t + s_{2\ell}^t < s_{1w}^t + s_{1\ell}^t$, $s_{2\ell}^t \geq s_{1w}^t$, $s_{2w}^t < s_{1\ell}^t$, and $s_{1\ell}^t \leq K_1 = \frac{18}{5}s_{2w}^t + \frac{9}{5}s_{2\ell}^t + \frac{3}{5}s_{1w}^t$; The optimal matching flows are the same as those in Scenario 1;
- Scenario 6: $s_{2w}^t + s_{2\ell}^t < s_{1w}^t + s_{1\ell}^t$, $s_{2\ell}^t \geq s_{1w}^t$, $s_{2w}^t < s_{1\ell}^t$, and $s_{1\ell}^t > K_2 = \frac{18}{5}s_{2w}^t + \frac{23}{5}s_{2\ell}^t - \frac{11}{5}s_{1w}^t$; The optimal matching flows are: $f_{2w,2\ell}^t = s_{2w}^t$, $f_{2\ell,1w}^t = f_{1w,2\ell}^t = s_{1w}^t$, $f_{2\ell,1\ell}^t = f_{1\ell,2\ell}^t = s_{2\ell}^t - s_{1w}^t$, and $f_{1\ell,1\ell}^t = s_{1\ell}^t - f_{2\ell,1\ell}^t$;
- Scenario 7: $s_{2w}^t + s_{2\ell}^t < s_{1w}^t + s_{1\ell}^t$, $s_{2\ell}^t \geq s_{1w}^t$, $s_{2w}^t < s_{1\ell}^t$, and $K_1 < s_{1\ell}^t \leq K_2$; The optimal matching flows are: $f_{2w,2\ell}^t = s_{2w}^t$, $f_{2\ell,1w}^t = f_{1w,2\ell}^t = s_{1w}^t$, $f_{2\ell,2\ell}^t = \frac{9}{7}s_{2w}^t + \frac{23}{14}s_{2\ell}^t - \frac{11}{14}s_{1w}^t - \frac{5}{14}s_{1\ell}^t$, $f_{2\ell,1\ell}^t = f_{1\ell,2\ell}^t = \frac{5}{14}s_{1\ell}^t - \frac{9}{7}s_{2w}^t - \frac{9}{14}s_{2\ell}^t - \frac{3}{14}s_{1w}^t$, and $f_{1\ell,1\ell}^t = s_{1\ell}^t - f_{2\ell,1\ell}^t$.

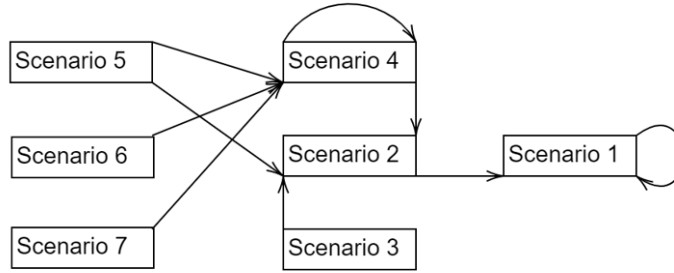


Figure EC.1 Evolution of Players' Demographics

Note that Scenarios 1 to 5 correspond to the third row of Table EC.5. In such cases, the optimal policy simply maximizes the matching between 2ℓ and $1w$ and does skill-based matching for the remaining players. Scenario 6 to 7 corresponds to the first two rows in Table EC.5. The reason we classify the aforementioned scenarios in this way is because of how the scenarios evolve over time, as shown in Figure EC.1. For example, at any period t , if we are in Scenario 1, then in the next period $t + 1$, one can verify that we always stay in the same state of players' demographics described by Scenario 1 under the proposed matching policy. The dual variables for any state in Scenario 1 are constant over time. Similarly, once we reach Scenario 2 at time t , we always transfer to Scenario 1 at $t + 1$.

For the rest of this proof, we show that the state evolves as Fig. EC.1, and our proposed matching policy is optimal in each scenario and its subsequent scenarios as the state demographics evolve. To be more specific, our proof goes through each scenario and shows the corresponding optimal matching flows in Table EC.5 are optimal. Note that by construction, the proposed matching policy in Table EC.5 is primal feasible, i.e., satisfying all constraints in (FB_S) and (ED_S) . We establish optimality by constructing dual variables

for each scenario and show that both complementary slackness and dual feasibility conditions hold (the primal feasibility can be easily verified for our proposed policy). When possible, we write the corresponding dual variables with closed-form expressions. Our proof starts with Scenario 1 as it is at the end of all transitions (does not transition to other scenarios), then works backward to parental scenarios. For each parental scenario, we readily construct the dual variables for all subsequent scenarios, representing all the following time periods. For example, when the demographic state is in Scenario 2, we show it will transition to Scenario 1 in one period under the proposed matching policy. Then we only need to construct the dual variables for the current period and use the dual variables constructed in Scenario 1 for all subsequent periods, which we already have, to establish optimality.

Scenario 1: $s_{2w}^t + s_{2\ell}^t \geq s_{1w}^t + s_{1\ell}^t$, $s_{2\ell}^t \geq s_{1w}^t$, and $s_{2w}^t \geq s_{1\ell}^t$; The optimal matching flows are: $f_{2w,2w}^t = s_{2w}^t$, $f_{2\ell,2\ell}^t = s_{2\ell}^t - s_{1w}^t$, $f_{1w,1w}^t = 0$, $f_{1\ell,1\ell}^t = s_{1\ell}^t$, and $f_{2\ell,1w}^t = f_{1w,2\ell}^t = s_{1w}^t$.

We first consider Scenario 1, and prove that it is the “end” of all scenarios in Figure EC.1. In other words, we shall show that once the state of players’ demographics falls in Scenario 1, it will remain in Scenario 1. To see this, for some $\mathbf{s}^t = \{s_{1w}^t, s_{1\ell}^t, s_{2w}^t, s_{2\ell}^t\}$ in Scenario 1, following the proposed policy, the state at $t+1$ is given by $s_{1w}^{t+1} = \frac{1}{2}s_{1\ell}^t$, $s_{1\ell}^{t+1} = s_{1w}^t$, $s_{2w}^{t+1} = \frac{1}{2}(s_{2w}^t + s_{2\ell}^t + s_{1w}^t)$, and $s_{2\ell}^{t+1} = \frac{1}{2}s_{2w}^t$. Then $s_{2\ell}^{t+1} \geq s_{1w}^{t+1}$ because $s_{2w}^t \geq s_{1\ell}^t$, and $s_{2w}^{t+1} \geq s_{1\ell}^{t+1}$ because $s_{2w}^{t+1} - s_{1\ell}^{t+1} = \frac{1}{2}(s_{2w}^t + s_{2\ell}^t - s_{1w}^t)$ and $s_{2\ell}^t - s_{1w}^t \geq 0$. Hence, $\mathbf{s}^{t+1}$ still belongs to Scenario 1.

Next, we show that the proposed policy in Theorem 1 is optimal for all subsequent periods once players’ demographics satisfy Scenario 1. The optimal solution in Theorem 1 suggests that in Scenario 1, only 4 variables are non-zero while all other flows are zero in each period. Therefore, we have

$$\begin{aligned} 1 &= \lambda_{2w}^t - \frac{1}{2}\lambda_{2w}^{t+1} - \frac{1}{2}\lambda_{2\ell}^{t+1}, \\ \frac{1}{2} &= \lambda_{2\ell}^t - \frac{1}{2}\lambda_{2w}^{t+1}, \\ 2 &= \lambda_{2\ell}^t + \lambda_{1w}^t - \lambda_{2w}^{t+1} - \lambda_{1\ell}^{t+1}, \\ \frac{1}{2} &= \lambda_{1\ell}^t - \frac{1}{2}\lambda_{1w}^{t+1}, \end{aligned} \tag{CS_1}$$

as the complementary conditions corresponding to primal non-zero variables $f_{2w,2w}^t$, $f_{2\ell,2\ell}^t$, $f_{2\ell,1w}^t$, and $f_{1\ell,1\ell}^t$, respectively, and have

$$\begin{aligned} 2 &\leq \lambda_{2w}^t + \lambda_{1w}^t - \lambda_{2w}^{t+1} - \lambda_{1\ell}^{t+1}, \\ 1 &\leq \lambda_{2w}^t + \lambda_{1\ell}^t - \lambda_{2w}^{t+1}, \\ 1 &\leq \lambda_{2\ell}^t + \lambda_{1\ell}^t - \lambda_{2w}^{t+1}, \\ 1 &\leq \lambda_{1w}^t - \frac{1}{2}\lambda_{1w}^{t+1} - \frac{1}{2}\lambda_{1\ell}^{t+1}, \end{aligned} \tag{DF_1}$$

as the dual feasibility conditions corresponding to variables, $f_{2w,1w}^t$, $f_{2w,1\ell}^t$, $f_{2\ell,1\ell}^t$, and $f_{1w,1w}^t$, that are zero in the primal problem, respectively.

The following dual solutions:

$$\lambda_{2w}^t = 5, \lambda_{2\ell}^t = 3, \lambda_{1w}^t = 9, \text{ and } \lambda_{1\ell}^t = 5, \quad \forall t, \tag{EC.8}$$

satisfies complementary slackness in (CS₁) and feasibility conditions in (DF₁). Thus, the proposed policy in Theorem 1 is optimal once the players’ demographics fall in scenario 1.

Scenario 2: $s_{2w}^t + s_{2\ell}^t \geq s_{1w}^t + s_{1\ell}^t$, $s_{2\ell}^t < s_{1w}^t$, and $s_{2w}^t \geq s_{1\ell}^t$; The optimal matching flows are: $f_{2w,2w}^t = s_{2w}^t$, $f_{2\ell,2\ell}^t = 0$, $f_{1w,1w}^t = s_{1w}^t - s_{2\ell}^t$, $f_{1\ell,1\ell}^t = s_{1\ell}^t$, and $f_{2\ell,1w}^t = f_{1w,2\ell}^t = s_{2\ell}^t$;

In the second step, we consider Scenario 2, which will transit to Scenario 1 after matching under the proposed policy as we have stated in Figure EC.1. To see that, for some $\mathbf{s}^t = \{s_{1w}^t, s_{1\ell}^t, s_{2w}^t, s_{2\ell}^t\}$ in Scenario 2, following the proposed policy, the state at $t+1$ is given by $s_{1w}^{t+1} = \frac{1}{2}(s_{1w}^t + s_{1\ell}^t - s_{2\ell}^t)$, $s_{1\ell}^{t+1} = \frac{1}{2}(s_{2\ell}^t + s_{1w}^t)$, $s_{2w}^{t+1} = \frac{1}{2}s_{2w}^t + s_{2\ell}^t$, and $s_{2\ell}^{t+1} = \frac{1}{2}s_{2w}^t$. Then $s_{2\ell}^{t+1} \geq s_{1w}^{t+1}$ because $s_{2\ell}^{t+1} - s_{1w}^{t+1} = \frac{1}{2}(s_{2w}^t + s_{2\ell}^t - s_{1w}^t - s_{1\ell}^t) \geq 0$. Also, $s_{2w}^{t+1} \geq s_{1\ell}^{t+1}$ because $s_{2w}^{t+1} - s_{1\ell}^{t+1} = \frac{1}{2}(s_{2w}^t + s_{2\ell}^t - s_{1w}^t) \geq s_{1\ell}^t \geq 0$. Hence, $\mathbf{s}^{t+1}$ belongs to Scenario 1.

Therefore, we only need to show that in any period t such that players' demographics satisfy Scenario 2, we can find solutions to dual variables, induced by the proposed policy in Theorem 1, which satisfy the complementary slackness conditions and dual feasibility conditions. Note that in period t under Scenario 2, the non-zero primal variables are $f_{2w,2w}^t$, $f_{2\ell,1w}^t$, $f_{1w,1w}^t$ and $f_{1\ell,1\ell}^t$. Therefore, by taking out the condition for $f_{2\ell,1\ell}^t$ and replacing it with the one for $f_{1w,1w}^t$, the complementary slackness conditions in (CS₁) change to

$$\begin{aligned} 1 &= \lambda_{2w}^t - \frac{1}{2}\lambda_{2w}^{t+1} - \frac{1}{2}\lambda_{2\ell}^{t+1}, \\ 2 &= \lambda_{2\ell}^t + \lambda_{1w}^t - \lambda_{2w}^{t+1} - \lambda_{1\ell}^{t+1}, \\ 1 &= \lambda_{1w}^t - \frac{1}{2}\lambda_{1w}^{t+1} - \frac{1}{2}\lambda_{1\ell}^{t+1}, \\ \frac{1}{2} &= \lambda_{1\ell}^t - \frac{1}{2}\lambda_{1w}^{t+1}, \end{aligned} \tag{CS₂}$$

in period t . Similarly, by taking out the condition for $f_{1w,1w}^t$ and replacing with the one for $f_{2\ell,1\ell}^t$, the dual feasibility conditions in (DF₁) turn into

$$\begin{aligned} 2 &\leq \lambda_{2w}^t + \lambda_{1w}^t - \lambda_{2w}^{t+1} - \lambda_{1\ell}^{t+1}, \\ 1 &\leq \lambda_{2w}^t + \lambda_{1\ell}^t - \lambda_{2w}^{t+1}, \\ \frac{1}{2} &\leq \lambda_{2\ell}^t - \frac{1}{2}\lambda_{2w}^{t+1}, \\ 1 &\leq \lambda_{2\ell}^t + \lambda_{1\ell}^t - \lambda_{2w}^{t+1}, \end{aligned} \tag{DF₂}$$

in period t . Note that the complementary slackness conditions and the dual feasibility conditions switch back to those in (CS₁) and (DF₁), starting period $t+1$, as player's demographics transit into Scenario 1. Therefore, we have

$$\lambda_{2w}^s = 5, \lambda_{2\ell}^s = 3, \lambda_{1w}^s = 9, \text{ and } \lambda_{1\ell}^s = 5, \quad \forall s = t+1, \dots, T-1,$$

from (EC.8) and only need to find λ_i^t , where $i \in \{2w, 2\ell, 1w, 1\ell\}$, satisfy conditions in (CS₂) and (DF₂). Indeed, such solutions exist and one can verify that

$$\lambda_{2w}^t = 5, \lambda_{2\ell}^t = 4, \lambda_{1w}^t = 8, \text{ and } \lambda_{1\ell}^t = 5, \tag{EC.9}$$

are the desired solution. Therefore, the proof for Scenario 2 is completed.

Scenario 3: $s_{2w}^t + s_{2\ell}^t \geq s_{1w}^t + s_{1\ell}^t$, $s_{2\ell}^t \geq s_{1w}^t$, and $s_{2w}^t < s_{1\ell}^t$; The optimal matching flows are: $f_{2w,2w}^t = s_{2w}^t$, $f_{2\ell,2\ell}^t = s_{2\ell}^t - s_{1w}^t$, $f_{1w,1w}^t = 0$, $f_{1\ell,1\ell}^t = s_{1\ell}^t$, and $f_{2\ell,1w}^t = f_{1w,2\ell}^t = s_{1w}^t$;

In the third step, we consider Scenario 3, which shall transit to Scenario 2 after matching is done under the proposed policy. To see this, for some $\mathbf{s}^t = \{s_{1w}^t, s_{1\ell}^t, s_{2w}^t, s_{2\ell}^t\}$ in Scenario 3, following the proposed policy, the state at $t + 1$ is given by $s_{2w}^{t+1} = \frac{1}{2}(s_{2w}^t + s_{2\ell}^t + s_{1w}^t)$, $s_{2\ell}^{t+1} = \frac{1}{2}s_{2w}^t$, $s_{1w}^{t+1} = \frac{1}{2}s_{1\ell}^t$, and $s_{1\ell}^{t+1} = s_{1w}^t$. Then $s_{2\ell}^{t+1} < s_{1w}^{t+1}$ because $s_{2w}^t < s_{1\ell}^t$, and $s_{2w}^{t+1} \geq s_{1\ell}^{t+1}$ because $s_{2w}^{t+1} - s_{1\ell}^{t+1} = \frac{1}{2}(s_{2w}^t + s_{2\ell}^t - s_{1w}^t)$ and $s_{2\ell}^t - s_{1w}^t \geq 0$. Finally, $s_{2w}^t + s_{2\ell}^t - s_{1w}^t - s_{1\ell}^t = \frac{1}{2}(2s_{2w}^t + s_{2\ell}^t - s_{1w}^t - s_{1\ell}^t) \geq 0$. Hence, $\mathbf{s}^{t+1}$ belongs to Scenario 2.

Thus, we only need to check we can find solutions to dual variables, which satisfy the complementary slackness conditions and dual feasibility conditions. According to the policy in Theorem 1, in period t under Scenario 3, primal variables $f_{2w,2w}^t$, $f_{2\ell,2\ell}^t$, $f_{2\ell,1w}^t$, and $f_{1\ell,1\ell}^t$ are non-zero. Thus, in period t , the complementary slackness and dual feasibility conditions are

$$\begin{aligned} 1 &= \lambda_{2w}^t - \frac{1}{2}\lambda_{2w}^{t+1} - \frac{1}{2}\lambda_{2\ell}^{t+1}, \\ \frac{1}{2} &= \lambda_{2\ell}^t - \frac{1}{2}\lambda_{2w}^{t+1}, \\ 2 &= \lambda_{2\ell}^t + \lambda_{1w}^t - \lambda_{2w}^{t+1} - \lambda_{1\ell}^{t+1}, \\ \frac{1}{2} &= \lambda_{1\ell}^t - \frac{1}{2}\lambda_{1w}^{t+1}, \end{aligned} \tag{CS_3}$$

and

$$\begin{aligned} 2 &\leq \lambda_{2w}^t + \lambda_{1w}^t - \lambda_{2w}^{t+1} - \lambda_{1\ell}^{t+1}, \\ 1 &\leq \lambda_{2w}^t + \lambda_{1\ell}^t - \lambda_{2w}^{t+1}, \\ 1 &\leq \lambda_{2\ell}^t + \lambda_{1\ell}^t - \lambda_{2w}^{t+1}, \\ 1 &\leq \lambda_{1w}^t - \frac{1}{2}\lambda_{1w}^{t+1} - \frac{1}{2}\lambda_{1\ell}^{t+1}, \end{aligned} \tag{DF_3}$$

respectively, where

$$\lambda_{2w}^{t+1} = 5, \lambda_{2\ell}^{t+1} = 4, \lambda_{1w}^{t+1} = 8, \text{ and } \lambda_{1\ell}^{t+1} = 5,$$

are from (EC.9), since players' demographic shall transit to Scenario 2 in the next period.

Finally, one can verify that

$$\lambda_{2w}^t = 5.5, \lambda_{2\ell}^t = 3, \lambda_{1w}^t = 9, \text{ and } \lambda_{1\ell}^t = 4.5, \tag{EC.10}$$

are the desired solutions we are looking for, satisfying (CS₃) and (DF₃). This completes the proof for Scenario 3.

Scenario 4: $s_{2w}^t + s_{2\ell}^t < s_{1w}^t + s_{1\ell}^t$ and $s_{2\ell}^t < s_{1w}^t$; The optimal matching flows are: $f_{2w,2w}^t = s_{2w}^t$, $f_{2\ell,2\ell}^t = 0$, $f_{1w,1w}^t = s_{1w}^t - s_{2\ell}^t$, $f_{1\ell,1\ell}^t = s_{1\ell}^t$, and $f_{2\ell,1w}^t = f_{1w,2\ell}^t = s_{2\ell}^t$.

In the fourth step, we consider Scenario 4. For some $\mathbf{s}^t = \{s_{1w}^t, s_{1\ell}^t, s_{2w}^t, s_{2\ell}^t\}$ in Scenario 4, following the proposed policy, the state at $t + 1$ is given by $s_{1w}^{t+1} = \frac{1}{2}(s_{1w}^t + s_{1\ell}^t - s_{2\ell}^t)$, $s_{1\ell}^{t+1} = \frac{1}{2}(s_{2\ell}^t + s_{1w}^t)$, $s_{2w}^{t+1} = \frac{1}{2}s_{2w}^t + s_{2\ell}^t$, and $s_{2\ell}^{t+1} = \frac{1}{2}s_{2w}^t$. Then $s_{2\ell}^{t+1} < s_{1w}^{t+1}$ because $s_{1w}^{t+1} - s_{2\ell}^{t+1} = \frac{1}{2}(s_{1w}^t + s_{1\ell}^t - s_{2w}^t - s_{2\ell}^t) > 0$. Hence, the state shall either transit to Scenario 2 or stay in Scenario 4 after a match. Suppose at time $t = s$, players' demographic is in Scenario 4. Denote $\tau := \min\{t \geq s \mid s_{2w}^t + s_{2\ell}^t \geq s_{1w}^t + s_{1\ell}^t\}$, representing the time period players' demographic transit to Scenario 2. First, we argue that $\tau < \infty$. Suppose otherwise, we have $\tau \rightarrow$

∞ , which implies that the players' demographic stays in Scenario 4 forever. Note that as long as players' demographic belongs to Scenario 4, no high-skilled players shall depart from the matching system since there are always enough low-skilled players to be matched with. Thus, we have

$$\sum_{t=s}^{\infty} (s_{2w}^t + s_{2\ell}^t) = \sum_{t=s}^{\infty} (s_{2w}^s + s_{2\ell}^s) \rightarrow \infty,$$

which contradicts Lemma EC.2. Therefore, we have $\tau < \infty$.

Next, under Scenario 2, according to the optimal policy in Theorem 1, we have $\{f_{2w,2w}^t\}_{t=s}^{\tau-1}$, $\{f_{2\ell,1w}^t\}_{t=s}^{\tau-1}$, $\{f_{1w,1w}^t\}_{t=s}^{\tau-1}$, and $\{f_{1\ell,1\ell}^t\}_{t=s}^{\tau-1}$ as the sequences that contain non-zero primal variables, whereas each elements in sequences $\{f_{2w,1w}^t\}_{t=s}^{\tau-1}$, $\{f_{2w,1\ell}^t\}_{t=s}^{\tau-1}$, $\{f_{2\ell,2\ell}^t\}_{t=s}^{\tau-1}$, and $\{f_{2\ell,1\ell}^t\}_{t=s}^{\tau-1}$ are zero.

To proof the proposed policy is optimal in Scenario 4, we verify that there exists sequences $\{\lambda_{2w}^t\}_{t=s}^{\tau}$, $\{\lambda_{2\ell}^t\}_{t=s}^{\tau}$, $\{\lambda_{1w}^t\}_{t=s}^{\tau}$, and $\{\lambda_{1\ell}^t\}_{t=s}^{\tau}$, such that

$$\lambda_{2w}^{\tau} = 5, \lambda_{2\ell}^{\tau} = 4, \lambda_{1w}^{\tau} = 8, \text{ and } \lambda_{1\ell}^{\tau} = 5, \quad (\text{EC.11})$$

and for all $s \leq t \leq \tau - 1$, we have

$$\begin{aligned} 1 &= \lambda_{2w}^t - \frac{1}{2}\lambda_{2w}^{t+1} - \frac{1}{2}\lambda_{2\ell}^{t+1}, & \lambda_{2w}^t &= 1 + \frac{1}{2}\lambda_{2w}^{t+1} + \frac{1}{2}\lambda_{2\ell}^{t+1} \\ 2 &= \lambda_{2\ell}^t + \lambda_{1w}^t - \lambda_{2w}^{t+1} - \lambda_{1\ell}^{t+1}, & \lambda_{2\ell}^t &= 1 + \lambda_{2w}^{t+1} - \frac{1}{2}\lambda_{1w}^{t+1} + \frac{1}{2}\lambda_{1\ell}^{t+1} \\ 1 &= \lambda_{1w}^t - \frac{1}{2}\lambda_{1w}^{t+1} - \frac{1}{2}\lambda_{1\ell}^{t+1}, & \lambda_{1w}^t &= 1 + \frac{1}{2}\lambda_{1w}^{t+1} + \frac{1}{2}\lambda_{1\ell}^{t+1} \\ \frac{1}{2} &= \lambda_{1\ell}^t - \frac{1}{2}\lambda_{1w}^{t+1}, & \lambda_{1\ell}^t &= \frac{1}{2} + \frac{1}{2}\lambda_{1w}^{t+1} \end{aligned} \quad (\text{CS}_4)$$

as complementary slackness conditions and

$$\begin{aligned} 2 &\leq \lambda_{2w}^t + \lambda_{1w}^t - \lambda_{2w}^{t+1} - \lambda_{1\ell}^{t+1}, \\ 1 &\leq \lambda_{2w}^t + \lambda_{1\ell}^t - \lambda_{2w}^{t+1}, \\ \frac{1}{2} &\leq \lambda_{2\ell}^t - \frac{1}{2}\lambda_{2w}^{t+1}, \\ 1 &\leq \lambda_{2\ell}^t + \lambda_{1\ell}^t - \lambda_{2w}^{t+1}, \end{aligned} \quad (\text{DF}_4)$$

as dual feasibility conditions.

For convenience, we also rearrange (CS₄) forwardly:

$$\lambda_{2w}^{t+1} = \lambda_{2\ell}^t - \lambda_{1w}^t + 2\lambda_{1\ell}^t - 1, \quad (\text{EC.12})$$

$$\lambda_{2\ell}^{t+1} = 2\lambda_{2w}^t - \lambda_{2\ell}^t + \lambda_{1w}^t - 2\lambda_{1\ell}^t - 1, \quad (\text{EC.13})$$

$$\lambda_{1w}^{t+1} = 2\lambda_{1\ell}^t - 1, \quad (\text{EC.14})$$

$$\lambda_{1\ell}^{t+1} = 2\lambda_{1w}^t - 2\lambda_{1\ell}^t - 1. \quad (\text{EC.15})$$

Equation (EC.14) follows the third equation of (CS₄). Equation (EC.15) follows the last equation in (CS₄) and (EC.14). Equation (EC.12) follows the second equation in (CS₄), (EC.14), and (EC.15). Finally, (EC.13) follows the first equation in (CS₄) and (EC.13).

Taking (EC.12)-(EC.15) into (DF₄), we can rewrite the dual feasibility conditions as

$$0 \leq \lambda_{2w}^t - \lambda_{2\ell}^t, \quad (\text{EC.16})$$

$$0 \leq \lambda_{2w}^t - \lambda_{2\ell}^t + \lambda_{1w}^t - \lambda_{1\ell}^t, \quad (\text{EC.17})$$

$$0 \leq \lambda_{2\ell}^t + \lambda_{1w}^t - 2\lambda_{1\ell}^t, \quad (\text{EC.18})$$

$$0 \leq \lambda_{1w}^t - \lambda_{1\ell}^t, \quad (\text{EC.19})$$

Note that (EC.17) is implied by summing up (EC.16) and (EC.19). For the rest of this step, we show that the updated dual feasibility conditions (EC.16), (EC.18), (EC.19) are satisfied for all $s \leq t \leq \tau$. We will prove a stronger result: (EC.16), (EC.18), (EC.19) together with the following (EC.20) are satisfied for all $s \leq t \leq \tau$.

$$0 \leq -\lambda_{2w}^t + \lambda_{2\ell}^t + \lambda_{1w}^t - \lambda_{1\ell}^t, \text{ and } -1 \leq \lambda_{2w}^t - \lambda_{1w}^t + \lambda_{1\ell}^t. \quad (\text{EC.20})$$

We prove this result by backward induction with the help of the following lemma.

LEMMA EC.5. *Consider the dual sequences $\{\lambda_{2w}^t\}_{t=s}^\tau$, $\{\lambda_{2\ell}^t\}_{t=s}^\tau$, $\{\lambda_{1w}^t\}_{t=s}^\tau$, and $\{\lambda_{1\ell}^t\}_{t=s}^\tau$ that is defined by Eq. (EC.11) and Eq. (CS₄). Then we have:*

- (1) $\lambda_{1w}^t \geq 5$, $\lambda_{1\ell}^t \geq 3$, for $t = s, \dots, \tau$.
- (2) $\lambda_{1w}^t \leq \lambda_{1w}^{t+1}$, $\lambda_{1\ell}^t \leq \lambda_{1\ell}^{t+1}$, for $t = s, \dots, \tau - 1$.
- (3) $\lambda_{2w}^t \geq \lambda_{2w}^{t+1}$, $\lambda_{2\ell}^t \geq \lambda_{2\ell}^{t+1}$, for $t = s, \dots, \tau - 1$.

For the base step, we first verify that the conditions are satisfied for both τ and $\tau - 1$. The solution for τ is given by $\lambda_{2w}^\tau = 5$, $\lambda_{2\ell}^\tau = 4$, $\lambda_{1w}^\tau = 8$, and $\lambda_{1\ell}^\tau = 5$, and the solution for $\tau - 1$ is given by $\lambda_{2w}^{\tau-1} = 5.5$, $\lambda_{2\ell}^{\tau-1} = 4.5$, $\lambda_{1w}^{\tau-1} = 7.5$, and $\lambda_{1\ell}^{\tau-1} = 4.5$, and one can easily verify that all the conditions are satisfied. As for the induction step, suppose inequalities in (EC.16), (EC.18), (EC.19) and (EC.20) hold for all periods $t = k + 1, \dots, \tau$. Consider period $k \leq \tau - 2$. For (EC.19), we have

$$\lambda_{1w}^k - \lambda_{1\ell}^k = (1 + \frac{1}{2}\lambda_{1w}^{k+1} + \frac{1}{2}\lambda_{1\ell}^{k+1}) - (\frac{1}{2} + \frac{1}{2}\lambda_{1w}^{k+1}) = \frac{1}{2}(1 + \lambda_{1\ell}^{k+1}) \geq 0,$$

where the first equality uses (CS₄) to expand λ_{1w}^k and $\lambda_{1\ell}^k$, and the inequality follows Lemma EC.5(1).

Next, we check the inequalities in (EC.16) and (EC.18). Using (CS₄) to expand the two conditions, we have

$$\begin{aligned} 0 &\leq \lambda_{2w}^k - \lambda_{2\ell}^k \\ \iff 0 &\leq (1 + \frac{1}{2}\lambda_{2w}^{k+1} + \frac{1}{2}\lambda_{2\ell}^{k+1}) - (1 + \lambda_{2w}^{k+1} - \frac{1}{2}\lambda_{1w}^{k+1} + \frac{1}{2}\lambda_{1\ell}^{k+1}) \\ \iff 0 &\leq -\lambda_{2w}^{k+1} + \lambda_{2\ell}^{k+1} + \lambda_{1w}^{k+1} - \lambda_{1\ell}^{k+1}, \end{aligned}$$

and

$$\begin{aligned} 0 &\leq \lambda_{2\ell}^k + \lambda_{1w}^k - 2\lambda_{1\ell}^k \\ \iff 0 &\leq (1 + \lambda_{2w}^{k+1} - \frac{1}{2}\lambda_{1w}^{k+1} + \frac{1}{2}\lambda_{1\ell}^{k+1}) + (1 + \frac{1}{2}\lambda_{1w}^{k+1} + \frac{1}{2}\lambda_{1\ell}^{k+1}) - 2(\frac{1}{2} + \frac{1}{2}\lambda_{1w}^{k+1}) \\ \iff -1 &\leq \lambda_{2w}^{k+1} - \lambda_{1w}^{k+1} + \lambda_{1\ell}^{k+1}. \end{aligned}$$

Note that these two inequalities are exactly (EC.20) in period $k+1$, which hold by our assumption.

Finally, we check (EC.20) for period k . Plugging in (CS₄) twice for period k and $k+1$, we can express $-\lambda_{2w}^k + \lambda_{2\ell}^k + \lambda_{1w}^k - \lambda_{1\ell}^k$ with variables from $k+2$:

$$\begin{aligned} -\lambda_{2w}^k + \lambda_{2\ell}^k + \lambda_{1w}^k - \lambda_{1\ell}^k &= \frac{1}{2}(1 + \lambda_{2w}^{k+1} - \lambda_{2\ell}^{k+1} - \lambda_{1w}^{k+1} + 2\lambda_{1\ell}^{k+1}) \\ &= \frac{1}{2} - \frac{1}{4}\lambda_{2w}^{k+2} + \frac{1}{4}\lambda_{2\ell}^{k+2} + \frac{1}{2}\lambda_{1w}^{k+2} - \frac{1}{2}\lambda_{1\ell}^{k+2} \\ &= \frac{1}{2} + \frac{1}{4}(-\lambda_{2w}^{k+2} + \lambda_{2\ell}^{k+2} + \lambda_{1w}^{k+2} - \lambda_{1\ell}^{k+2}) + \frac{1}{4}(\lambda_{1w}^{k+2} - \lambda_{1\ell}^{k+2}) \geq 0 \end{aligned}$$

where the inequality follows (EC.20) and (EC.19) in period $k+2$. Similarly, we also have

$$\begin{aligned} 1 + \lambda_{2w}^k - \lambda_{1w}^k + \lambda_{1\ell}^k &= \frac{1}{2}(\lambda_{2w}^{k+1} + \lambda_{2\ell}^{k+1} - \lambda_{1\ell}^{k+1}) + \frac{3}{2} \\ &\geq \frac{1}{2}(\lambda_{2w}^{\tau-1} + \lambda_{2\ell}^{\tau-1} - \lambda_{1\ell}^{\tau-1}) + \frac{3}{2} = 4.25 > 0, \end{aligned}$$

where the inequality uses the decreasing property of λ_{2w}^t , $\lambda_{2\ell}^t$ and the increasing property of $\lambda_{1\ell}^t$ from Lemma EC.5. Thus, all dual feasible conditions in (EC.16), (EC.18), (EC.19), and (EC.20) are satisfied, which imply that conditions in (DF₄) also hold for all $s \leq t \leq \tau$. This completes the proof for Scenario 4.

Scenario 5: $s_{2w}^t + s_{2\ell}^t < s_{1w}^t + s_{1\ell}^t$, $s_{2\ell}^t \geq s_{1w}^t$, $s_{2w}^t < s_{1\ell}^t$, and $s_{1\ell}^t \leq K_1 = \frac{18}{5}s_{2w}^t + \frac{9}{5}s_{2\ell}^t + \frac{3}{5}s_{1w}^t$; The optimal matching flows are: $f_{2w,2w}^t = s_{2w}^t$, $f_{2\ell,2\ell}^t = s_{2\ell}^t - s_{1w}^t$, $f_{1\ell,1\ell}^t = s_{1\ell}^t$, and $f_{2\ell,1w}^t = f_{1w,2\ell}^t = s_{1w}^t$;

According to the proposed matching flows above, the state of demographics either goes to Scenario 2 or Scenario 4 in period $t+1$. If it goes to Scenario 4, following the optimal solution in Scenario 4, players' demographic eventually evolves to Scenario 2 at period $\tau \leq t+4$. That is, we summarize players' demographics in period $k = t+1, \dots, \tau$ when $\tau = t+4$ in the next table. Note that in period $t+4$, we have

Table EC.6 Players' Demographics for Scenario 5

k	s_{2w}^k	$s_{2\ell}^k$	s_{1w}^k	$s_{1\ell}^k$
$t+1$	$\frac{1}{2}(s_{2w}^t + s_{2\ell}^t + s_{1w}^t)$	$\frac{1}{2}s_{2w}^t$	$\frac{1}{2}s_{1\ell}^t$	s_{1w}^t
$t+2$	$\frac{1}{4}(3s_{2w}^t + s_{2\ell}^t + s_{1w}^t)$	$\frac{1}{4}(s_{2w}^t + s_{2\ell}^t + s_{1w}^t)$	$\frac{1}{4}(-s_{2w}^t + 2s_{1w}^t + s_{1\ell}^t)$	$\frac{1}{4}(s_{1\ell}^t + s_{2w}^t)$
$t+3$	$\frac{1}{8}(5s_{2w}^t + 3s_{2\ell}^t + 3s_{1w}^t)$	$\frac{1}{8}(3s_{2w}^t + s_{2\ell}^t + s_{1w}^t)$	$\frac{1}{8}(-s_{2w}^t - s_{2\ell}^t + s_{1w}^t + 2s_{1\ell}^t)$	$\frac{1}{8}(s_{2\ell}^t + 3s_{1w}^t + s_{1\ell}^t)$
$t+4$	$\frac{1}{16}(11s_{2w}^t + 5s_{2\ell}^t + 5s_{1w}^t)$	$\frac{1}{16}(5s_{2w}^t + 3s_{2\ell}^t + 3s_{1w}^t)$	$\frac{1}{16}(-4s_{2w}^t - s_{2\ell}^t + 3s_{1w}^t + 3s_{1\ell}^t)$	$\frac{1}{8}(s_{2w}^t + s_{1w}^t + s_{1\ell}^t)$

$$s_{2w}^{t+4} + s_{2\ell}^{t+4} - s_{1w}^{t+4} - s_{1\ell}^{t+4} = \frac{1}{16}(18s_{2w}^t + 9s_{2\ell}^t + 3s_{1w}^t - 5s_{1\ell}^t) \geq 0,$$

where the inequality follows $s_{1\ell}^t \leq \frac{18}{5}s_{2w}^t + \frac{9}{5}s_{2\ell}^t + \frac{3}{5}s_{1w}^t$. Therefore, we have $\tau \leq t+4$.

In Table EC.7, we list the dual variables when $\tau = t+1, \dots, t+4$ respectively, and one can easily verify that the proposed dual variables satisfy complementary slackness and dual feasible conditions.

Scenario 6: $s_{2w}^t + s_{2\ell}^t < s_{1w}^t + s_{1\ell}^t$, $s_{2\ell}^t \geq s_{1w}^t$, $s_{2w}^t < s_{1\ell}^t$, and $s_{1\ell}^t > K_2 = \frac{18}{5}s_{2w}^t + \frac{23}{5}s_{2\ell}^t - \frac{11}{5}s_{1w}^t$; The optimal matching flows are: $f_{2w,2w}^t = s_{2w}^t$, $f_{2\ell,1w}^t = f_{1w,2\ell}^t = s_{1w}^t$, $f_{2\ell,1\ell}^t = f_{1\ell,2\ell}^t = s_{2\ell}^t - s_{1w}^t$, and $f_{1\ell,1\ell}^t = s_{1\ell}^t - f_{2\ell,1\ell}^t$.

Table EC.7 Dual Variables for Scenario 5

$\tau = t + 1$	$k = t$	$k = t + 1$			
λ_{2w}^k	5.5	5			
$\lambda_{2\ell}^k$	3	4			
λ_{1w}^k	9	8			
$\lambda_{1\ell}^k$	4.5	5			
$\tau = t + 2$	$k = t$	$k = t + 1$	$k = t + 2$		
λ_{2w}^k	6	5.5	5		
$\lambda_{2\ell}^k$	3.25	4.5	4		
λ_{1w}^k	8.75	7.5	8		
$\lambda_{1\ell}^k$	4.25	4.5	5		
$\tau = t + 3$	$k = t$	$k = t + 1$	$k = t + 2$	$k = t + 3$	
λ_{2w}^k	6.5	6	5.5	5	
$\lambda_{2\ell}^k$	3.5	5	4.5	4	
λ_{1w}^k	8.75	7	7.5	8	
$\lambda_{1\ell}^k$	4	4.25	4.5	5	
$\tau = t + 4$	$k = t$	$k = t + 1$	$k = t + 2$	$k = t + 3$	$k = t + 4$
λ_{2w}^k	7.0625	6.5	6	5.5	5
$\lambda_{2\ell}^k$	3.75	5.625	5	4.5	4
λ_{1w}^k	8.75	6.625	7	7.5	8
$\lambda_{1\ell}^k$	3.8125	4	4.25	4.5	5

Following the policy above, the state of demographics in the next period is given by $s_{1w}^{t+1} = \frac{1}{2}(s_{1\ell}^t - (s_{2\ell}^t - s_{1w}^t))$, $s_{1\ell}^{t+1} = s_{1w}^t$, $s_{2w}^{t+1} = \frac{1}{2}s_{2w}^t + s_{2\ell}^t$, $s_{2\ell}^{t+1} = \frac{1}{2}s_{2w}^t$. This corresponds to Scenario 4 since

$$\begin{aligned}
s_{2w}^{t+1} + s_{2\ell}^{t+1} - s_{1w}^{t+1} - s_{1\ell}^{t+1} &= s_{2w}^t + \frac{3}{2}s_{2\ell}^t - \frac{3}{2}s_{1w}^t - \frac{1}{2}s_{1\ell}^t \\
&< s_{2w}^t + \frac{3}{2}s_{2\ell}^t - \frac{3}{2}s_{1w}^t - \frac{1}{2}K_2 \\
&= -\frac{4}{5}s_{2w}^t - \frac{4}{5}s_{2\ell}^t - \frac{2}{5}s_{1w}^t < 0.
\end{aligned}$$

Also, $s_{1w}^{t+1} - s_{2\ell}^{t+1} = \frac{1}{2}(s_{1w}^t + s_{1\ell}^t - s_{2w}^t - s_{2\ell}^t) < 0$.

Since $t + 1$, we follow the proposed policy in Scenario 4 until we reach Scenario 2 at τ . One can verify that $\tau > t + 5$, because following the proposed solution in Scenario 4, we have $s_{1w}^{t+5} = -\frac{1}{4}s_{2w}^t - \frac{1}{16}s_{2\ell}^t + \frac{3}{16}s_{1w}^t + \frac{3}{16}s_{1\ell}^t$, $s_{1\ell}^{t+5} = \frac{1}{8}s_{2w}^t + \frac{1}{8}s_{1w}^t + \frac{1}{8}s_{1\ell}^t$, $s_{2w}^{t+5} = \frac{11}{16}s_{2w}^t + \frac{5}{8}s_{2\ell}^t$, $s_{2\ell}^{t+5} = \frac{5}{16}s_{2w}^t + \frac{3}{8}s_{2\ell}^t$. To check if it still belongs to Scenario 4,

$$\begin{aligned}
s_{2w}^{t+5} + s_{2\ell}^{t+5} - s_{1w}^{t+5} - s_{1\ell}^{t+5} &= \frac{9}{8}s_{2w}^t + \frac{23}{16}s_{2\ell}^t - \frac{11}{16}s_{1w}^t - \frac{5}{16}s_{1\ell}^t \\
&< \frac{9}{8}s_{2w}^t + \frac{23}{16}s_{2\ell}^t - \frac{11}{16}s_{1w}^t - \frac{5}{16}K_2 = 0.
\end{aligned}$$

For periods $t + 1$ and forward, we use the dual variables proposed in Scenario 4, 2, and 1. Hence, we only need to show that the proposed policy as well as the corresponding dual variables at period t satisfy complementary slackness and dual feasibility in order to establish optimality. By complementary slackness, we have

$$\lambda_{2w}^t = 1 + \frac{1}{2}\lambda_{2w}^{t+1} + \frac{1}{2}\lambda_{2\ell}^{t+1} \quad (\text{EC.21})$$

$$\lambda_{2\ell}^t = \frac{1}{2} + \lambda_{2w}^{t+1} - \frac{1}{2}\lambda_{1w}^{t+1}, \quad (\text{EC.22})$$

$$\lambda_{1w}^t = \frac{3}{2} + \lambda_{1\ell}^{t+1} + \frac{1}{2}\lambda_{1w}^{t+1}, \quad (\text{EC.23})$$

$$\lambda_{1\ell}^t = \frac{1}{2} + \frac{1}{2}\lambda_{1w}^{t+1}. \quad (\text{EC.24})$$

Then, we need to validate the dual feasibility condition corresponding to $f_{2\ell,2\ell}^t, f_{1w,1w}^t, f_{2w,1w}^t, f_{2w,1\ell}^t$, which are given by:

$$\lambda_{2\ell}^t - 0.5\lambda_{2w}^{t+1} \geq 0.5, \quad (\text{EC.25})$$

$$\lambda_{1w}^t - 0.5\lambda_{1w}^{t+1} - 0.5\lambda_{1\ell}^{t+1} \geq 1, \quad (\text{EC.26})$$

$$\lambda_{2w}^t + \lambda_{1w}^t - \lambda_{2w}^{t+1} - \lambda_{1\ell}^{t+1} \geq 2, \quad (\text{EC.27})$$

$$\lambda_{2w}^t + \lambda_{1\ell}^t - \lambda_{2w}^{t+1} \geq 1. \quad (\text{EC.28})$$

Taking (EC.21)-(EC.24) into the above inequalities, it is equivalent to validate

$$\lambda_{2w}^{t+1} \geq \lambda_{1w}^{t+1}, \quad (\text{EC.29})$$

$$1.5 + \lambda_{1\ell}^{t+1} \geq 1, \quad (\text{EC.30})$$

$$0.5 + 0.5\lambda_{2\ell}^{t+1} + 0.5\lambda_{1w}^{t+1} \geq 0.5\lambda_{2w}^{t+1}, \quad (\text{EC.31})$$

$$1.5 + 0.5\lambda_{2\ell}^{t+1} + 0.5\lambda_{1w}^{t+1} \geq 0.5\lambda_{2w}^{t+1}. \quad (\text{EC.32})$$

Among them, (EC.30) is trivially true, because $\lambda_{1\ell}^{t+1}$ is in Scenario 4 and is greater than 5 by Lemma EC.5. (EC.31) and (EC.32) are directly from (EC.20). Thus, we only need to validate (EC.29). Note that in Scenario 6, we have $\tau \geq t + 5$. At $k = \tau - 4 \geq t + 1$, we have

$$\lambda_{2w}^{\tau-4} = 7.0625, \lambda_{2\ell}^{\tau-4} = 6.1875, \lambda_{1w}^{\tau-4} = 6.3125, \text{ and } \lambda_{1\ell}^{\tau-4} = 3.8125, \quad (\text{EC.33})$$

by using the complementary slackness conditions recursively. By Lemma EC.5, we know that $\lambda_{2w}^{t+1} \geq \lambda_{2w}^{\tau-4} \geq \lambda_{1w}^{\tau-4} \geq \lambda_{1w}^{t+1}$, which completes the proof.

Scenario 7: $s_{2w}^t + s_{2\ell}^t < s_{1w}^t + s_{1\ell}^t$, $s_{2\ell}^t \geq s_{1w}^t$, $s_{2w}^t < s_{1\ell}^t$, and $K_1 < s_{1\ell}^t \leq K_2$; The optimal matching flows are: $f_{2w,2w}^t = s_{2w}^t$, $f_{2\ell,1w}^t = f_{1w,2\ell}^t = s_{1w}^t$, $f_{2\ell,2\ell}^t = \frac{9}{7}s_{2w}^t + \frac{23}{14}s_{2\ell}^t - \frac{11}{14}s_{1w}^t - \frac{5}{14}s_{1\ell}^t$, $f_{2\ell,1\ell}^t = f_{1\ell,2\ell}^t = \frac{5}{14}s_{1\ell}^t - \frac{9}{7}s_{2w}^t - \frac{9}{14}s_{2\ell}^t - \frac{3}{14}s_{1w}^t$, and $f_{1\ell,1\ell}^t = s_{1\ell}^t - f_{2\ell,1\ell}^t$.

In this scenario, the state of demographics will transit to Scenario 4 in the second period. Further, one can verify that the system reaches Scenario 2 in period $k = t + 4$, with $s_{2w}^{t+4} + s_{2\ell}^{t+4} = s_{1w}^{t+4} + s_{1\ell}^{t+4}$. Further, in period $k = t + 5$, the system goes to Scenario 1 with $s_{2\ell}^{t+5} = s_{1w}^{t+5}$. Hence, in period $k = t + 5$, we reach a degenerate case with $f_{2\ell,2\ell}^{t+5} = 0$. From the view of the simplex method, under the solution of Scenario 5.1, the reduced cost of $f_{2\ell,2\ell}^t$ is positive, so we take it into the basic feasible solution, and move $f_{2\ell,2\ell}^{t+5}$ out of the basis. The positiveness of all the other flows remains. To see if the solution is optimal, we list out the dual variables for the first $t + 6$ periods in Table EC.8, and for periods $k > t + 6$, we always have

$$\lambda_{2w}^k = 5, \lambda_{2\ell}^k = 3, \lambda_{1w}^k = 9, \text{ and } \lambda_{1\ell}^k = 5.$$

Then one can easily verify that the proposed dual variables satisfy complementary slackness and dual feasibility. $\square$

Table EC.8 Dual Variables for Scenario 7

k	t	$t+1$	$t+2$	$t+3$	$t+4$	$t+5$	$t+6$
λ_{2w}^k	50/7	737/112	85/14	39/7	71/14	5	5
$\lambda_{2\ell}^k$	849/224	639/112	285/56	32/7	57/14	22/7	4
λ_{1w}^k	1963/224	737/112	389/56	52/7	111/14	62/7	8
$\lambda_{1\ell}^k$	849/224	445/112	59/14	125/28	69/14	5	5
Primal	Scenario 7	Scenario 4	Scenario 4	Scenario 4	Scenario 2	Scenario 1	Scenario 1

Proof of Lemma EC.5 (1) We prove the statements by backward induction from period τ .

First, we show that $\lambda_{1w}^t \geq 5$, $\lambda_{1\ell}^t \geq 3$ for all $t = s, \dots, \tau$. This is true for τ according to (EC.11), which completes the base step.

The induction hypothesis is that $\lambda_{1w}^k \geq 5$, $\lambda_{1\ell}^k \geq 3$ for all $k = t, \dots, \tau$. Then for period $t-1$, from Eq. (CS₄) we have

$$\lambda_{1w}^{t-1} = 1 + 0.5\lambda_{1w}^t + 0.5\lambda_{1\ell}^t \geq 1 + 0.5 \cdot 5 + 0.5 \cdot 3 = 5,$$

$$\lambda_{1\ell}^{t-1} = 0.5 + 0.5\lambda_{1w}^t \geq 0.5 + 0.5 \cdot 5 = 3,$$

and the induction step is completed.

(2) We show that $\lambda_{1w}^t \leq \lambda_{1w}^{t+1}$, $\lambda_{1\ell}^t \leq \lambda_{1\ell}^{t+1}$, for $t = s, \dots, \tau-1$. For the induction base step, we have $\lambda_{1w}^\tau = 7.5 \leq 8 = \lambda_{1w}^{\tau+1}$ and $\lambda_{1\ell}^\tau = 4.5 \leq 5 = \lambda_{1\ell}^{\tau+1}$. Our induction hypothesis is that for all $k = t, \dots, \tau-1$, we have $\lambda_{1w}^k \leq \lambda_{1w}^{k+1}$ and $\lambda_{1\ell}^k \leq \lambda_{1\ell}^{k+1}$. Now, consider period $t-1$. Take the difference between λ_{1w}^{t-1} and λ_{1w}^t , we have

$$\begin{aligned}
& \lambda_{1w}^{t-1} - \lambda_{1w}^t \\
&= 1 + 0.5\lambda_{1w}^t + 0.5\lambda_{1\ell}^t - \lambda_{1w}^t \\
&= 1 + 0.5\lambda_{1\ell}^t - 0.5\lambda_{1w}^t \\
&= 0.75 - 0.25\lambda_{1\ell}^{t+1} \\
&\leq 0.75 - 0.25 \cdot 3 = 0,
\end{aligned} \tag{EC.34}$$

where the first equality follows Eq. (CS₄), the third equality follows (EC.15), and the last inequality follows the fact that $\lambda_{1\ell}^t \geq 3$. Next, we show that $\lambda_{1\ell}^{t-1} \leq \lambda_{1\ell}^t$. Note that from the last equation in (CS₄), we have $\lambda_{1\ell}^{t-1} = 0.5 + 0.5\lambda_{1w}^t$. Since $\lambda_{1w}^t \leq \lambda_{1w}^{t+1}$, we have $\lambda_{1\ell}^{t-1} \leq 0.5 + 0.5\lambda_{1w}^{t+1} = \lambda_{1\ell}^t$.

(3) Finally, we show that $\lambda_{2w}^t \geq \lambda_{2w}^{t+1}$, $\lambda_{2\ell}^t \geq \lambda_{2\ell}^{t+1}$. For the base step, we have $\lambda_{2w}^{\tau-1} = 5.5 \geq 5 = \lambda_{2w}^\tau$ and $\lambda_{2\ell}^{\tau-1} = 4.5 \geq 4 = \lambda_{2\ell}^\tau$. The induction hypothesis is that for all $k = t, \dots, \tau-1$, we have $\lambda_{2w}^k \geq \lambda_{2w}^{k+1}$ and $\lambda_{2\ell}^k \geq \lambda_{2\ell}^{k+1}$. Now, consider period $t-1$. We have

$$\lambda_{2w}^{t-1} = 1 + 0.5\lambda_{2w}^t + 0.5\lambda_{2\ell}^t \geq 1 + 0.5\lambda_{2w}^{t+1} + 0.5\lambda_{2\ell}^{t+1} = \lambda_{2w}^t,$$

where the equality on the two sides follows the first equation of (CS₄). Next, consider $\lambda_{2\ell}^{t-1}$. We have

$$\begin{aligned}
& \lambda_{2\ell}^{t-1} = 1 + \lambda_{2w}^t - 0.5\lambda_{1w}^t + 0.5\lambda_{1\ell}^t \\
&= 1 + \lambda_{2w}^t - 0.5(1 + 0.5\lambda_{1w}^{t+1} + 0.5\lambda_{1\ell}^{t+1}) + 0.5(0.5 + 0.5\lambda_{1w}^{t+1}) \\
&= 1 + \lambda_{2w}^t - 0.25 - 0.25\lambda_{1\ell}^{t+1} \\
&\geq 1 + \lambda_{2w}^{t+1} - 0.25 - 0.25\lambda_{1\ell}^{t+2} \\
&= \lambda_{2\ell}^t,
\end{aligned}$$

where the first equality follows the second equation of (CS₄), the second equality follows the third and fourth equations of (CS₄), the first inequality follows the fact that $\lambda_{1\ell}^{t+1} \leq \lambda_{1\ell}^{t+2}$ and $\lambda_{2w}^t \geq \lambda_{2w}^{t+1}$. Thus, the induction step is completed and this completes the proof. $\square$

Proof of Theorem 2. (a) In order to prove the first statement, we consider the linear program for the one-shot matching problem in (P₁) and use the same simplification tricks we used in Appendix B.1. Without loss of generality, we will set the engagement level of SBMM in the next period to be 1, which is equivalent to the constraint $1 = s_{2w} + \frac{1}{2}s_{2\ell} + s_{1w} + \frac{1}{2}s_{1\ell}$. Thus, the following optimization problem selects the initial state of the demographics $\mathbf{s}$ to maximize the ratio of the optimal policy to SBMM for the one-period problem (we drop all the superscripts, representing time periods, since it is a one-shot problem):

$$\begin{aligned} \max_{\mathbf{f}, \mathbf{s}} \quad & f_{2w,2w} + \frac{1}{2}f_{2\ell,2\ell} + f_{1w,1w} + \frac{1}{2}f_{1\ell,1\ell} + 2f_{2w,1w} + f_{2\ell,1\ell} + f_{2w,1\ell} + 2f_{2\ell,1w} \\ \text{s.t.} \quad & \\ & 1 = s_{2w} + \frac{1}{2}s_{2\ell} + s_{1w} + \frac{1}{2}s_{1\ell} \\ & s_{2w} = f_{2w,2w} + f_{2w,1w} + f_{2w,1\ell}, \\ & s_{2\ell} = f_{2\ell,2\ell} + f_{2\ell,1w} + f_{2\ell,1\ell}, \\ & s_{1w} = f_{1w,1w} + f_{2\ell,1w} + f_{2w,1w}, \\ & s_{1\ell} = f_{1\ell,1\ell} + f_{2w,1\ell} + f_{2\ell,1\ell}. \end{aligned}$$

which is the linear program for the one-shot matching problem in (P₁), maximizing over the initial demographics and matching flows, with an additional constraint. Without loss of generality, the additional constraint $1 = s_{2w} + \frac{1}{2}s_{2\ell} + s_{1w} + \frac{1}{2}s_{1\ell}$, normalizes the value of one-shot SBMM to 1.

We verify that the optimal solution to the above optimization problem is $s_{2\ell} = s_{1w} = 2/3$, $f_{2\ell,1w} = 2/3$, and all other matching flows are 0. The objective value is $4/3$, which is the desired ratio. Denote λ_0 as the dual variable corresponding to the constraint normalizing the engagement for SBMM to be 1. We verify the proposed solution using complementary slackness conditions:

$$\begin{aligned} 0 &= \frac{1}{2}\lambda_0 - \lambda_{2\ell}, \\ 0 &= \lambda_0 - \lambda_{1w}, \\ 2 &= \lambda_{2\ell} + \lambda_{1w}, \end{aligned}$$

where the conditions correspond to primal non-zero variables $s_{2\ell}$, s_{1w} , and $f_{2\ell,1w}$, respectively. There is a unique solution of dual variables solving the complementary slackness conditions: $\lambda_0 = 4/3$, $\lambda_{2\ell} = 2/3$, and $\lambda_{1w} = 4/3$. To complete the proof, we need to check dual feasibility conditions:

$$\begin{aligned} 0 &\leq \lambda_0 - \lambda_{2w}, \quad 0 \leq \frac{1}{2}\lambda_0 - \lambda_{1\ell}, \quad 1 \leq \lambda_{2w}, \quad \frac{1}{2} \leq \lambda_{2\ell}, \quad 1 \leq \lambda_{1w}, \quad \frac{1}{2} \leq \lambda_{1\ell}, \\ 2 &\leq \lambda_{2w} + \lambda_{1w}, \quad 1 \leq \lambda_{2\ell} + \lambda_{1\ell}, \quad 1 \leq \lambda_{2w} + \lambda_{1\ell}, \quad 1 \leq \lambda_{2\ell} + \lambda_{1w}, \end{aligned}$$

representing zero state variables $(s_{2w}, s_{1\ell})$ and zero matching flows $(f_{2w,2w}, f_{2\ell,2\ell}, f_{1w,1w}, f_{1\ell,1\ell}, f_{2w,1w}, f_{2\ell,1\ell}, f_{2w,1\ell})$, respectively.

(b) Next, we turn our attention to the infinite horizon problem in (P'). Using a similar idea, we can solve an optimization problem to find the maximum ratio between the optimal matching policy and SBMM. Using Lemma 1, the value function of the baseline model under SBMM is

$$V^{\text{SBMM}}(\mathbf{s}^0) = 5(s_{2w}^t + s_{1w}^t) + 3(s_{2\ell}^t + s_{1\ell}^t), \quad t = 1, 2, \dots, \quad (\text{EC.35})$$

which we normalize to 1 without loss of generality. Thus, the following optimization problem selects the initial state of the demographics $\mathbf{s}^0$ to maximize the ratio of the optimal policy to SBMM for the infinite-horizon problem (we set $t = 0$ without loss of generality):

$$\max_{\{\mathbf{f}^t\}, \{\mathbf{s}^0\}} \sum_{t=0}^{\infty} \left(f_{2w,2w}^t + \frac{1}{2}f_{2\ell,2\ell}^t + f_{1w,1w}^t + \frac{1}{2}f_{1\ell,1\ell}^t + 2f_{2w,1w}^t + f_{2\ell,1\ell}^t + f_{2w,1\ell}^t + 2f_{2\ell,1w}^t \right) \quad (\text{EC.36})$$

$$(\text{EC.37})$$

s.t.

$$1 = 5(s_{2w}^0 + s_{1w}^0) + 3(s_{2\ell}^0 + s_{1\ell}^0), \quad (\text{EC.38})$$

$$0 = f_{2w,2w}^0 + f_{2w,1w}^0 + f_{2w,1\ell}^0 - s_{2w}^0,$$

$$0 = f_{2\ell,2\ell}^0 + f_{2\ell,1w}^0 + f_{2\ell,1\ell}^0 - s_{2\ell}^0,$$

$$0 = f_{1w,1w}^0 + f_{2\ell,1w}^0 + f_{2w,1w}^0 - s_{1w}^0,$$

$$0 = f_{1\ell,1\ell}^0 + f_{2w,1\ell}^0 + f_{2\ell,1\ell}^0 - s_{1\ell}^0,$$

and for all $t = 1, 2, \dots$,

$$f_{2w,2w}^t + f_{2w,1w}^t + f_{2w,1\ell}^t = \frac{1}{2} (f_{2w,2w}^{t-1} + f_{2\ell,2\ell}^{t-1}) + f_{2w,1w}^{t-1} + f_{2w,1\ell}^{t-1} + f_{2\ell,1w}^{t-1} + f_{2\ell,1\ell}^{t-1},$$

$$f_{2\ell,2\ell}^t + f_{2\ell,1w}^t + f_{2\ell,1\ell}^t = \frac{1}{2} f_{2w,2w}^{t-1},$$

$$f_{1w,1w}^t + f_{2\ell,1w}^t + f_{2w,1w}^t = \frac{1}{2} (f_{1w,1w}^{t-1} + f_{1\ell,1\ell}^{t-1}),$$

$$f_{1\ell,1\ell}^t + f_{2w,1\ell}^t + f_{2\ell,1\ell}^t = \frac{1}{2} f_{1w,1w}^{t-1} + f_{2w,1w}^{t-1} + f_{2\ell,1w}^{t-1},$$

$$f_{i,j}^t \geq 0, \forall i, j \in \{2w, 2\ell, 1w, 1\ell\}.$$

Again, we use dual complementary slackness and feasibility conditions verifying the initial state $\mathbf{s}^0 = \{0, 1/8, 1/8, 0\}$, along with the optimal matching flows in Theorem 1, is the optimal solution to the above maximization problem. The objective value is $3/2$, which is the desired ratio.

With slight abuse of notation, denote the dual variable to the new constraint (EC.38) as λ_0 . We show that under the optimal initial state where $s_{2\ell} = s_{1w} = 1/8$, we have $\lambda_0 = 3/2$, representing the maximum ratio. The proposed initial state is in Scenario 1. Based on the transition in Fig. EC.1, we shall always stay in Scenario 1. We list the states in periods 0, 1, and 2 in Table EC.9.

Note that in period 0, we reach a degenerate period with only one positive flow $f_{2\ell,1w}^0$, and in period 1 we reach a degenerate case with only two positive flows $f_{2w,2w}^1$ and $f_{1\ell,1\ell}^1$. Thus, for these two periods, we can use only part of the equations in (CS₁). To be specific, given our proposed primal solution, the complementary slackness equations are given by

$$\begin{aligned}
0 &= 3\lambda_0 - \lambda_{2\ell}^0, \\
0 &= 5\lambda_0 - \lambda_{1w}^0, \\
2 &= \lambda_{2\ell}^0 + \lambda_{1w}^0 - \lambda_{2w}^1 - \lambda_{1\ell}^1, \\
1 &= \lambda_{2w}^1 - \frac{1}{2}\lambda_{2w}^2 - \frac{1}{2}\lambda_{2\ell}^2, \\
\frac{1}{2} &= \lambda_{1\ell}^1 - \frac{1}{2}\lambda_{1w}^2, \\
&(\text{CS}_1), t = 2, \dots
\end{aligned}$$

and the dual feasibility constraints we need to check is

$$\begin{aligned}
0 &\leq 5\lambda_0 - \lambda_{2w}^0, \quad 0 \leq 3\lambda_0 - \lambda_{1w}^0, \quad 1 \leq \lambda_{2w}^0 - \frac{1}{2}\lambda_{2w}^1 - \frac{1}{2}\lambda_{2\ell}^1, \\
\frac{1}{2} &\leq \lambda_{2\ell}^0 - \frac{1}{2}\lambda_{2w}^1, \quad \frac{1}{2} \leq \lambda_{1\ell}^0 - \frac{1}{2}\lambda_{1w}^1, \quad \frac{1}{2} \leq \lambda_{1\ell}^1 - \frac{1}{2}\lambda_{1w}^2, \\
2 &\leq \lambda_{2\ell}^1 + \lambda_{1w}^1 - \lambda_{2w}^2 - \lambda_{1\ell}^2, \quad (\text{DF}_1) \text{ for } t = 0, \dots
\end{aligned}$$

We then list out the dual variables in periods 0, 1, and 2 in Table EC.9. For $t > 2$, we use $\lambda_{2w}^0 = 5$, $\lambda_{2\ell}^0 = 3$, $\lambda_{1w} = 9$, $\lambda_{1\ell} = 5$. Together with $\lambda_0 = 3/2$, one can verify that the proposed dual solution satisfies complementary slackness and dual feasibility constraints.

Table EC.9 Primal States and Dual Variables in Period 0, 1, and 2.

	s_{2w}^t	$s_{2\ell}^t$	s_{1w}^t	$s_{1\ell}^t$	λ_{2w}^t	$\lambda_{2\ell}^t$	λ_{1w}^t	$\lambda_{1\ell}^t$
$t = 0$	0	1/8	1/8	0	11/2	9/2	15/2	9/2
$t = 1$	1/8	0	0	1/8	5	4	8	5
$t = 2$	1/16	1/16	1/16	0	5	3	9	5

□

Proof of Corollary 1. Denote the winrate between high- and low-skilled players as $\alpha \in [0.5, 1]$. In this part, we show that for $\alpha < 1$, the maximum power of optimal matching is non-increasing when α decreases. Note that under this setting, one unit of cross-level matching that induces high-skilled players to win with probability α can be decomposed into the following three matching flows: 1) $2\alpha - 1$ unit of perfect cross-level matching flow, where high-skilled players always win against low-skilled players; 2) $2(1 - \alpha)$ unit of high-skilled players matching with each other; and 3) low-skilled players matching with each other. Take the flow between 2ℓ and $1w$ as an example. One can easily verify that it can be decomposed into $(2\alpha - 1)f_{2\ell, 1w}$, $2(1 - \alpha)f_{2\ell, 2\ell}$, and $2(1 - \alpha)f_{1w, 1w}$, which induce the same player demographics in the next period. As a result, to have a unit of cross-level flow in this setting, we also need $\frac{2\alpha - 1}{2(1 - \alpha)}$ unit of SBMM among both high- and low-skilled level players. Therefore, when solving we need to include the following additional constraints when solving the linear program in (EC.37) in the proof of part (b):

$$f_{i,j}^t \leq \frac{2\alpha - 1}{2(1 - \alpha)} f_{i,i}^t, \text{ and } f_{i,j}^t \leq \frac{2\alpha - 1}{2(1 - \alpha)} f_{i,i}^t, \quad \forall i \in \{2w, 2\ell\}, j \in \{1w, 1\ell\}, t = 0, 1, 2, \dots \quad (\text{EC.39})$$

Clearly, $\frac{2\alpha - 1}{2(1 - \alpha)}$ is a increasing function in α . Therefore, as α decreases, the feasible region also shrinks, leading to non-increasing optimal objective values. Lastly, note that when $\alpha = 1/2$, the problem reduces to pure SBMM and this completes the proof. □

B.1.3. Proof of Theorem 3 Before showing the results of the interplay between PTW strategies and matchmaking, we update the flow balancing and demographic evolution constraints, then formally state the matchmaker's problem when facing PTW.

In any period t , the flow balance constraints are now

$$\begin{aligned} \sum_j f_{i,j}^t &= s_i^t, \forall i \in \bar{\mathcal{P}}, \\ \sum_i f_{i,j}^t &= s_j^t, \forall j \in \bar{\mathcal{P}}, \\ f_{i,j}^t &= f_{j,i}^t, \forall i \neq j, i, j \in \bar{\mathcal{P}}, \\ f_{i,j}^t &\geq 0, \forall i \neq j, i, j \in \bar{\mathcal{P}}, \end{aligned} \tag{FB}_{\text{ptw}}$$

where $\bar{\mathcal{P}} := \{\bar{2}w, \bar{2}\ell, \underline{2}w, \underline{2}\ell, \underline{1}w, \underline{1}\ell\}$, and evolution of demographics are now

$$\begin{aligned} s_{\bar{2}w}^{t+1} &= \frac{1}{2}(f_{\bar{2}w, \bar{2}w}^t + f_{\bar{2}w, \bar{2}\ell}^t + f_{\bar{2}\ell, \bar{2}w}^t + f_{\bar{2}\ell, \bar{2}\ell}^t + f_{\bar{2}w, \underline{2}w}^t + f_{\bar{2}w, \underline{2}\ell}^t + f_{\bar{2}\ell, \underline{2}w}^t + f_{\bar{2}\ell, \underline{2}\ell}^t) + f_{\bar{2}w, \underline{1}w}^t + f_{\bar{2}w, \underline{1}\ell}^t + f_{\bar{2}\ell, \underline{1}w}^t + f_{\bar{2}\ell, \underline{1}\ell}^t, \\ s_{\bar{2}\ell}^{t+1} &= \frac{1}{2}(f_{\bar{2}w, \bar{2}w}^t + f_{\bar{2}w, \bar{2}\ell}^t + f_{\bar{2}\ell, \bar{2}w}^t + f_{\bar{2}\ell, \bar{2}\ell}^t) + \beta(\frac{1}{2}f_{\underline{1}w, \underline{1}w}^t + \frac{1}{2}f_{\underline{1}w, \underline{1}\ell}^t + f_{\underline{1}w, \underline{2}w}^t + f_{\underline{1}w, \underline{2}\ell}^t + f_{\underline{1}w, \bar{2}w}^t + f_{\underline{1}w, \bar{2}\ell}^t), \\ s_{\underline{2}w}^{t+1} &= \frac{1}{2}(f_{\underline{2}w, \underline{2}w}^t + f_{\underline{2}w, \underline{2}\ell}^t + f_{\underline{2}\ell, \underline{2}w}^t + f_{\underline{2}\ell, \underline{2}\ell}^t + f_{\underline{2}w, \bar{2}w}^t + f_{\underline{2}w, \bar{2}\ell}^t + f_{\underline{2}\ell, \bar{2}w}^t + f_{\underline{2}\ell, \bar{2}\ell}^t) + f_{\underline{2}w, \underline{1}w}^t + f_{\underline{2}w, \underline{1}\ell}^t + f_{\underline{2}\ell, \underline{1}w}^t + f_{\underline{2}\ell, \underline{1}\ell}^t, \\ s_{\underline{2}\ell}^{t+1} &= \frac{1}{2}(f_{\underline{2}w, \underline{2}w}^t + f_{\underline{2}w, \underline{2}\ell}^t + f_{\underline{2}\ell, \underline{2}w}^t + f_{\underline{2}\ell, \underline{2}\ell}^t), \\ s_{\underline{1}w}^{t+1} &= \frac{1}{2}(f_{\underline{1}w, \underline{1}w}^t + f_{\underline{1}w, \underline{1}\ell}^t + f_{\underline{1}\ell, \underline{1}w}^t + f_{\underline{1}\ell, \underline{1}\ell}^t), \\ s_{\underline{1}\ell}^{t+1} &= (1 - \beta)(\frac{1}{2}f_{\underline{1}w, \underline{1}w}^t + \frac{1}{2}f_{\underline{1}w, \underline{1}\ell}^t + f_{\underline{1}w, \underline{2}w}^t + f_{\underline{1}w, \underline{2}\ell}^t + f_{\underline{1}w, \bar{2}w}^t + f_{\underline{1}w, \bar{2}\ell}^t). \end{aligned} \tag{ED}_{\text{ptw}}$$

From (FB_{ptw}) and (ED_{ptw}) conditions, we can see that players in the new demographics with skill level $\bar{2}$ behave the same as those with skill level $\underline{2}\ell$. However, the matchmaker's objective is largely different from the one in (P). The matchmaker also needs to consider the revenue generated by paid subscriptions. Given β , r , and the initial demographics $\mathbf{s}^0 = \{s_{\bar{2}w}^0, s_{\bar{2}\ell}^0, s_{\underline{1}w}^0, s_{\underline{1}\ell}^0\}$, the matchmaker's problem is

$$\begin{aligned} V^*(\beta, r, \mathbf{s}^0) &:= \max_{\{f_{i,j}^t\}_{t=1}^{\infty}} \sum_{t=1}^{\infty} \sum_i s_i^t + \sum_{t=1}^{\infty} r(s_{\bar{2}w}^t + s_{\bar{2}\ell}^t) = \max_{\{f_{i,j}^t\}_{t=1}^{\infty}} \text{ENG}(\beta, r, \mathbf{s}^0) + \text{REV}(\beta, r, \mathbf{s}^0) \tag{P}_{\text{ptw}} \\ \text{s.t. } s_{\bar{2}w}^0 &= 0, \\ s_{\bar{2}\ell}^0 &= 0, \\ s_{\underline{2}w}^0 &= s_{\bar{2}w}^0, \\ s_{\underline{2}\ell}^0 &= s_{\bar{2}\ell}^0, \\ s_{\underline{1}w}^0 &= s_{\bar{2}w}^0, \\ s_{\underline{1}\ell}^0 &= s_{\bar{2}\ell}^0, \\ (\text{FB}_{\text{ptw}}) \text{ and } (\text{ED}_{\text{ptw}}) &\forall t = 0, 1, 2, \dots, \text{ and } i, j \in \bar{\mathcal{P}}, \end{aligned}$$

where the objective function is to maximize the sum of player engagement $\text{ENG}(\beta, r, \mathbf{s}^0)$ and revenue from paid subscription $\text{REV}(\beta, r, \mathbf{s}^0)$.

Proof of Theorem 3. We prove the three parts separately.

(a) We show that unsubscribed high-skilled players in $\underline{2}w$ ($\underline{2}\ell$) would only be matched with any unsubscribed low-skilled players after all the subscribed players in $\bar{2}w$ ($\bar{2}\ell$) have matched with unsubscribed low-skilled players.

We prove the statement above by contradiction. Suppose on the optimal trajectory, the flow between high-skilled non-paying players in $\underline{2}i$ and low-skilled unsubscribed players $\underline{1}j$ are positive for some $i, j = w, \ell$, while there exists subscribed players $\bar{2}i$ who are matched by skill levels. Then by matching $\bar{2}i$ with $\underline{1}j$, we can collect strictly more rewards in the current period, and a player in $\bar{2}i$ would replace a player $\underline{2}i$ in all the subsequent periods. Hence, the solution cannot be optimal.

(b) We show that if $s_{2w}^0 + s_{2\ell}^0 \geq s_{1w}^0 + s_{1\ell}^0$, then $ENG(\beta, r, \mathbf{s}^0) < V^*(0, 0, \mathbf{s}^0)$. Furthermore, there exists a threshold $\bar{r}$ such that $V^*(\beta, r, \mathbf{s}^0) \geq V^*(0, 0, \mathbf{s}^0)$ if and only if $r \geq \bar{r}$.

We first show that $ENG(\beta, r, \mathbf{s}^0) \leq V^*(0, 0, \mathbf{s}^0)$. The engagement $ENG(\beta, r, \mathbf{s}^0)$ is at most $V^*(\beta, 0, \mathbf{s}^0)$, which is the optimal engagement with the same demographic but without revenue. We now show that $V^*(\beta, 0, \mathbf{s}^0) \leq V^*(0, 0, \mathbf{s}^0)$, and the equality only holds when we do not have low players at all.

To proceed, we will first show that when $\beta > 0$, $r = 0$, the optimal matching policy (for engagement maximization) is exactly the same as the baseline model. Recall that when $s_{2w}^0 + s_{2\ell}^0 \geq s_{1w}^0 + s_{1\ell}^0$, the state space can be decomposed into three scenarios. We shall show that these three scenarios (as well as the scenario transition) are the same as the baseline model, and provide the shadow prices. Then, with the help of shadow prices, we shall show that for any initial states such that $s_{2w}^0 + s_{2\ell}^0 \geq s_{1w}^0 + s_{1\ell}^0$, the optimal engagement decreases with β .

Step 1: Our first step is to show that the optimal policy remains the same, and provide closed-form shadow prices for Scenario 1-3. When $r = 0$, we can merge level $\bar{2}$ and $\underline{2}$. Compared with (P'), a fraction of 1ℓ players will transit to state 2ℓ . The resulting LP formulation is different from (P') only on two constraints. That is, we replace

$$\begin{aligned} f_{2\ell, 2\ell}^t + f_{2\ell, 1w}^t + f_{2\ell, 1\ell}^t &= \frac{1}{2} f_{2w, 2w}^{t-1}, \\ f_{1\ell, 1\ell}^t + f_{2w, 1\ell}^t + f_{2\ell, 1\ell}^t &= \frac{1}{2} f_{1w, 1w}^{t-1} + f_{2w, 1w}^{t-1} + f_{2\ell, 1w}^{t-1}, \end{aligned}$$

with

$$\begin{aligned} f_{2\ell, 2\ell}^t + f_{2\ell, 1w}^t + f_{2\ell, 1\ell}^t &= \frac{1}{2} f_{2w, 2w}^{t-1} + \beta \left(\frac{1}{2} f_{1w, 1w}^{t-1} + f_{2w, 1w}^{t-1} + f_{2\ell, 1w}^{t-1} \right), \\ f_{1\ell, 1\ell}^t + f_{2w, 1\ell}^t + f_{2\ell, 1\ell}^t &= (1 - \beta) \frac{1}{2} f_{1w, 1w}^{t-1} + f_{2w, 1w}^{t-1} + f_{2\ell, 1w}^{t-1}. \end{aligned}$$

Next, we shall show that for Scenario 1-3 in Figure EC.1, the transition is still valid, and the proposed matching policy still satisfies dual feasibility and complementary slackness.

Scenario 1: $s_{2w}^t + s_{2\ell}^t \geq s_{1w}^t + s_{1\ell}^t$, $s_{2\ell}^t \geq s_{1w}^t$, and $s_{2w}^t \geq s_{1\ell}^t$; The optimal matching flows are: $f_{2w, 2w}^t = s_{2w}^t$, $f_{2\ell, 2\ell}^t = s_{2\ell}^t - s_{1w}^t$, $f_{1w, 1w}^t = 0$, $f_{1\ell, 1\ell}^t = s_{1\ell}^t$, and $f_{2\ell, 1w}^t = f_{1w, 2\ell}^t = s_{1w}^t$.

We first show that once the state of players' demographics falls in Scenario 1, it will remain in Scenario 1. To see this, for some $\mathbf{s}^t = \{s_{1w}^t, s_{1\ell}^t, s_{2w}^t, s_{2\ell}^t\}$ in Scenario 1, following the proposed policy, the state at $t+1$ is given by $s_{1w}^{t+1} = \frac{1}{2} s_{1\ell}^t$, $s_{1\ell}^{t+1} = (1 - \beta) s_{1w}^t$, $s_{2w}^{t+1} = \frac{1}{2} (s_{2w}^t + s_{2\ell}^t + s_{1w}^t)$, and $s_{2\ell}^{t+1} = \frac{1}{2} s_{2w}^t + \beta s_{1w}^t$. Then $s_{2\ell}^{t+1} \geq s_{1w}^{t+1}$.

because $s_{2w}^t \geq s_{1\ell}^t$, and $s_{2w}^{t+1} \geq s_{1\ell}^{t+1}$ because $s_{2w}^{t+1} - s_{1\ell}^{t+1} = \frac{1}{2}(s_{2w}^t + s_{2\ell}^t - s_{1w}^t) + \beta s_{1w}^t$ and $s_{2\ell}^t - s_{1w}^t \geq 0$. Hence, $\mathbf{s}^{t+1}$ still belongs to Scenario 1.

Next, we show that the proposed policy in Theorem 1 is optimal for all subsequent periods once players' demographics satisfy Scenario 1. The optimal solution in Theorem 1 suggests that in Scenario 1, only 4 variables are non-zero while all other flows are zero in each period. Therefore, we have

$$\begin{aligned} 1 &= \lambda_{2w}^t - \frac{1}{2}\lambda_{2w}^{t+1} - \frac{1}{2}\lambda_{2\ell}^{t+1}, \\ \frac{1}{2} &= \lambda_{2\ell}^t - \frac{1}{2}\lambda_{2w}^{t+1}, \\ 2 &= \lambda_{2\ell}^t + \lambda_{1w}^t - \lambda_{2w}^{t+1} - (1 - \beta)\lambda_{1\ell}^{t+1} - \beta\lambda_{2\ell}, \\ \frac{1}{2} &= \lambda_{1\ell}^t - \frac{1}{2}\lambda_{1w}^{t+1}, \end{aligned}$$

as the complementary conditions corresponding to primal non-zero variables $f_{2w,2w}^t$, $f_{2\ell,2\ell}^t$, $f_{2\ell,1w}^t$, and $f_{1\ell,1\ell}^t$, respectively. The following dual solutions:

$$\lambda_{2w}^t = 5, \lambda_{2\ell}^t = 3, \lambda_{1w}^t = \frac{9+5\beta}{1+\beta}, \text{ and } \lambda_{1\ell}^t = \frac{5+3\beta}{1+\beta}, \quad \forall t, \quad (\text{EC.40})$$

satisfies complementary slackness. The dual feasibility conditions corresponding to variables, $f_{2w,1w}^t$, $f_{2w,1\ell}^t$, $f_{2\ell,1\ell}^t$, and $f_{1w,1w}^t$, that are zero in the primal problem, are given by

$$\begin{aligned} \lambda_{2w}^t + \lambda_{1w}^t - \lambda_{2w}^{t+1} - (1 - \beta)\lambda_{1\ell}^{t+1} - \beta\lambda_{2\ell} &\geq 2, \iff 2 \geq 0 \\ \lambda_{2w}^t + \lambda_{1\ell}^t - \lambda_{2w}^{t+1} &\geq 1, \iff \frac{4+2\beta}{1+\beta} \geq 0 \\ \lambda_{2\ell}^t + \lambda_{1\ell}^t - \lambda_{2w}^{t+1} &\geq 1, \iff \frac{2}{1+\beta} \geq 0 \\ \lambda_{1w}^t - \frac{1}{2}\lambda_{1w}^{t+1} - \frac{1}{2}(1 - \beta)\lambda_{1\ell}^{t+1} - \frac{1}{2}\beta\lambda_{2\ell}^{t+1} &\geq 1, \iff 1 \geq 0, \end{aligned}$$

where the right-hand-side is by plugging in (EC.40), and is obviously true for $\beta > 0$. Thus, the proposed policy in Theorem 1 is optimal once the players' demographics fall in scenario 1.

Scenario 2: $s_{2w}^t + s_{2\ell}^t \geq s_{1w}^t + s_{1\ell}^t$, $s_{2\ell}^t < s_{1w}^t$, and $s_{2w}^t \geq s_{1\ell}^t$; The optimal matching flows are: $f_{2w,2w}^t = s_{2w}^t$, $f_{2\ell,2\ell}^t = 0$, $f_{1w,1w}^t = s_{1w}^t - s_{2\ell}^t$, $f_{1\ell,1\ell}^t = s_{1\ell}^t$, and $f_{2\ell,1w}^t = f_{1w,2\ell}^t = s_{2\ell}^t$;

In the second step, we consider Scenario 2, which will transit to Scenario 1 after matching under the proposed policy as we have stated in Figure EC.1. To see that, for some $\mathbf{s}^t = \{s_{1w}^t, s_{1\ell}^t, s_{2w}^t, s_{2\ell}^t\}$ in Scenario 2, following the proposed policy, the state at $t+1$ is given by $s_{1w}^{t+1} = \frac{1}{2}(s_{1w}^t + s_{1\ell}^t - s_{2\ell}^t)$, $s_{1\ell}^{t+1} = \frac{1}{2}(1 - \beta)(s_{2\ell}^t + s_{1w}^t)$, $s_{2w}^{t+1} = \frac{1}{2}s_{2w}^t + s_{2\ell}^t$, and $s_{2\ell}^{t+1} = \frac{1}{2}s_{2w}^t + \frac{1}{2}\beta(s_{2\ell}^t + s_{1w}^t)$. Then $s_{2\ell}^{t+1} \geq s_{1w}^{t+1}$ because $s_{2\ell}^{t+1} - s_{1w}^{t+1} = \frac{1}{2}(s_{2w}^t + s_{2\ell}^t - s_{1w}^t - s_{1\ell}^t) + \frac{1}{2}\beta(s_{2\ell}^t + s_{1w}^t) \geq 0$. Also, $s_{2w}^{t+1} \geq s_{1\ell}^{t+1}$ because $s_{2w}^{t+1} - s_{1\ell}^{t+1} = \frac{1}{2}(s_{2w}^t + s_{2\ell}^t - s_{1w}^t) + \frac{1}{2}\beta(s_{2\ell}^t + s_{1w}^t) \geq s_{1\ell}^t \geq 0$. Hence, $\mathbf{s}^{t+1}$ belongs to Scenario 1.

Therefore, we only need to show that in any period t such that players' demographics satisfy Scenario 2, we can find solutions to dual variables, induced by the proposed policy in Theorem 1, which satisfy the complementary slackness conditions and dual feasibility conditions. Note that in period t under Scenario 2,

the non-zero primal variables are $f_{2w,2w}^t$, $f_{2\ell,1w}^t$, $f_{1w,1w}^t$ and $f_{1\ell,1\ell}^t$. The complementary slackness condition is given by

$$\begin{aligned} 1 &= \lambda_{2w}^t - \frac{1}{2}\lambda_{2w}^{t+1} - \frac{1}{2}\lambda_{2\ell}^{t+1}, \\ 2 &= \lambda_{2\ell}^t + \lambda_{1w}^t - \lambda_{2w}^{t+1} - (1-\beta)\lambda_{1\ell}^{t+1} - \beta\lambda_{2\ell}^{t+1}, \\ 1 &= \lambda_{1w}^t - \frac{1}{2}\lambda_{1w}^{t+1} - \frac{1}{2}(1-\beta)\lambda_{1\ell}^{t+1} - \frac{1}{2}\beta\lambda_{2\ell}^{t+1}, \\ \frac{1}{2} &= \lambda_{1\ell}^t - \frac{1}{2}\lambda_{1w}^{t+1}, \end{aligned}$$

in period t . By plugging in λ_i^{t+1} from (EC.40), the shadow prices that satisfy the complementary slackness in Scenario 2 are given by

$$\lambda_{2w}^t = 5, \lambda_{2\ell}^t = 4, \lambda_{1w}^t = \frac{8+4\beta}{1+\beta}, \text{ and } \lambda_{1\ell}^t = \frac{5+3\beta}{1+\beta}. \quad (\text{EC.41})$$

The dual feasibility conditions are given by

$$\begin{aligned} \lambda_{2w}^t + \lambda_{1w}^t - \lambda_{2w}^{t+1} - (1-\beta)\lambda_{1\ell}^{t+1} - \beta\lambda_{2\ell}^{t+1} &\geq 2, \iff 1 \geq 0, \\ \lambda_{2w}^t + \lambda_{1\ell}^t - \lambda_{2w}^{t+1} &\geq 1, \iff \frac{4+2\beta}{1+\beta} \geq 0, \\ \lambda_{2\ell}^t - \frac{1}{2}\lambda_{2w}^{t+1} &\geq \frac{1}{2}, \iff 1 \geq 0, \\ \lambda_{2\ell}^t + \lambda_{1\ell}^t - \lambda_{2w}^{t+1} &\geq 1, \iff \frac{3+\beta}{1+\beta} \geq 0, \end{aligned}$$

where the right-hand-side is by plugging in (EC.40) and (EC.41). Therefore, the proof for Scenario 2 is complete.

Scenario 3: $s_{2w}^t + s_{2\ell}^t \geq s_{1w}^t + s_{1\ell}^t$, $s_{2\ell}^t \geq s_{1w}^t$, and $s_{2w}^t < s_{1\ell}^t$; The optimal matching flows are: $f_{2w,2w}^t = s_{2w}^t$, $f_{2\ell,2\ell}^t = s_{2\ell}^t - s_{1w}^t$, $f_{1w,1w}^t = 0$, $f_{1\ell,1\ell}^t = s_{1\ell}^t$, and $f_{2\ell,1w}^t = f_{1w,2\ell}^t = s_{1w}^t$;

In the third step, we consider Scenario 3. Different from the baseline model, Scenario 3 shall transit to either Scenario 2 or Scenario 1 after matching is done under the proposed policy. To see this, for some $\mathbf{s}^t = \{s_{1w}^t, s_{1\ell}^t, s_{2w}^t, s_{2\ell}^t\}$ in Scenario 3, following the proposed policy, the state at $t+1$ is given by $s_{2w}^{t+1} = \frac{1}{2}(s_{2w}^t + s_{2\ell}^t + s_{1w}^t)$, $s_{2\ell}^{t+1} = \frac{1}{2}s_{2w}^t + \beta s_{1w}^t$, $s_{1w}^{t+1} = \frac{1}{2}s_{1\ell}^t$, and $s_{1\ell}^{t+1} = (1-\beta)s_{1w}^t$. Then $s_{2w}^{t+1} \geq s_{1\ell}^{t+1}$ because $s_{2w}^{t+1} - s_{1\ell}^{t+1} = \frac{1}{2}(s_{2w}^t + s_{2\ell}^t - s_{1w}^t) + \beta s_{1w}^t$ and $s_{2\ell}^t - s_{1w}^t \geq 0$. Also, $s_{2w}^{t+1} + s_{2\ell}^{t+1} - s_{1w}^{t+1} - s_{1\ell}^{t+1} = \frac{1}{2}(2s_{2w}^t + s_{2\ell}^t - s_{1w}^t - s_{1\ell}^t) + 2\beta s_{1w}^t \geq 0$. However, the relation between $s_{2\ell}^{t+1}$ and s_{1w}^{t+1} rely on the value of β . Hence, $\mathbf{s}^{t+1}$ belongs to either Scenario 2 or Scenario 1.

We first consider the case when $\mathbf{s}^{t+1}$ belongs to Scenario 2. According to the proposed policy, in period t under Scenario 3, primal variables $f_{2w,2w}^t$, $f_{2\ell,2\ell}^t$, $f_{2\ell,1w}^t$, and $f_{1\ell,1\ell}^t$ are non-zero. Thus, in period t , the complementary slackness conditions are given by

$$\begin{aligned} 1 &= \lambda_{2w}^t - \frac{1}{2}\lambda_{2w}^{t+1} - \frac{1}{2}\lambda_{2\ell}^{t+1}, \\ \frac{1}{2} &= \lambda_{2\ell}^t - \frac{1}{2}\lambda_{2w}^{t+1}, \\ 2 &= \lambda_{2\ell}^t + \lambda_{1w}^t - \lambda_{2w}^{t+1} - (1-\beta)\lambda_{1\ell}^{t+1} - \beta\lambda_{2\ell}^{t+1}, \\ \frac{1}{2} &= \lambda_{1\ell}^t - \frac{1}{2}\lambda_{1w}^{t+1}. \end{aligned}$$

By plugging in (EC.41) for period $t+1$, the corresponding shadow prices are

$$\lambda_{2w}^t = 5.5, \lambda_{2\ell}^t = 3, \lambda_{1w}^t = \frac{(3+\beta)^2}{1+\beta}, \text{ and } \lambda_{1\ell}^t = \frac{9+5\beta}{2(1+\beta)}. \quad (\text{EC.42})$$

The dual feasibility conditions are

$$\begin{aligned} \lambda_{2w}^t + \lambda_{1w}^t - \lambda_{2w}^{t+1} - (1-\beta)\lambda_{1\ell}^{t+1} - \beta\lambda_{2\ell}^{t+1} &\geq 2, \iff 2.5 \geq 0, \\ \lambda_{2w}^t + \lambda_{1\ell}^t - \lambda_{2w}^{t+1} &\geq 1, \iff \frac{4+2\beta}{1+\beta} \geq 0, \\ \lambda_{2\ell}^t + \lambda_{1\ell}^t - \lambda_{2w}^{t+1} &\geq 1, \iff \frac{3-\beta}{2+2\beta} \geq 0, \\ \lambda_{1w}^t - \frac{1}{2}\lambda_{1w}^{t+1} - \frac{1}{2}\lambda_{1\ell}^{t+1} &\geq 1, \iff \frac{3+3\beta+2\beta^2}{2+2\beta} \geq 0, \end{aligned}$$

respectively, where the right-hand-side is by plugging in (EC.41) and (EC.42). It is straightforward to see that the proposed shadow prices are dual feasible for $\beta \in [0, 1]$.

Next, consider the case $\mathbf{s}^{t+1}$ belongs to Scenario 1. In period t , the complementary slackness conditions are given by

$$\begin{aligned} 1 &= \lambda_{2w}^t - \frac{1}{2}\lambda_{2w}^{t+1} - \frac{1}{2}\lambda_{2\ell}^{t+1}, \\ \frac{1}{2} &= \lambda_{2\ell}^t - \frac{1}{2}\lambda_{2w}^{t+1}, \\ 2 &= \lambda_{2\ell}^t + \lambda_{1w}^t - \lambda_{2w}^{t+1} - (1-\beta)\lambda_{1\ell}^{t+1} - \beta\lambda_{2\ell}^{t+1}, \\ \frac{1}{2} &= \lambda_{1\ell}^t - \frac{1}{2}\lambda_{1w}^{t+1}. \end{aligned}$$

By plugging in (EC.40) for period $t+1$, the corresponding shadow prices are

$$\lambda_{2w}^t = 5, \lambda_{2\ell}^t = 3, \lambda_{1w}^t = \frac{9+5\beta}{1+\beta}, \text{ and } \lambda_{1\ell}^t = \frac{5+3\beta}{1+\beta}. \quad (\text{EC.43})$$

The dual feasibility conditions are

$$\begin{aligned} \lambda_{2w}^t + \lambda_{1w}^t - \lambda_{2w}^{t+1} - (1-\beta)\lambda_{1\ell}^{t+1} - \beta\lambda_{2\ell}^{t+1} &\geq 2, \iff 2 \geq 0, \\ \lambda_{2w}^t + \lambda_{1\ell}^t - \lambda_{2w}^{t+1} &\geq 1, \iff \frac{4+2\beta}{1+\beta} \geq 0, \\ \lambda_{2\ell}^t + \lambda_{1\ell}^t - \lambda_{2w}^{t+1} &\geq 1, \iff \frac{2}{1+\beta} \geq 0, \\ \lambda_{1w}^t - \frac{1}{2}\lambda_{1w}^{t+1} - \frac{1}{2}\lambda_{1\ell}^{t+1} &\geq 1, \iff \frac{1}{1+\beta} \geq 0, \end{aligned}$$

respectively, where the right-hand-side is by plugging in (EC.40) and (EC.43). It is straightforward to see that the proposed shadow prices are dual feasible for $\beta \in [0, 1]$.

Step 2: With Step 1, we can compute the maximum engagement for any β , as long as the initial level 2 players are more than level 1 players. We now show that for $\beta > 0$, the engagement is always more than the case when $\beta = 0$.

If $\mathbf{s}^0$ belongs to Scenario 1, then the engagement without PTW is $5s_{2w}^0 + 3s_{2\ell}^0 + 9s_{1w}^0 + 5s_{1\ell}^0$. The engagement with PTW is $5s_{2w}^0 + 3s_{2\ell}^0 + \frac{9+5\beta}{1+\beta}s_{1w}^0 + \frac{5+3\beta}{1+\beta}s_{1\ell}^0$. The difference is $\frac{4\beta}{1+\beta}s_{1w}^0 + \frac{2\beta}{1+\beta}s_{1\ell}^0$, which is always positive for $\beta > 0$. Thus, the engagement must decrease.

If $\mathbf{s}^0$ belongs to Scenario 2, then the engagement without PTW is $5s_{2w}^0 + 4s_{2\ell}^0 + 8s_{1w}^0 + 5s_{1\ell}^0$. The engagement with PTW is $5s_{2w}^0 + 4s_{2\ell}^0 + \frac{8+4\beta}{1+\beta}s_{1w}^0 + \frac{5+3\beta}{1+\beta}s_{1\ell}^0$. The difference is $\frac{4\beta}{1+\beta}s_{1w}^0 + \frac{2\beta}{1+\beta}s_{1\ell}^0$, which is always positive for $\beta > 0$. Thus, the engagement must decrease.

If $\mathbf{s}^0$ belongs to Scenario 3, then the engagement without PTW is $5.5s_{2w}^0 + 3s_{2\ell}^0 + 9s_{1w}^0 + 4.5s_{1\ell}^0$. With PTW, if it transits to Scenario 2 in the next period, then the engagement is $5.5s_{2w}^0 + 3s_{2\ell}^0 + \frac{(3+\beta)^2}{1+\beta}s_{1w}^0 + \frac{9+5\beta}{2(1+\beta)}s_{1\ell}^0$. The difference is $\frac{6\beta-\beta^2}{1+\beta}s_{1w}^0 + \frac{2\beta}{1+\beta}s_{1\ell}^0$, which is always positive for $\beta \in (0, 1)$. Thus, the engagement must decrease. If it transits to Scenario 1 in the next period, then the engagement is $5s_{2w}^0 + 3s_{2\ell}^0 + \frac{9+5\beta}{1+\beta}s_{1w}^0 + \frac{5+3\beta}{1+\beta}s_{1\ell}^0$. The difference is $\frac{1}{2}s_{2w}^0 + \frac{4\beta}{1+\beta}s_{1w}^0 + \frac{1.5\beta-0.5}{1+\beta}s_{1\ell}^0$. To see that this value is positive, note that if $\mathbf{s}^1$ belongs to Scenario 1, then we have $s_{2\ell}^1 \geq s_{1w}^1 \iff \frac{1}{2}s_{2w}^0 + \beta s_{1w}^0 \geq \frac{1}{2}s_{1\ell}^0 \iff 2\beta s_{1w}^0 \geq s_{1\ell}^0 - s_{2w}^0$. Thus,

$$\begin{aligned} & \frac{1}{2}s_{2w}^0 + \frac{4\beta}{1+\beta}s_{1w}^0 + \frac{1.5\beta-0.5}{1+\beta}s_{1\ell}^0 \\ & \geq \frac{1}{2}s_{2w}^0 + \left(\frac{2}{1+\beta}s_{1\ell}^0 - \frac{2}{1+\beta}s_{2w}^0\right) + \frac{1.5\beta-0.5}{1+\beta}s_{1\ell}^0, \\ & = 1.5s_{1\ell}^0 - \frac{1.5-0.5\beta}{1+\beta}s_{2w}^0, \end{aligned}$$

where the inequality is by plugging in $2\beta s_{1w}^0 \geq s_{1\ell}^0 - s_{2w}^0$. Because $\frac{1.5-0.5\beta}{1+\beta} < 1.5$ and $s_{2w}^0 < s_{1\ell}^0$, which value is positive. Thus, the engagement always decreases in the presence of PTW.

(c) Fix $s_{2w}^0/s_{2\ell}^0$ and $s_{1w}^0/s_{1\ell}^0$ and vary $(s_{2w}^0 + s_{2\ell}^0)/(s_{1w}^0 + s_{1\ell}^0)$. Consider $\mathbf{s}^0$ such that $(s_{2w}^0 + s_{2\ell}^0)/(s_{1w}^0 + s_{1\ell}^0) = 0$, i.e., there are only low players. Then the optimal matching is simply SBMM. When $r = 0$, in the presence of PTW, some of the low player now becomes high player, which enables cross-level matchmaking, and we must have $ENG(\beta, 0, \mathbf{s}^0) = V^*(\beta, 0, \mathbf{s}^0) > V^*(0, 0, \mathbf{s}^0)$. That said, when there is no revenue, the engagement is higher thanks to the change in demographic distribution. Because $ENG^*(\beta, r, \mathbf{s}^0)$ is a continuous function of r , for r that is sufficiently close to 0, the engagement is still higher than SBMM. $\square$

B.1.4. Proof of Proposition 1 and Theorem 4

Proof of Proposition 1. Under SBMM, players are only matched with others within the same skill level. Thus, the only losses of the system are from losing players with high and low skills, respectively, and the two skill levels do not affect each other. As a result, in order to maximize engagement by adjusting the bot ratio, we need to minimize the churning rate of losing players. In other words, we need to maximize the winrate of losing players.

The winrate of losing players in either skill level can be written as the following:

$$\alpha(1 - p(\alpha)) + \frac{1 - \alpha}{2},$$

where the first term represents that with probability α , a losing player shall be matched with a bot, and she has probability $1 - p(\alpha)$ to stay after being matched with a bot; the second term represents that with probability $1 - \alpha$, a losing player is still matched with another human player with the same skill level, and thus, she has 50% chance to win and stay in the system for subsequent matchings. This completes the proof of the desired result. $\square$

Proof of Theorem 4. To better facilitate this proof, we first present an equivalent primal formulation in the presence of bots.

Note that when matched with a bot, a losing player still has a probability $(1 - p(\alpha))$ to leave the system. As a result, we can split the flow that involves bots into two: one to a “perfect bot” that guarantees the return of the human player in the next period; one to a “skill-based bot” that perfectly mimics a human

player with the same skill level. In other words, an imperfect bot (that induces churning) can be viewed as a combination of a perfect bot (that does not induce churning) and SBMM.

Mathematically, for one unit of flow of losing players there is $(1 - \alpha)$ fraction shall participate in regular matching without bots. Among the α fraction that is matched with bots, there is $2p(\alpha)$ fraction matched to perfect bots, and $1 - 2p(\alpha)$ fraction matched based on SBMM. One can easily verify that we have $2p(\alpha) + 1 - 2p(\alpha) = 1$, representing the decomposition induces the same amount of total flow, and $1 - p(\alpha) = \frac{1}{2}2p(\alpha) + (1 - 2p(\alpha))$, representing the decomposition gives the same non-churning players.

Using the flow decomposition above, we introduce $f_{i,PB}^t$, where $i \in \{2\ell, 1\ell\}$ and PB abbreviated for *perfect bots*, to denote the flow of losing players who are matched to perfect bots. Then the engagement maximization problem is

$$\max_{\alpha} P_{bot}(\alpha), \quad (\text{EC.44})$$

where

$$P_{bot}(\alpha) := \max_{f^t} \sum_{t=0}^{\infty} \left(f_{2w,2w}^t + \frac{1}{2}f_{2\ell,2\ell}^t + f_{1w,1w}^t + \frac{1}{2}f_{1\ell,1\ell}^t + 2f_{2w,1w}^t + f_{2\ell,1\ell}^t + f_{2w,1\ell}^t + 2f_{2\ell,1w}^t + f_{2\ell,PB}^t + f_{1\ell,PB}^t \right)$$

such that

$$\begin{aligned} s_{2w}^0 &= f_{2w,2w}^0 + f_{2w,1w}^0 + f_{2w,1\ell}^0, \\ s_{2\ell}^0 &= f_{2\ell,2\ell}^0 + f_{2\ell,1w}^0 + f_{2\ell,1\ell}^0 + f_{2\ell,PB}^0, \\ s_{1w}^0 &= f_{1w,1w}^0 + f_{2\ell,1w}^0 + f_{2w,1w}^0, \\ s_{1\ell}^0 &= f_{1\ell,1\ell}^0 + f_{2w,1\ell}^0 + f_{2\ell,1\ell}^0 + f_{1\ell,PB}^0, \end{aligned}$$

and for all $t = 1, 2, \dots$

$$\begin{aligned} s_{2w}^t &= f_{2w,2w}^t + f_{2w,1w}^t + f_{2w,1\ell}^t = \frac{1}{2}f_{2w,2w}^{t-1} + \frac{1}{2}f_{2\ell,2\ell}^{t-1} + f_{2\ell,PB}^{t-1} + f_{2w,1w}^{t-1} + f_{2w,1\ell}^{t-1} + f_{2\ell,1w}^{t-1} + f_{2\ell,1\ell}^{t-1}, \\ s_{2\ell}^t &= f_{2\ell,2\ell}^t + f_{2\ell,1w}^t + f_{2\ell,1\ell}^t + f_{2\ell,PB}^t = \frac{1}{2}f_{2w,2w}^{t-1}, \\ s_{1w}^t &= f_{1w,1w}^t + f_{2\ell,1w}^t + f_{2w,1w}^t = f_{1\ell,PB}^{t-1} + \frac{1}{2}(f_{1w,1w}^{t-1} + f_{1\ell,1\ell}^{t-1}) \\ s_{1\ell}^t &= f_{1\ell,1\ell}^t + f_{2w,1\ell}^t + f_{2\ell,1\ell}^t + f_{1\ell,PB}^t = \frac{1}{2}f_{1w,1w}^{t-1} + f_{2w,1w}^{t-1} + f_{2\ell,1w}^{t-1}, \end{aligned}$$

and for all $t = 0, 1, 2, \dots$

$$\begin{aligned} f_{2\ell,PB}^t &\leq \alpha(1 - 2p(\alpha))s_{2\ell}^t, \\ f_{2\ell,1\ell}^t + f_{2\ell,1w}^t &\leq (1 - \alpha)s_{2\ell}^t, \\ f_{1\ell,PB}^t &\leq \alpha(1 - 2p(\alpha))s_{1\ell}^t, \\ f_{2w,1\ell}^t + f_{2\ell,1\ell}^t &\leq (1 - \alpha)s_{1\ell}^t. \end{aligned} \quad (\text{EC.45})$$

where constraints in (EC.45) represent the flow decomposition to perfect bots (fraction $\alpha(1 - 2p(\alpha))$) and regular matching (fraction $1 - \alpha$, and the rests are to SBMM).

Immediately, we observe that α does not impact the objective in $P_{bot}(\alpha)$, but the constraints, in particular, constraints in (EC.45).

Next, we show that any $\alpha > \alpha^{SBMM}$ cannot be optimal. When $\alpha(1 - p(\alpha)) + \frac{1}{2}(1 - \alpha)$ is concave, it renders a unique maximizer α^{SBMM} , which solves

$$\frac{1}{2} - p(\alpha) - \alpha p'(\alpha) = 0. \quad (\text{EC.46})$$

Consider any $\alpha > \alpha^{SBMM}$, we must have

$$\left. \frac{d[\alpha(1 - 2p(\alpha))]}{d\alpha} \right|_{\alpha=\alpha'} = 1 - 2p(\alpha) - 2\alpha p'(\alpha) < 0,$$

according to concavity. Therefore, function $\alpha(1 - 2p(\alpha))$ decreases in α for all $\alpha > \alpha^{SBMM}$. Furthermore, $1 - \alpha$ also decreases in α . Thus, the feasible region described by (EC.45) strictly decreases when increasing α for any $\alpha > \alpha^{SBMM}$. As a result, for any $\alpha < \alpha'$ where $\alpha' > \alpha^{SBMM}$, we have $P_{bot}(\alpha) \geq P_{bot}(\alpha')$. This completes the proof. $\square$

B.2. Omitted Proofs from Appendix A

Proof of Lemma EC.1. Denote $\mathbf{v}_k = \{v_{kg}\}_{g \in \bar{\mathcal{G}}}$ as the vector of value (engagement) that 1 unit of players in level k can create. Note that v_{kg} is the average active time starting from state g in the absorbing Markov chain M_{kk} before absorption. By Theorem 3.2.4 in Kemeny et al. (1960), we have

$$\begin{aligned} \mathbf{v}_k &= (\bar{M}_{kk} + \gamma \bar{M}_{kk}^2 + \gamma^2 \bar{M}_{kk}^3 + \cdots) \mathbf{1} \\ &= \left(\frac{1}{\gamma} (I + \gamma \bar{M}_{kk} + \gamma^2 \bar{M}_{kk}^2 + \cdots) - \frac{1}{\gamma} I \right) \mathbf{1} \\ &= \gamma^{-1} \left((I - \gamma \bar{M}_{kk})^{-1} - I \right) \mathbf{1}. \end{aligned}$$

Note that by Theorem 3.2.1 in Kemeny et al. (1960), $(I - \gamma \bar{M}_{kk})^{-1}$ always exists. Summing up players at all levels, we have

$$V^{SBMM}(\mathbf{s}^0) = \sum_{k=1}^K \mathbf{v}_k^\top \mathbf{s}_k^0.$$

$\square$

Proof of Lemma EC.2. We prove this lemma by induction when $\gamma = 1$. For the base case, consider the engagement of level 1 players. Because their winrate is at most 0.5, a positive proportion of level 1 players would experience $m + 1$ consecutive losses and quit, for every $m + 1$ rounds. Let $\epsilon > 0$ be the probability that a player quits after $m + 1$ consecutive losses. After $m + 1$ rounds of matches, a level 1 player starting from any state has at least $2^{-(m+1)}\epsilon$ probability to quit. Hence, a player's engagement is bounded by

$$(m+1)(1 - 2^{-(m+1)}\epsilon) + (m+1)(1 - 2^{-(m+1)}\epsilon)^2 + (m+1)(1 - 2^{-(m+1)}\epsilon)^3 + \cdots = \frac{(m+1)(1 - 2^{-(m+1)}\epsilon)}{2^{-(m+1)}\epsilon}. \quad (\text{EC.47})$$

The induction hypothesis is that the engagement is finite for players of level 1 to level k . We then show that the engagement is also finite for players of level $k + 1$. From the induction hypothesis, the engagement from matches with players from levels 1 to k must be finite. Thus, we only need to consider engagement generated from matches with players of level $k + 1$ to K , and show it is also finite. Note that level $k + 1$ players' winrate is at most 0.5 since they are matched with players with at least the same skill level. Thus,

following the exact same argument of the base case with players of level 1, we can show that a player's engagement is bounded by the same expression in (EC.47), which is finite.

Finally, when $\gamma < 1$, the claim is obvious since the amount of active players is non-increasing over time.

□

Proof of Lemma EC.3. Our proof relies on Theorem 2.3 in Ghate (2015), which requires showing that our LP formulation (EC.2) satisfies the following five hypotheses. Let $X \subseteq \mathcal{R}^{\mathbb{N}}$ be a linear subspace. For an infinite primal LP

$$V(P) = \sup \sum_{j=1}^{\infty} c_j x_j \quad (\text{EC.48})$$

$$\sum_{j=1}^{\infty} a_{ij} x_j = b_i, \quad i = 1, 2, \dots \quad (\text{EC.49})$$

$$x_j \geq 0, \quad j = 1, 2, \dots \quad (\text{EC.50})$$

$$x \in X, \quad (\text{EC.51})$$

we assume for any $x \in X$

$$\text{H1. } \sum_{j=1}^{\infty} c_j x_j < \infty.$$

$$\text{H2. } \sum_{j=1}^{\infty} a_{ij} x_j < \infty, \quad j = 1, 2, \dots$$

Further, let $Y \subseteq \mathcal{R}^{\mathbb{N}}$ be the subset of all $y \in \mathcal{R}^{\mathbb{N}}$ such that

$$\text{H3. } \sum_{i=1}^{\infty} b_i y_i < \infty.$$

$$\text{H4. For every } x \in X, \sum_{j=1}^{\infty} |a_{ij} x_j y_i| \text{ converges to some limit } L_i(x, y_i) \text{ for } i = 1, 2, \dots, \text{ and}$$

$$\text{H5. The above limits } L_i(x, y_i) \text{ have the property that } \sum_{i=1}^{\infty} L_i(x, y_i) < \infty.$$

Then consider the dual problem

$$V(D) = \inf \sum_{i=1}^{\infty} b_i y_i \quad (\text{EC.52})$$

$$\sum_{i=1}^{\infty} a_{ij} y_i \geq c_j, \quad j = 1, 2, \dots \quad (\text{EC.53})$$

$$y \in Y. \quad (\text{EC.54})$$

By Theorem 2.3 in Ghate (2015), suppose $x \in X$ and $y \in Y$ are feasible to the primal and dual problems and are complementary ($x_j(c_j - \sum_{i=1}^{\infty} a_{ij} y_i) = 0$ for all j). Then x and y are optimal solutions to the primal and dual problems, and $V(P) = V(D)$.

For our problem in (EC.2), let X to be the l_1 sequence space. Because of Lemma EC.2, letting X be the l_1 sequence space is without of generality. We check hypotheses H1 to H5, respectively. For dual variables, we only consider y from l_{∞} space. As we will show, any y from l_{∞} space satisfy H3 to H5, and we only use such y in the following proofs.

Hypothesis H1 is satisfied because X is the l_1 space. Hypothesis H2 and H4 are satisfied since we have finitely many primal variables in each constraint of problem EC.2 and X is the l_1 space. Hypothesis H3 is satisfied since only the constraints associate with the initial period ($t = 0$) leads to nonzero values (thus

finite), and y is in l_∞ space. Finally, for hypothesis H5, let A_i be the set of nonzero columns for row i . Note that we have

$$\sum_{i=1}^{\infty} L_i(x, y_i) = \sum_{i=1}^{\infty} \sum_{j \in A_i} |a_{ij} x_j y_i| \leq \sum_{i=1}^{\infty} \sup_{i \in \mathbb{N}} |y_i| \sum_{j \in A_i} x_j = \sup_{i \in \mathbb{N}} |y_i| \sum_{i=1}^{\infty} \sum_{j \in A_i} x_j \leq 2|\bar{\mathcal{G}}| \sup_{i \in \mathbb{N}} |y_i| \sum_{j \in \mathbb{N}} |x_j| < \infty,$$

where the first inequality follows the fact $a_{ij} \in [0, 1]$ and Y is l_∞ space, the second inequality follows the fact that each primal variable x_j appears in two periods, so it shows up in at most $2|\bar{\mathcal{G}}|$ constraints in problem (EC.2), and the last inequality follows from the fact that X is the l_1 space. $\square$